44.1 1. A linear space containing the relevant curvature hull

Fix a finite physical torus, or a fixed compact coordinate patch after multiplication by a smooth cutoff. Let $0 < s < 1$. For a matrix-valued one-form define

$$\|A\|_{\mathcal{X}_s} = \|A\|_{W^{1,\infty}} + \|A\|_{H^2} + \|dA\|_{H^{1+s}}. \tag{FH1}$$

This is a norm on a linear space. For the usual curvature

$$F(A) = dA + A \wedge A,$$

one has

$$\|A \wedge B\|_{H^2} \le C\big(\|A\|_{W^{1,\infty}}\|B\|_{H^2} + \|B\|_{W^{1,\infty}}\|A\|_{H^2}\big). \tag{FH2}$$

Indeed, the second product derivative consists of a bounded factor times a second derivative and two first-derivative factors, one of which is bounded. The zero- and first-order estimates are easier. Since H^2 embeds continuously into H^{1+s}, (FH2) proves that

$$A \in W^{1,\infty} \cap H^2, \quad F(A) \in H^{1+s} \quad \Longleftrightarrow \quad A \in \mathcal{X}_s. \tag{FH3}$$

The curvature map is a continuous quadratic map from $\mathcal{X}_s$ to H^{1+s}. In particular, it is holomorphic on its complexification. Its first two derivatives are

$$DF(A)B = dB + A \wedge B + B \wedge A, \qquad D^2F(A)[B,C] = B \wedge C + C \wedge B. \tag{FH4}$$

No independent-current domain is being redefined by (FH1). It is an auxiliary space for the actual evaluated fields.

44.1.1 1.1 Scalar interpolation

For any real or complex number t, the exact identity

$$F(tA) = tF(A) + (t^2 - t)A \wedge A \tag{FH5}$$

gives

$$\|F(tA)\|_{H^{1+s}} \le |t|\|F(A)\|_{H^{1+s}} + C|t^2 - t|\|A\|_{W^{1,\infty}}\|A\|_{H^2}. \tag{FH6}$$

Thus loss of the nonlinear Euler equation along tA does not imply loss of the fractional curvature norm. On every fixed bounded t-disc, that norm stays bounded by the displayed quantities.

More generally, a finite affine combination of fields in $\mathcal{X}_s$ stays in that space. The curvature cross terms are controlled by (FH2). This is the relevant distinction between a critical submanifold and a linear regularity hull containing it.

44.1.2 1.2 A smooth spatial cutoff

For a fixed smooth scalar function χ,

$$F(\chi A) = \chi F(A) + d\chi \wedge A + (\chi^2 - \chi)A \wedge A. \tag{FH7}$$

Multiplication by $\chi \in C^3$ is bounded in the norms appearing here. Therefore

$$\|\chi A\|_{\mathcal{X}_s} \le C_s(1 + \|\chi\|_{C^3})\|A\|_{\mathcal{X}_s}. \tag{FH8}$$

The corresponding curvature estimate contains the quadratic term from (FH7). If $\chi(x) = \chi_0(x/\rho)$, its derivative costs must be restored as powers of ρ^{-1}. A cutoff at the fine lattice spacing is not a fixed smooth cutoff in (FH8).

The same product estimates hold for compatible lattice difference norms. The exact difference-product rule contains translated factors, whose L^p norms are unchanged, and gives (FH2) with constants independent of the mesh. This observation is about algebraic products in a controlled gauge chart; it does not identify arbitrary random-walk localization with multiplication by a scalar cutoff.

44.1.3 1.3 Parameter derivatives

At a fixed cutoff, let $A(z)$ be analytic on $|z_i| < \rho_i$ and suppose $\sup_z \|A(z)\|_{\mathcal{X}_s} \le K$. Applying the scalar Cauchy formula componentwise and then using the Banach norm yields

$$\|\partial^\alpha A(0)\|_{\mathcal{X}_s} \le K\,\alpha! \prod_i \rho_i^{-\alpha_i}. \tag{FH9}$$

All constants are uniform in the cutoff when the same K and radii work there. Analyticity only in a weaker norm does not supply the premise of (FH9).

44.2 2. A protected local equation supplies a fractional hull

The protected-seed parametrix identity in the local-residual chapter is useful precisely because it can give an actual local equation even when the global auxiliary field is not a minimizer. Here is a simple rigorous consequence in a continuum gauge chart.

Suppose $A \in W^{1,\infty} \cap H^2$ on a ball B_2, it satisfies the local Coulomb condition $\partial_\mu A_\mu = 0$, and its actual Euler equation is

$$D_\mu F_{\mu\nu}(A) = J_\nu \tag{FH10}$$

there, with $J \in H^s(B_2)$, $0 < s < 1$. Expanding (FH10) gives

$$\Delta A_\nu = J_\nu + N_\nu(A, \partial A), \tag{FH11}$$

where N is a fixed sum of quadratic terms $A\partial A$ and cubic terms in A, with signs depending on the convention for D. The exact coefficients do not affect the following norm estimate. The ordinary product rule gives

$$\|N(A, \partial A)\|_{H^1(B_2)} \le C(K + K^2)\|A\|_{H^2(B_2)}, \quad K = \|A\|_{W^{1,\infty}(B_2)}. \tag{FH12}$$

Choose $\chi \in C_c^\infty(B_2)$ with $\chi = 1$ on B_1. The equation for χA has right-hand side $\chi J + \chi N$ plus the two cutoff commutators. The latter belong to H^1, since $A \in H^2$. Fourier inversion of the ordinary Laplacian, retaining the harmless L^2 term at zero frequency, gives

$$\|A\|_{H^{2+s}(B_1)} \le C_s \left[\|J\|_{H^s(B_2)} + (1 + K + K^2)\|A\|_{H^2(B_2)} \right]. \tag{FH13}$$

It follows that $F(A) \in H^{1+s}(B_1)$, with the corresponding bound. This is an elementary local implication, and it works for complex matrix coefficients as well.

For the actual lattice Wilson field the more appropriate statement is the exact cube/flux Hodge argument in the dimension-six chapter, (D6.22)–(D6.33). That argument avoids imposing an unverified local Coulomb equation on the original projected gauge. The present calculation explains how a verified protected local equation could be used; it does not assert that all independently localized propagators possess such an equation.

44.3 3. Uniform Sobolev stability of flat box reconstruction

We now prove an all-depth flat estimate without multiplying a one-step Sobolev constant at every scale.

Let the fine spacing be a, let the retained spacing be $r = na$, and let Q_r be normalized averaging over disjoint n^d-site boxes. One may also compose it with a normalized length-n line average in a fixed coordinate direction. Use physical volume pairings. On the mean-zero subspace let

$$L_a = -\Delta_a, \quad C_r = Q_r L_a^{-1} Q_r^*, \quad H_r = L_a^{-1} Q_r^* C_r^{-1}, \quad P_r = H_r Q_r. \tag{FH14}$$

Constants can instead be retained by using $L_a = m^2 - \Delta_a$, with $m > 0$. The map P_r is the flat constrained energy minimizer with the same block data as its argument.

Flat reconstruction theorem. For every fixed $0 < s < 1/2$,

$$\|P_r A\|_{H_a^{2+s}} \le C_{s,d} \|A\|_{H_a^{2+s}}, \tag{FH15}$$

uniformly in a, r/a, and the number of intermediate blockings. The constant is also uniform in the physical torus volume under the usual Fourier normalization. The statement is on the nonzero modes when $m = 0$.

44.3.1 3.1 The exact alias-fibre calculation

Fix a coarse momentum θ in its Brillouin zone. Its fine aliases are $p_\nu = \theta + 2\pi\nu/r$, reduced into the fine Brillouin zone. Put

$$\lambda_\nu = m^2 + \widehat{\Delta}_a(p_\nu), \qquad q_\nu = \widehat{Q}_r(p_\nu), \qquad C(\theta) = \sum_\nu \frac{|q_\nu|^2}{\lambda_\nu}. \tag{FH16}$$

Phases from cell centres do not affect the estimates. On this finite alias fibre,

$$(P_r u)_\nu = \frac{\overline{q}_\nu}{\lambda_\nu C(\theta)} \sum_\omega q_\omega u_\omega. \tag{FH17}$$

The squared operator norm in the weighted space with norm $\sum_\nu \lambda_\nu^{2+s} |u_\nu|^2$ is exactly

$$\frac{\left(\sum_\nu |q_\nu|^2 \lambda_\nu^s\right)\left(\sum_\nu |q_\nu|^2 \lambda_\nu^{-2-s}\right)}{\left(\sum_\nu |q_\nu|^2 \lambda_\nu^{-1}\right)^2}. \tag{FH18}$$

This formula retains the same aliases in the numerator and denominator; separate crude inverse powers of the retained spacing would lose the estimate. When $m = 0$, it initially gives the homogeneous Sobolev norm. The same rank-one calculation with zero Sobolev exponent gives a uniform L^2 bound: its numerator is at most λ_0^{-2} and its denominator at least $c_d^4 \lambda_0^{-2}$. Combining the homogeneous and L^2 bounds gives the usual inhomogeneous H^{2+s} norm, uniformly in volume.

Let $\lambda_0 = m^2 + \widehat{\Delta}_a(\theta)$. Then

$$\lambda_\nu \geq \lambda_0, \quad |q_0| \geq c_d > 0, \quad \sum_\nu |q_\nu|^2 \leq 1. \tag{FH19}$$

For pure box averaging the last quantity is exactly one, because its adjoint is the isometric block hold. A subsequent line average is a contraction. The lower bound on q_0 follows in each coordinate from $\sin z/z \geq 2/\pi$ for $|z| \leq \pi/2$.

The only nontrivial upper moment is

$$\sum_\nu |q_\nu|^2 \lambda_\nu^s \leq C_{s,d} \lambda_0^s. \tag{FH20}$$

Here is a direct proof. In the continuum alias notation $z = r\theta \in [-\pi, \pi]^d$, the box factor in one coordinate is

$$b_k(z) = \frac{2\sin(z/2)}{z + 2\pi k}$$

up to a unit phase, with its continuous value at zero. Its squared sum is one. For $0 < s < 1/2$,

$$\sum_{k\in\mathbb{Z}} |b_k(z)|^2 |z + 2\pi k|^{2s} \leq |z|^{2s} + C\sin^2(z/2) \sum_{k\neq 0} |k|^{-2+2s} \leq C_s |z|^{2s}. \tag{FH21}$$

The series is the reason for the strict threshold $s < 1/2$. In d dimensions use $|\xi|^{2s} \leq \sum_j |\xi_j|^{2s}$, factor the remaining squared sums, and add the mass term. This proves (FH20) for continuum box factors.

For the finite lattice factor, the denominator $rp/2$ is replaced by $n\sin(ap/2)$. On the fine Brillouin zone, this changes its absolute value by a factor between one and $\pi/2$. The lattice Laplacian symbol is similarly comparable to $|p|^2$, with constants independent of a. Restricting the convergent alias sum to the finite fine fibre therefore proves (FH20) with a uniform constant. A line-average factor has absolute value at most one and preserves the lower bound at the principal alias.

Equations (FH19)–(FH20) imply

$$\sum |q_\nu|^2 \lambda_\nu^{-2-s} \leq \lambda_0^{-2-s}, \qquad C(\theta) \geq c_d^2 \lambda_0^{-1}.$$

Substitution in (FH18) gives a constant independent of the fibre and of both meshes. Summing the Fourier fibres proves (FH15).

44.3.2 3.2 A coarse-to-fine fractional lift bound

The same calculation, using only (FH20), proves

$$\|Q_r^*\lambda\|_{H_a^s} \le C_{s,d}\|\lambda\|_{H_r^s}, \qquad 0 < s < 1/2. \tag{FH22}$$

The coarse norm on the right uses its physical lattice Laplacian. Its symbol is uniformly comparable to the principal fine symbol on the coarse Brillouin zone. In the massless case, combine the homogeneous estimate with the elementary L^2 contraction of the hold/line map to obtain the inhomogeneous norm. This is a direct all-depth estimate for the sharp hold, including its faces.

The direct box/line averages also compose exactly at different integer scales. In one coordinate the elementary geometric-sum identity is

$$\left(\frac{1}{n}\sum_{j=0}^{n-1} z^j\right)\left(\frac{1}{m}\sum_{k=0}^{m-1} z^{nk}\right) = \frac{1}{mn}\sum_{v=0}^{mn-1} z^v. \tag{FH23}$$

Apply it to every box coordinate and to the extra line factor. Thus (FH22) bounds a whole composed flat prolongation, with one constant, rather than a product of constants.

44.3.3 3.3 The flat gauge-field version

For a flat one-form, use Q_μ equal to box averaging followed by line averaging in direction μ. Its exact chain identity is

$$Qd_a = d_r R, \tag{FH24}$$

where R is the site box average. A raw contour average differs from this Q by a coarse exact gradient, which disappears against a coarse co-closed multiplier.

On a coarse fibre let Π be the orthogonal projection onto coarse co-closed one-forms. Replace the scalar $C(\theta)$ by

$$C_\Pi(\theta) = \Pi\left(\sum_\nu Q_\nu \lambda_\nu^{-1} Q_\nu^*\right)\Pi\Big|_{\operatorname{ran}\Pi}. \tag{FH25}$$

Here Q_ν is the diagonal matrix of component box/line factors. The principal alias gives

$$c_d^2\lambda_0^{-1} I \le C_\Pi(\theta) \le \lambda_0^{-1} I. \tag{FH26}$$

The upper and lower moment estimates hold componentwise. Consequently the two factors mapping into and out of the coarse fibre have norms bounded by $C_s\lambda_0^{-1-s/2}$ and $C_s\lambda_0^{s/2}$, respectively, while $\|C_\Pi^{-1}\| \le C\lambda_0$. Their product is bounded. This proves (FH15) for the gauge-fixed flat reconstruction modulo gauge.

By (FH24), $Q^*\lambda$ is fine co-closed whenever λ is coarse co-closed. The inverse Laplacian then agrees with the curl-energy inverse on that subspace. Thus this is the physical flat Maxwell reconstruction, not an inverse imposed on a gauge zero mode. Constant harmonic modes can be retained separately; their curvature is zero. In particular,

$$\|d_a P_r A\|_{H_a^{1+s}} \le C_s \|A\|_{H_a^{2+s}}. \tag{FH27}$$

There is a useful version requiring only the fractional curl part of $\mathcal{X}_s$. Decompose A into its transverse, exact-gradient and harmonic parts. The gradient part drops out by (FH24), and the harmonic part has zero reconstructed curvature. The transverse representative satisfies $\|A_T\|_{H^{2+s}} \le C_s(\|A\|_2 + \|d_a A\|_{H^{1+s}})$: the L^2 term controls frequencies below one, and the curl controls the remaining transverse modes. Applying (FH15) therefore gives

$$\|d_a P_r A\|_{H_a^{1+s}} \le C_s \left(\|d_a A\|_{H_a^{1+s}} + \|A\|_2 \right). \tag{FH27a}$$

The constant is volume-uniform under the same Fourier normalization. No embedding $H^{2+s} \subset W^{1,\infty}$ in four dimensions is asserted. The separate low-regularity domain theorem must still supply the $W^{1,\infty}$ part of a nonlinear $\mathcal{X}_s$ claim.

44.4 4. A valid telescoping comparison for nonlinear currents

The one-step physical-current estimate in the physical-germs chapter decomposes the actual adjoint only on an input satisfying its actual Ward equation. This restriction prevents an unrestricted nonlinear Duhamel argument. There is, however, a valid comparison in which every propagated error uses only the flat operator.

Let $h_{j+1} = Lh_j$, $h_k = r$, and let λ_j be the complete actual current obtained from λ_{j+1}. Work first on a physical torus with compatible nested lattices and one coherent periodic gauge, and assume:

- every actual fine/coarse potential in this chart has physical sup norm at most K;
- the physical plaquette curvature is bounded by F_*;
- the terminal current obeys its actual Ward equation, so every complete current does also;
- the physical-volume L^2 norms and the complete Ward equation are taken on that same torus.

These are explicit chart hypotheses. The nonunitarity bound proved elsewhere controls path norms, but does not by itself give a bounded physical logarithm in a chosen gauge. A local application must first obtain a compatible extension or carry its actual boundary terms. Multiplying the current by a cutoff before using the Ward equation is not permitted: that multiplication creates a divergence term. One may localize the resulting complete-current estimate afterward.

Let T_j^0 be the flat box/line prolongation from h_{j+1} to h_j. The actual one-step decomposition and the elementary path comparison imply

$$\lambda_j = T_j^0 \lambda_{j+1} + e_j, \qquad \|e_j\|_2 \le C_L(Kh_{j+1} + F_* h_{j+1}^2)\|\lambda_{j+1}\|_2. \tag{FH28}$$

The first term in the error compares the positive transported hold/line map to its flat value; a path of length Ch_{j+1} differs from the identity by at most $CKh_{j+1}e^{CKh_{j+1}}$. The second is the already proved Ward-restricted actual-map error. A fixed small-field factor is included in C_L. Write the full adjoint as $T_U - r_U^* d_V^* + \widetilde{E}_U$ before restricting to the actual input. The endpoint term then vanishes on that input, while the separate operator $\widetilde{E}_U$ has full row and column bounds. Its Schur estimate, combined with the positive reference map, gives $\sup_j \|\lambda_j\|_2 \le C\|\lambda_k\|_2$.

At spacing $h_j = h_{j+1}/L$, the elementary Fourier bound is

$$\|e_j\|_{H^s_{h_j}} \le C_s(1 + h_j^{-s})\|e_j\|_2. \tag{FH29}$$

For $r \le 1$, this gives

$$\|e_j\|_{H^s_{h_j}} \le C_{s,L}(Kh_{j+1}^{1-s} + F_* h_{j+1}^{2-s})\|\lambda_k\|_2. \tag{FH30}$$

Unrolling the exact recurrence (FH28) yields

$$\lambda_0 - (T_0^0 \cdots T_{k-1}^0)\lambda_k = \sum_{j=0}^{k-1} (T_0^0 \cdots T_{j-1}^0) e_j. \tag{FH31}$$

Every tail here is a *flat* direct prolongation. Equations (FH22)–(FH23) give a uniform H^s norm for the tail, regardless of whether e_j is co-closed. Geometric summation therefore proves

$$\boxed{\|\lambda_0 - Q^*_{r,\mathrm{flat}}\lambda_k\|_{H^s_{h_0}} \le C_{s,L}(Kr^{1-s} + F_* r^{2-s})\|\lambda_k\|_2, \quad 0 < s < \tfrac{1}{2}.} \tag{FH32}$$

No actual-current estimate has been applied to an isolated e_j. The Ward equation is used only to establish (FH28) for the complete λ input before the recurrence is unrolled.

Combining (FH22) with (FH32) gives the useful sufficient bound

$$\|\lambda_0\|_{H^s_{h_0}} \le C_s \left[\|\lambda_k\|_{H^s_r} + (Kr^{1-s} + F_* r^{2-s})\|\lambda_k\|_2\right]. \tag{FH33}$$

This retains the terminal coarse H^s norm. Replacing it by the inverse estimate $Cr^{-s}\|\lambda_k\|_2$ would lose precisely the useful scale uniformity for arbitrary old readouts.

44.5 5. The nonlinear reconstruction estimate still required

For the actual unlocalized minimizer, nested constraints give an exact identity. If U_k minimizes the fine action subject to the composed k-scale constraint, then it also minimizes the action subject to the stronger condition $M^j U = M^j U_k$. Indeed, every competitor for the latter is a competitor for the former. On the unique regular branch,

$$U_j(M^j U_k) = U_k. \tag{FH34}$$

This is associativity of constrained minimization. It is not associativity of the nonlinear group mean. It allows a true coarse-k field to retain its k-scale current estimate when it is written as an inner j-scale reconstruction.

For an arbitrary interpolated outer field A, (FH34) is unavailable. The flat theorem (FH15) proves the appropriate stable-projection statement at zero background. The corresponding remaining nonlinear assertion is a bound of the form

$$\|\mathscr{R}_{r,U}(A)\|_{\mathcal{X}_s(\Omega)} + \|\lambda_{r,U}(A)\|_{H^s_r(\Omega')} \le C_s\, \mathcal{P}\Big(\|A\|_{\mathcal{X}_s(\Omega'')} + \text{actual boundary data}\Big), \tag{FH35}$$

with fixed nested physical collars and constants independent of the retained old spacing r, in the common complex parameter domain used by the original formation map. Here $\mathscr{R}$ is the actual

constrained reconstruction and λ its actual multiplier. The polynomial $\mathcal{P}$, gauge choice, small-field radii and boundary norm must be specified by such a theorem; (FH35) is a target, not a result proved in this chapter.

The actual-current comparison (FH33) addresses the adjoint-lift part of that task. The nonlinear coarse multiplier estimate and every mixed determining-set or wall-localized reconstruction still need verification. The protected-seed identity can identify a local equation only after its typed operator factors have been matched to the original maps. These are concrete restrictions on the new fractional route, and prevent the flat result from being silently promoted to the full interacting construction.

45.1 1. The actual mean and the exact reflection

Let the fine spacing be a, the next spacing be $\ell = La$, and let blocks be centered as in the original construction. Use anti-Hermitian fine potentials at bond midpoints:

$$U_\nu(x, x + ae_\nu) = e^{aA_\nu(x+ae_\nu/2)}.$$

The field on a reversed bond has the negative logarithm. Fix a regular chart with $\ell\|A\|_\infty$ sufficiently small.

For each coarse bond, write the actual outer operation as

$$\mathcal{M}(\{P_i\}, R) = \exp\left(\sum_i w_i \log(P_i R^{-1})\right) R, \qquad w_i \geq 0, \quad \sum_i w_i = 1. \tag{CM1}$$

Here $R = U(c)$ is the specified reference transport. The P_i are the original center-to-center contours, or the corresponding products with the inner averaged endpoint contours of (0.11). All variables, including R, depend on the fine field.

The exact identity

$$\mathcal{M}(\{P_i^{-1}\}, R^{-1}) = \mathcal{M}(\{P_i\}, R)^{-1} \tag{CM2}$$

follows because $P_i^{-1}R = R^{-1}(P_iR^{-1})^{-1}R$. The principal logarithm respects inversion and conjugation in the common regular domain. The contour sets are closed under reversal and under lattice reflections. The inner endpoint average has the same inversion and permutation symmetries. Therefore full inversion about the midpoint c of a coarse bond has the exact action

$$q_{\ell,\mu}(\iota_c^* A)(c) = -q_{\ell,\mu}(A)(c), \qquad q_{\ell,\mu}(A) = \ell^{-1} \log V_\mu, \tag{CM3}$$

where on fine bond midpoints

$$(\iota_c^* A)_\nu(z) = -A_\nu(2c - z).$$

The coarse bond reverses and its midpoint is fixed. The centered block partition is carried to itself with its endpoints exchanged; odd L preserves the fine/coarse midpoint conventions.

In particular a spatially constant input remains constant on the coarse lattice, and

$$q_\ell(-C) = -q_\ell(C) \tag{CM4}$$

for every sufficiently small constant collection $C = (C_1, \dots, C_d)$, with no commutativity assumption on its matrices. Every even field-degree Taylor coefficient therefore vanishes on constant data.

45.1.1 1.1 The reference term really cancels at quadratic order

Set $P_i = e^{X_i}$, $R = e^Y$, and $\overline{X} = \sum_i w_i X_i$. BCH through degree two gives

$$\sum_i w_i \log(e^{X_i} e^{-Y}) = \overline{X} - Y - \tfrac{1}{2}[\overline{X}, Y] + O(3).$$

Recombining the outside reference factor in (CM1) gives

$$\log \mathcal{M}(\{e^{X_i}\}, e^Y) = \overline{X} + O(3). \tag{CM5}$$

Thus Y, including its field dependence, cancels through quadratic order in the complete outer expression. The X_i themselves still have quadratic path/contour terms, which must be retained. Dropping the derivative of R before recombination would not give (CM5).

This direct calculation is consistent with, but weaker than, the exact symmetry (CM4). It does not require nonlinear associativity of the means.

45.2 2. The analytic finite kernel and its zero constant moment

At fixed L, each output depends on a fixed finite set of fine bonds in a patch of diameter $C_L \ell$. In the scaled variables ℓA, the products, logarithms, inner mean and outer mean are analytic on one fixed polydisc. Consequently

$$q_\ell(A) = Q_{\ell,\mathrm{raw}} A + \sum_{n \geq 2} \ell^{n-1} B_n(A, \dots, A), \tag{CM6}$$

where B_n is a translation-covariant finite multilinear kernel and its absolute coefficient sum is bounded by C_L^n, after choosing a fixed smaller chart. The Taylor factorial is included in B_n. Matrix multiplication order is retained inside these multilinear maps; no scalar commutativity is used.

The quadratic kernel is the symmetric second Fréchet coefficient. Equation (CM4) gives

$$B_2(C, C) = 0 \tag{CM7}$$

for every constant collection C. Polarization yields $B_2(C, D) = 0$ for any two constant collections. This is the exact zero constant moment, including all component/matrix signs.

For an output midpoint c, choose, for each component, one fine bond midpoint c_ν in the determining patch. Define the constant collection $C_\nu = A_\nu(c_\nu)$. Then the exact algebra is

$$B_2(A, A) = B_2(A - C, A) + B_2(C, A - C). \tag{CM8}$$

Every nonzero term now contains a spatial difference of A over distance at most $C_L\ell$. No evaluation of A_ν at a point outside its actual fine component lattice is needed.

In particular, with $M_0 = \|A\|_\infty$, $M_1 = \|\nabla_a A\|_\infty$,

$$\|q_\ell(A) - Q_{\ell,\mathrm{raw}}A\|_\infty \le C_L\ell^2(M_0M_1 + M_0^3) \tag{CM9}$$

provided $C_L\ell M_0$ is small. The degree-two contribution uses (CM8). The degree $n \ge 3$ terms are summed as a convergent geometric series and cost $\ell^2 M_0^3$.

45.3 3. Coarse sampling has a fixed-L Sobolev bound

Use the physical discrete Bessel norm

$$\|u\|_{H_b^t}^2 = \sum_p (1 + \widehat{\Delta}_b(p))^t |\widehat{u}(p)|^2$$

at spacing b, with physical-volume Fourier normalization. Translates and the fixed phases of bond-centered components have unchanged norms.

Let $S_\zeta u$ retain one fixed fine offset in each coarse block. Its alias fibre has L^d entries. On that fibre the coarse symbol obeys

$$1 + \widehat{\Delta}_\ell(\theta) \le 1 + \widehat{\Delta}_a(p_\nu) \tag{CM10}$$

for EVERY alias $p_\nu = \theta + 2\pi\nu/\ell$. This follows coordinatewise from $|\sin Lt| \le L|\sin t|$. Alias Cauchy-Schwarz therefore proves, for any $t \ge 0$,

$$\|S_\zeta u\|_{H_\ell^t} \le L^{d/2}\|u\|_{H_a^t}. \tag{CM11}$$

The constant depends on the fixed blocking factor, not on either physical spacing. This is a literal lattice sampling statement. No continuum trace of a second derivative along a line is being assumed.

A fine path between two offsets contains $O_L(1)$ nearest-neighbor steps. Telescoping gives

$$\|S_\zeta A - S_{\zeta'}A\|_{H_\ell^s} \le C_L\ell\|\nabla_a A\|_{H_a^s}, \qquad \|S_\zeta A - S_{\zeta'}A\|_\infty \le C_L\ell M_1. \tag{CM12}$$

For $0 < s < 1$, the discrete fractional product inequality is uniform in the mesh:

$$\|uv\|_{H_b^s} \le C_s\{\|u\|_\infty\|v\|_{H_b^s} + \|v\|_\infty\|u\|_{H_b^s}\}. \tag{CM13}$$

It follows directly from the fractional translation differences, or from the equivalent discrete Fourier norm. Finite-dimensional matrix products obey the same estimate.

Apply (CM11)–(CM13) to (CM8). The quadratic contribution in (CM6) is bounded by

$$C_{s,L}\ell^2\{M_0\|\nabla_a A\|_{H_a^s} + M_1\|A\|_{H_a^s}\}.$$

The analytic terms of degree at least three have bound $C_{s,L}\ell^2 M_0^2\|A\|_{H_a^s}$. We have therefore proved the actual one-step estimate

$$\boxed{\|N_\ell(A)\|_{H_\ell^s} \le C_{s,L}\ell^2\left[M_0\|\nabla_a A\|_{H_a^s} + (M_1 + M_0^2)\|A\|_{H_a^s}\right], \quad N_\ell = q_\ell - Q_{\ell,\mathrm{raw}}.} \tag{CM14}$$

The same proof works for complex fields in the common regular chart. Analytic composition estimates retain the actual matrix order. It gives the usual mixed Moser bound for a parameter derivative, including the term with the derivative's $W^{1,\infty}$ norm; no small Lipschitz constant in a weak norm alone is claimed.

45.4 4. Exact coarse gauge straightening and its sign

At the flat point the actual raw derivative is

$$Q_{\ell,\mathrm{raw}}A = Q_{\ell,\mathrm{bl}}A - d_\ell r_\ell(A). \tag{CM15}$$

Here $Q_{\ell,\mathrm{bl}}$ is normalized box averaging followed by the normalized line average in the bond direction. The dimensionless site field $r_\ell(A)(y)$ is the normalized average of the fine line integrals from the coarse center y to points of its centered block, averaged also over the original endpoint contour permutations. It has physical size $O_L(\ell\|A\|_\infty)$. For a constant collection C,

$$r_\ell(C) = 0 \tag{CM16}$$

because the average endpoint displacement is zero.

Define the EXPLICIT coarse gauge transformation

$$g_\ell(A)(y) = e^{-r_\ell(A)(y)}$$

and the analyzed coarse connection

$$V_\mu^{\mathrm{str}}(y) = g_\ell(A)(y)V_\mu(y)g_\ell(A)(y+\ell e_\mu)^{-1}, \qquad q_\ell^{\mathrm{str}}(A) = \ell^{-1}\log V^{\mathrm{str}}. \tag{CM17}$$

The exponent r_ℓ is dimensionless. At first order the additional potential is $+d_\ell r_\ell(A)$, so

$$Dq_\ell^{\mathrm{str}}(0) = Q_{\ell,\mathrm{bl}}. \tag{CM18}$$

This fixes the sign. There is no division of the gauge exponent by ℓ.

For constant C, (CM16) makes $q_\ell^{\mathrm{str}}(C) = q_\ell(C)$. Hence the exact constant-field oddness and quadratic moment cancellation also hold for the straightened map. It is still a fixed finite analytic local function of ℓA. The proof of (CM9), (CM14) therefore applies to

$$N_\ell^{\mathrm{str}} = q_\ell^{\mathrm{str}} - Q_{\ell,\mathrm{bl}} \tag{CM19}$$

with the same type of constants. It includes every nonlinear term of the gauge exponentiation.

Straightening is not needed for the raw ONE-STEP cancellation (CM14). It is useful when composing steps because the flat box/line maps have a uniform full-tail Sobolev bound.

Equation (CM17) is an exact coarse gauge transformation of the actual averaged field. Gauge covariance of the original map shows inductively that its iterated outputs are gauge equivalent to the outputs of the original unstraightened averaging chain. This is a statement about coarse gauge ORBITS. It is not equality of the two connection matrices in one fixed original coarse gauge, nor yet a fine-gauge identification of a nonlinear constrained minimizer with prescribed target data. That inverse-constraint issue must remain explicit.

45.5 5. Uniform composition of the nonlinear coarsening defect

Assume an initially small global potential chart on a periodic torus, or a common buffered chart containing all determining paths under consideration. Let $a_{j+1} = La_j = \ell_j$, $a_k = r \le 1$, and iterate

$$A_{j+1} = q_{\ell_j}^{\mathrm{str}}(A_j).$$

Write $M_j = \|A_j\|_\infty + \|\nabla_{a_j} A_j\|_\infty$ and $K_j = \|A_j\|_{H^{1+s}_{a_j}}$. All statements in this section concern this explicit analyzed connection chart.

The flat box/line map has exact L^∞ contraction. A coarse difference is a positive line average of fine differences:

$$\nabla_{\ell,\nu} Q_{\ell,\mathrm{bl}} A = Q_{\ell,\mathrm{bl}} L_{\nu,L} \nabla_{a,\nu} A. \tag{CM20}$$

Thus it contracts the displayed first-difference sup norm as well.

In the Bessel norms of Section 3 it is also a contraction for every $t \geq 0$. On each alias fibre its coefficients satisfy $\sum_\nu |q_\nu|^2 \leq 1$. Together with (CM10), weighted Cauchy-Schwarz gives

$$(1 + \widehat{\Delta}_\ell(\theta))^t \left| \sum_\nu q_\nu u_\nu \right|^2 \leq \sum_\nu (1 + \widehat{\Delta}_a(p_\nu))^t |u_\nu|^2. \tag{CM21}$$

No per-step equivalence constant is multiplied.

The coarse inverse inequalities and (CM9), (CM14) imply

$$\begin{aligned} M_{j+1} &\leq M_j + C_{s,L} \ell_j (M_j^2 + M_j^3), \\ K_{j+1} &\leq \{1 + C_{s,L} \ell_j (M_j + M_j^2)\} K_j. \end{aligned} \tag{CM22}$$

For the second line use $\|N\|_{H^{1+s}_\ell} \leq C\ell^{-1} \|N\|_{H^s_\ell}$. For the first use $\|\nabla_\ell N\|_\infty \leq C\ell^{-1} \|N\|_\infty$. The factor ℓ^2 in (CM14) is what leaves a summable ℓ in these two recurrences.

Choose $C_{s,L}(M_0 + M_0^2) r$ sufficiently small. Since $\sum_j \ell_j \leq r/(1 - L^{-1})$, a first-exit argument gives

$$M_j \leq 2M_0, \qquad K_j \leq e^{C_{s,L}(M_0 + M_0^2) r} K_0. \tag{CM23}$$

This also keeps every logarithm and mean inside the same fixed analytic chart.

Unroll the exact recurrence against the ENTIRE flat tail:

$$A_k - Q_{r,\mathrm{bl}} A_0 = \sum_{j<k} (Q_{k-1,\mathrm{bl}} \cdots Q_{j+1,\mathrm{bl}}) N^{\mathrm{str}}_{\ell_j}(A_j). \tag{CM24}$$

The tail is the direct box/line average, by the geometric-sum filter identity. Equations (CM14), (CM21), (CM23), and $\sum_j \ell_j^2 \leq r^2/(1 - L^{-2})$ prove

$$\boxed{\|A_k - Q_{r,\mathrm{bl}} A_0\|_{H^s_r} \leq C_{s,L} r^2 (M_0 + M_0^2) e^{C_{s,L}(M_0 + M_0^2) r} \|A_0\|_{H^{1+s}_{a_0}}.} \tag{CM25}$$

This is uniform in the microscopic mesh and in the number of intermediate steps.

If equivalent Sobolev norms are used instead of the explicit Bessel norm, retain a SINGLE uniform whole-flat-tail bound in (CM24). The resulting Volterra inequality has $K_n \leq CK_0 + C \sum_{j<n} \ell_j (M_j + M_j^2) K_j$; discrete Gronwall gives the same conclusion. Multiplying unrelated norm-equivalence constants at every step would not.

45.6 6. What this does and does not settle for dimension six

The exact quadratic cancellation removes the dangerous estimate $N_r = O(rA^2)$ in the weak Sobolev norm. Formula (CM25) instead has an r^2 factor. Thus a coarse inverse of size r^{-2}, such as the flat Schur operator in the reconstruction problem, need not generate a negative power of r from this particular constraint defect.

There are still real qualifications before concluding nonlinear inner-reconstruction stability.

- The statement is in the explicit regular potential chart and coarse gauge (CM17). The prescribed fine/coarse gauge and target orbit of the original constrained minimization must be matched.
- A current-dependent or wall-localized map is not identified with this genuine averaging chain by the estimate. Its exact typed equations and domain must be used.
- The mixed Moser derivative bound must be applied with the actual directional $W^{1,\infty}$ and Sobolev margins. Holomorphy in a weaker norm is insufficient.
- The complete dimension-six center, its descendants and lower-dimensional mixing have not been normalized here.

These qualifications are separate from the proved fixed-L moment cancellation, the coarse sampling bound, and the uniform coarsening-defect estimate.

45.7 7. A strong bound for the complete raw nonlinear remainder

The gauge-compatibility chapter proves the weaker-norm raw transverse estimate needed by the KKT argument. One can also estimate the FULL raw nonlinear remainder in H^{1+s}, provided its actual flat raw derivative is subtracted. The omitted gradient at first order is then retained inside that derivative.

Use the notation of Section 5, with final spacing r, and let

$$A_j^0 = Q_{j-1,\mathrm{bl}} \cdots Q_{0,\mathrm{bl}} A_0, \qquad E_j = A_j - A_j^0, \qquad m = M_0 + M_0^2, \quad K = \|A_0\|_{H^{1+s}}.$$

Apply (CM25) at the intermediate final spacing a_j, followed by one inverse difference on that lattice. The corresponding sup estimate follows from (CM9) and the same whole-flat-tail argument. Thus

$$\begin{aligned} \|E_j\|_{H^s_{a_j}} &\le C a_j^2 m K, \qquad \|E_j\|_{H^{1+s}_{a_j}} \le C a_j m K, \\ \|E_j\|_\infty &\le C a_j^2 M_0 m, \quad \|\nabla_{a_j} E_j\|_\infty \le C a_j M_0 m. \end{aligned} \tag{CM26}$$

The estimates are exact zero at $j = 0$; no inverse inequality is applied to a nonzero initial error. The fixed small-radius exponential in (CM23) is included in C.

At a final node, the accumulated straightening gauge is the ordered product

$$h_k(A_0) = \prod_{j<k}^{\leftarrow} \exp[-b_{j,r}], \qquad b_{j,r} = S_{\ell_j,r} r_{\ell_j}(A_j).$$

Let $b^0_{j,r} = S_{\ell_j,r} r_{\ell_j}(A_j^0)$. The sparse restriction theorem GQ10, applied to E_j, gives

$$\begin{aligned} \sum_{j<k} \|b_{j,r} - b^0_{j,r}\|_{H^s_r} &\le C \sum_{j<k} r^2 (\ell_j/r)^s a_j m K \\ &\le C r^3 m K. \end{aligned} \tag{CM27}$$

The pointwise zero-moment estimate GQ14 and (CM26) likewise give

$$\sum_{j<k} \|b_{j,r} - b^0_{j,r}\|_\infty \le C r^3 M_0 m. \tag{CM28}$$

Here $a_j = \ell_j/L$, and the geometric sum has exponent $s+1$. It is not proportional to the number of levels.

The following ordered-product estimate retains the matrix ordering and has a constant independent of the number of factors. If

$$B_\infty = \sum_j \|b_j\|_\infty < c, \qquad B_s = \sum_j \|b_j\|_{H^s},$$

then

$$\left\| \log\left(\prod_j^{\leftarrow} e^{-b_j} \right) + \sum_j b_j \right\|_{H^s} \le C_s B_\infty B_s, \qquad \left\| \log\left(\prod_j^{\leftarrow} e^{-b_j} \right) + \sum_j b_j \right\|_{\infty} \le C B_\infty^2. \tag{CM29}$$

One proof expands the ordered product and then the principal logarithm. After its linear part is removed, every word has at least two factors. Bound all but one of them in the sup norm and use the exact telescoping identity for a fractional translation difference. The sum of absolute product coefficients of total degree n is at most $B_\infty^n/n!$, and differentiating a word contributes at most n choices. The logarithm is summed on the fixed smaller matrix ball. Thus there is one C_s, rather than one Sobolev product constant at every level.

Set

$$\kappa_k(A_0) = -\log h_k(A_0), \qquad \kappa_k^{(1)}(A_0) = \sum_{j<k} b_{j,r}^0.$$

This is the actual first Fréchet coefficient of κ_k. Equations GQ16 and (CM27)–(CM29) prove

$$\begin{aligned} \|\kappa_k - \kappa_k^{(1)}\|_{H_r^s} &\le C(r^3 mK + r^4 M_0 K) \le C r^3 mK, \\ \|\kappa_k - \kappa_k^{(1)}\|_{\infty} &\le C(r^3 M_0 m + r^4 M_0^2) \le C r^3 M_0 m. \end{aligned} \tag{CM30}$$

The extra power r, compared with GQ18, comes from subtracting the complete linear accumulated gauge.

Let $q_{r,\mathrm{raw}}^{(k)}$ be the genuine raw composed physical potential, and write

$$Q_{r,\mathrm{raw}}^{(k)} = D q_{r,\mathrm{raw}}^{(k)}(0) = Q_{r,\mathrm{bl}} - d_r \kappa_k^{(1)}. \tag{CM31}$$

The exact last equality follows from the actual coarse gauge relation; it is not an identification with a differently chosen direct nonlinear average. Its action on gradients is

$$Q_{r,\mathrm{raw}}^{(k)} d_a = d_r S_{a,r}, \tag{CM32}$$

where $S_{a,r}$ restricts a fine site gauge parameter to the final coarse centers. This follows by differentiating the genuine raw gauge covariance.

The complete raw remainder is

$$\begin{aligned} N_{r,\mathrm{raw}}(A_0) &= q_{r,\mathrm{raw}}^{(k)}(A_0) - Q_{r,\mathrm{raw}}^{(k)} A_0 \\ &= (A_k - Q_{r,\mathrm{bl}} A_0) - d_r(\kappa_k - \kappa_k^{(1)}) + \mathcal{E}_r(\kappa_k, A_k), \end{aligned} \tag{CM33}$$

where $\mathcal{E}_r$ is the full nonlinear coarse gauge remainder in GQ20–22. Every term in this exact identity is retained.

The first term has H^{1+s} norm $CrmK$ by (CM26). The second costs two coarse inverse derivatives relative to H^s, and (CM30) supplies r^3. The last term has H^s norm $Cr^2 mK$ and costs one inverse derivative. Therefore

$$\boxed{\begin{aligned} \|N_{r,\mathrm{raw}}(A_0)\|_{H_r^s} &\le C_{s,L} r^2 (M_0 + M_0^2) \|A_0\|_{H_a^{1+s}}, \\ \|N_{r,\mathrm{raw}}(A_0)\|_{H_r^{1+s}} &\le C_{s,L} r (M_0 + M_0^2) \|A_0\|_{H_a^{1+s}}. \end{aligned}} \tag{CM34}$$

In the sup norms,

$$\|N_{r,\mathrm{raw}}\|_\infty \le C r^2 M_0 m, \qquad \|\nabla_r N_{r,\mathrm{raw}}\|_\infty \le C r M_0 m. \tag{CM35}$$

All constants are uniform in depth and microscopic mesh on the same small chart.

Finally, linearizing the full raw bound GQ25 at zero gives

$$\|Q^{(k)}_{r,\mathrm{raw}}A\|_{H^{1+s}_r} \le C_{s,L}\|A\|_{H^{1+s}_a}. \tag{CM36}$$

The full chain is bounded once. No product of a per-step raw norm appears.

Equations (CM32), (CM34), and (CM36) can be used in an actual projected-gauge closure: the target controls the longitudinal coarse data through the true center restriction, while the nonlinear part has a small $r(M_0 + M_0^2)$ factor. They do not themselves identify the original background-dependent Landau projector or prove the corresponding fine scalar reconstruction theorem.

45.8 Primary map reference

T. Balaban, *Renormalization group approach to lattice gauge field theories. I*, Commun. Math. Phys. **109** (1987), 249–301: primary PDF. The relevant exact maps are (0.4), (0.11), (0.12), with the group-mean symmetries (0.5)–(0.9). The proof above keeps the outer reference-log operation distinct from the inner endpoint contour mean.

46.1 1. The exact quotient identity

Write the gauge action as

$$U^g(x,x') = g(x)U(x,x')g(x')^{-1}. \tag{GQ1}$$

Thus $(U^g)^h = U^{hg}$. Let M_j be the actual raw block map at level j. Its covariance is

$$M_j(U^g) = M_j(U)^{S_j g}, \tag{GQ2}$$

where S_j restricts a fine gauge to the next block centers. Suppose an analyzed map differs from it by an actual coarse gauge transformation:

$$q_j(U) = M_j(U)^{v_j(U)}. \tag{GQ3}$$

Define $R_0 = W_0 = U$, $R_{j+1} = M_j(R_j)$, and $W_{j+1} = q_j(W_j)$. Then

$$W_j = R_j^{h_j(U)}, \qquad h_0 = 1, \qquad h_{j+1}(U) = v_j(W_j) S_j h_j(U). \tag{GQ4}$$

This follows by one substitution into (GQ2). In particular, the maps induced on gauge orbits are identical at every depth. No associativity of the nonlinear group means is used.

For a general field-dependent v_j, the gauge action on q_j is the cocycle

$$q_j(U^g) = q_j(U)^{\sigma_j(g,U)}, \qquad \sigma_j(g,U) = v_j(U^g) S_j g\, v_j(U)^{-1}. \tag{GQ5}$$

Surjectivity of S_j alone would not prove that a fixed representative of the q_j output is reachable. A map that completely fixes the output gauge supplies a simple counterexample to that general inference.

46.1.1 1.1 Exact representative reachability for the internal-path straightening

For the particular map in the centered-mean chapter,

$$v_j(U)(y) = \exp[-r_{\ell_j}(A_j)(y)], \qquad U(b) = \exp(a_j A_j(b)), \qquad \ell_j = L a_j, \tag{GQ6}$$

the linear dimensionless site field r_{ℓ_j} uses only center-to-site paths contained in the block of y. Let $H_j g$ be the fine site gauge that equals $g(y)$ throughout that block. Internal link logarithms then transform by the same conjugation. Consequently,

$$r_{\ell_j}(A_j^{H_j g})(y) = \mathrm{Ad}_{g(y)}\, r_{\ell_j}(A_j)(y), \qquad q_j(U^{H_j g}) = q_j(U)^g. \tag{GQ7}$$

The second identity uses both the first identity and the exact raw covariance. It is an equality of group-valued maps. For complex variables it is asserted on the common logarithmic branch; for real compact gauges the internal branches are preserved by conjugation.

The blocks are nested. A fine gauge constant on each final block remains constant on the appropriate intermediate blocks. Thus (GQ7) iterates and makes the final fixed representative reachable as well. This statement is about gauge orbits and action values. A blockwise constant gauge can have sharp jumps in a chosen physical potential chart. It is not a bounded map in the fine $W^{1,\infty}$ or higher Sobolev norms.

On a gauge-saturated admissible domain, a gauge-invariant action has the same minimizing fine orbits under the raw fixed-representative constraint, the straightened orbit constraint, and, using (GQ7), the straightened fixed-representative constraint. If the original construction uses a particular Landau representative, its regularity is still a theorem about that representative; it is not supplied by the discontinuous block hold.

46.1.2 1.2 Physical multiplier forces do not differentiate the gauge correction

Let $V = M(U)$ and $q(U) = V^{v(U)}$. Use physical left-trivialized link tangents and the corresponding volume pairings. Write $\mathcal{A}_v$ for adjoint transport of a coarse link tangent and Ξ_U for the site tangent $Dv(U)v(U)^{-1}$. With

$$(d_q \xi)(y,\mu) = \ell^{-1}(\mathrm{Ad}_{q_\mu(y)}\, \xi(y + \ell e_\mu) - \xi(y)),$$

direct differentiation of (GQ1) gives

$$Dq = \mathcal{A}_v DM - d_q \Xi_U. \tag{GQ8}$$

Therefore every coarse Ward covector λ, satisfying $d_q^* \lambda = 0$, obeys

$$Dq^* \lambda = DM^* \mathcal{A}_v^* \lambda. \tag{GQ9}$$

The transformed covector satisfies the raw Ward equation. This is an exact identification of the physical Lagrange-multiplier force on the quotient constraint. All derivatives of the field-dependent coarse gauge disappear because they are vertical gauge tangents. For complex parameters, the transpose is for the invariant bilinear color pairing; estimates are subsequently taken in an absolute Hermitian norm.

Equation (GQ9) does not identify a full gauge-fixed determinant without its coordinate, gauge, and coarea factors. It identifies the actual first variation and the physical constrained orbit, which are the facts needed below.

46.2 2. A sparse node restriction estimate in four dimensions

Fix $0 < s < 1$, an odd fixed L, and physical spacings

$$a = \ell/L, \qquad r = L^m \ell \le 1.$$

All lattices are periodic and nested. The same statement applies to a common buffered chart after an extension with the displayed norms. Let r_ℓ be any translation-covariant finite linear node kernel with support diameter at most $C_L \ell$, absolute coefficient sum at most $C_L \ell$, and zero response to every constant one-form. These are precisely the elementary properties of the internal connector in (GQ6).

Let $S_{\ell,r}$ restrict its node output to the final r-lattice. In physical Bessel norms,

$$\boxed{\|S_{\ell,r} r_\ell A\|_{H_r^s} \le C_{s,L}\, r^2 (\ell/r)^s \|A\|_{H_a^{1+s}}.} \tag{GQ10}$$

The constant is independent of both meshes, the ratio r/a, and the physical torus volume. The input and output can have finitely many components, including Lie-algebra components.

Proof. Write $\rho_\ell(p)$ for the Fourier multiplier before restricting to the final nodes. Subtracting its zero constant moment gives

$$|\rho_\ell(p)| \le C_L \ell^2 |p|, \qquad |\rho_\ell(p)| \le C_L \ell, \qquad |p| \le C_L/\ell. \tag{GQ11}$$

The phases of bond-centered components do not change these estimates. In an alias fibre over a final momentum θ, let

$$p_\nu = \theta + 2\pi\nu/r$$

run over the representatives in the fine Brillouin zone. Weighted Cauchy–Schwarz bounds the squared operator norm of that fibre by

$$(1+\widehat{\Delta}_r(\theta))^s \sum_{\nu} |\rho_\ell(p_\nu)|^2 (1+\widehat{\Delta}_a(p_\nu))^{-1-s}. \tag{GQ12}$$

The lattice Laplacian symbols are uniformly comparable to the corresponding squared momenta within their own Brillouin zones. For the principal alias, (GQ11) makes (GQ12) at most $C\ell^4$. For every other alias,

$$|p_\nu| \geq c|\nu|/r, \qquad 1 \leq |\nu| \leq C_L r/\ell, \qquad (1+\widehat{\Delta}_r(\theta))^s \leq C_s r^{-2s}.$$

The nonprincipal part is therefore at most

$$\begin{aligned} C_{s,L}\ell^4 r^{-2s} \sum_{1\leq|\nu|\leq C_L r/\ell} (|\nu|/r)^{-2s} &\leq C_{s,L}\ell^4 (r/\ell)^{4-2s} \\ &= C_{s,L} r^4 (\ell/r)^{2s}. \end{aligned} \tag{GQ13}$$

Here the exponent 4 is the spatial dimension. Since $s < 1 < 2$ and $\ell \leq r$, the principal contribution ℓ^4 satisfies the same bound. Taking the square root and summing the alias fibres proves (GQ10).

The estimate concerns a dimensionless node factor, not an arbitrary point evaluation of an H^{1+s} function. The factor ℓ^2 and the zero constant moment in (GQ11) are essential. The summable factor $(\ell/r)^s$ would be absent at $s = 0$.

The elementary pointwise counterpart is

$$\|r_\ell A\|_\infty \leq C_L \ell^2 \|\nabla_a A\|_\infty. \tag{GQ14}$$

It follows by subtracting one local constant collection and telescoping along the internal paths.

46.3 3. Uniform bounds for the accumulated node gauge

Let A_j be the iterated straightened potentials in the centered-mean chapter, with $a_{j+1} = \ell_j = La_j$ and $a_k = r$. Put

$$M = \|A_0\|_\infty + \|\nabla_{a_0} A_0\|_\infty, \qquad K = \|A_0\|_{H^{1+s}_{a_0}}.$$

On its common small chart, that chapter proves

$$\|A_j\|_{W^{1,\infty}} \leq 2M, \qquad \|A_j\|_{H^{1+s}_{a_j}} \leq e^{C_{s,L}(M+M^2)r} K. \tag{GQ15}$$

At a final node, (GQ4) is an ordered product of the intermediate node factors evaluated at that same node. Define

$$b_{j,r} = S_{\ell_j,r} r_{\ell_j}(A_j).$$

Equations (GQ10), (GQ14), and (GQ15) give

$$\sum_{j<k} \|b_{j,r}\|_\infty \leq C_L r^2 M, \qquad \sum_{j<k} \|b_{j,r}\|_{H^s_r} \leq C_{s,L} r^2 e^{C_{s,L}(M+M^2)r} K. \tag{GQ16}$$

The second sum uses $\sum_j (\ell_j/r)^s \le (1-L^{-s})^{-1}$, not a bound proportional to the number of scales.

For completeness, a repeated generic Sobolev product constant must not be multiplied once per factor. Instead, use the exact identity for the difference of two ordered products. For matrices $F_j = e^{-b_{j,r}}$, it gives

$$\|\tau_z \prod_j F_j - \prod_j F_j\|_2 \le e^{C\sum_j \|b_{j,r}\|_\infty} \sum_j \|\tau_z b_{j,r} - b_{j,r}\|_2. \tag{GQ17}$$

The corresponding undifferenced estimate holds after subtracting the identity. Apply Minkowski in the fractional translation-difference norm, which is uniformly equivalent to the discrete Bessel norm for fixed $0 < s < 1$. The order of all matrix factors is retained. Thus the accumulated gauge h_k, and its inverse, satisfy

$$\begin{aligned} \|h_k - 1\|_\infty + \|h_k^{-1} - 1\|_\infty &\le C_L r^2 M, \\ \|h_k - 1\|_{H^s_r} + \|h_k^{-1} - 1\|_{H^s_r} &\le C_{s,L} r^2 K, \end{aligned} \tag{GQ18}$$

after fixing a sufficiently small common radius. Constants may include the bounded exponentials in (GQ16). The principal logarithm $\kappa_k = \log h_k$ has the same bounds. These statements hold on the common complex chart as well, using absolute matrix norms.

46.4 4. Keeping the complete nonlinear coarse gauge action

At spacing r, let B be a physical coarse potential and $g = e^\kappa$ a site gauge. Its exact transformed potential is

$$\mathcal{G}_r(\kappa, B)_\mu(y) = r^{-1} \log\left(e^{\kappa(y)} e^{rB_\mu(y)} e^{-\kappa(y+re_\mu)}\right). \tag{GQ19}$$

Define the forward coarse difference $d_r\kappa$. Then

$$\mathcal{G}_r(\kappa, B) = B - d_r\kappa + \mathcal{E}_r(\kappa, B). \tag{GQ20}$$

Assume

$$\|B\|_\infty \le CM, \quad \|B\|_{H^s_r} \le CK, \quad \|\kappa\|_\infty \le Cr^2 M, \quad \|\kappa\|_{H^s_r} \le Cr^2 K.$$

For $r \le 1$ and a fixed small potential radius,

$$\|\mathcal{E}_r(\kappa, B)\|_{H^s_r} \le C_{s,L} r^2 (M + M^2) K, \qquad \|\mathcal{E}_r(\kappa, B)\|_\infty \le C_L r^2 (M^2 + M^3). \tag{GQ21}$$

To verify the scale without discarding a gauge term, set

$$a = \kappa(y), \qquad c = rB_\mu(y), \qquad \delta = \kappa(y + re_\mu) - \kappa(y).$$

The analytic function

$$\Psi(a, c, \delta) = \log(e^a e^c e^{-(a+\delta)}) - (c - \delta)$$

has no constant or linear terms and vanishes identically when $a = \delta = 0$. Thus every monomial has degree at least two and contains a or δ. In particular there is no pure c^2 term that would cost only one power of r. The fractional product inequality, summed on a smaller analytic ball, bounds $r^{-1}\Psi$ by

$$\begin{aligned}\frac{C}{r}\Big[&(\|a\|_\infty + \|\delta\|_\infty)(\|c\|_{H^s} + \|a\|_{H^s} + \|\delta\|_{H^s}) \\ &+ (\|a\|_{H^s} + \|\delta\|_{H^s})\|c\|_\infty\Big].\end{aligned} \tag{GQ22}$$

Now a, δ have sup size $O(r^2M)$ and H^s size $O(r^2K)$, whereas c has size $O(rM)$ and $O(rK)$. This proves (GQ21). The pointwise bound follows in the same way. The exact gradient in (GQ20) is the only term removed by a flat transverse projection.

46.5 5. The actual raw transverse constraint theorem

Let

$$q^{(k)}_{r,\mathrm{raw}}(A) = r^{-1}\log M_{k-1}\cdots M_0(e^{a_0 A})$$

denote the physical potential of the actual raw iterated average. Let Q_r be the direct flat box-and-line average. Let Π_r be the orthogonal projection away from coarse gradients, retaining any harmonic modes explicitly. On the periodic small potential chart of (GQ15),

$$\boxed{\|\Pi_r(q^{(k)}_{r,\mathrm{raw}}(A) - Q_r A)\|_{H^s_r} \le C_{s,L} r^2 (M + M^2)\|A\|_{H^{1+s}_{a_0}}.} \tag{GQ23}$$

The constants are independent of the microscopic mesh and the number of intermediate averages.

Proof. The centered-mean theorem writes the straightened output as

$$B_k = Q_r A + N_k(A), \qquad \|N_k(A)\|_{H^s_r} \le C_{s,L} r^2 (M + M^2)K.$$

Its exact relation to the raw output is $M^k(e^{a_0 A}) = (e^{rB_k})^{h_k^{-1}}$. Use (GQ18) and (GQ20)–(GQ21) with $\kappa = -\log h_k$. This gives

$$q^{(k)}_{r,\mathrm{raw}}(A) = Q_r A - d_r\kappa + \mathcal{R}_r(A), \qquad \|\mathcal{R}_r(A)\|_{H^s_r} \le C_{s,L} r^2 (M + M^2)K. \tag{GQ24}$$

Applying Π_r kills the exact displayed gradient and has norm at most one in the Fourier Bessel norm. It does not remove $\mathcal{E}_r$; that complete nonlinear gauge remainder was estimated in (GQ21). This proves (GQ23).

The same estimates also recover bounded raw intermediate charts. Coarse inverse inequalities applied to (GQ18) give

$$\|d_r\kappa\|_{H^{1+s}_r} \le Cr^{-2}\|\kappa\|_{H^s_r} \le CK, \qquad \|d_r\kappa\|_{W^{1,\infty}} \le Cr^{-2}\|\kappa\|_\infty \le CM.$$

Use a single coarse inverse derivative for $\mathcal{E}_r$ and the already controlled straightened chart. At every retained scale,

$$\|q^{(k)}_{r,\mathrm{raw}}(A)\|_{H^{1+s}_r} \le C_{s,L}K, \qquad \|q^{(k)}_{r,\mathrm{raw}}(A)\|_{W^{1,\infty}} \le C_L M. \tag{GQ25}$$

There is one fixed constant, not a product of constants over depths. Decreasing the initial radius by this fixed constant keeps all raw logarithms inside the required common branch.

46.5.1 5.1 The consequence for the original constraint

Suppose two fields u, A are in this same type of fine potential chart and obey the exact raw fixed-representative constraint

$$M^k(e^{a_0 u}) = M^k(e^{a_0 A}).$$

Subtract (GQ23). With M_* a common bound on their $W^{1,\infty}$ norms,

$$\boxed{\|\Pi_r Q_r(u-A)\|_{H^s_r} \le C_{s,L} r^2 (M_* + M_*^2)(\|u\|_{H^{1+s}_{a_0}} + \|A\|_{H^{1+s}_{a_0}}).} \tag{GQ26}$$

Thus a transverse flat Schur inverse of size r^{-2} does not lose a negative power of r from this actual nonlinear constraint defect. Equation (GQ26) is derived in the original fine gauges. It neither changes the fine field by a discontinuous block hold nor assumes a smooth inverse for the nonlinear gauge cocycle.

46.7 Sources and relation to the other chapters

The actual outer reference-log map, the inner endpoint contour mean, and their lattice covariance are from T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, Commun.

Math. Phys. **109** (1987), 249–301, especially (0.4), (0.11), and (0.12). The proof uses the explicit internal-path connector and fixed-L estimates proved in the centered-mean chapter; it does not import an associativity identity for a nonlinear group mean.

The distinction between adapted constraints, Landau gauge, and the alternative covariant-Laplacian equation is explicit in T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Commun. Math. Phys. **102** (1985), 277–309, particularly (20), (44)–(49), (123)–(143), and (174)–(180). The local-residual chapter records these equations and the separate physical-gauge argument needed between the two representations.

47.1 1. Definitions and physical normalization

Let the fine spacing be a, the coarse spacing be $r = na$, and the physical torus have an integer number of coarse cells in each direction. The integer n may be arbitrarily large. Centered odd n is convenient for the original lattice symmetries; the estimates are unchanged by translating the block convention.

Use physical scalar products

$$\langle f, g\rangle_a = a^d \sum_x \overline{f(x)} g(x), \qquad \langle u, v\rangle_r = r^d \sum_y \overline{u(y)} v(y). \tag{SG1}$$

All constructions act componentwise on a fixed finite-dimensional matrix or Lie-algebra space.

Let R_r be normalized averaging over disjoint site blocks:

$$(R_r f)(y) = n^{-d} \sum_{x \in B_r(y)} f(x), \qquad H_r = R_r^*. \tag{SG2}$$

Thus H_r is the block hold, $R_r H_r = I$, and H_r is an isometry in physical L^2. It preserves the L^∞ norm and the physical L^1 norm.

On the fine lattice define the normalized line filters

$$F_{n,j} = \frac{1}{n} \sum_{v=0}^{n-1} \tau_{vae_j}, \qquad T_r = \prod_{j=1}^{d} F_{n,j}. \tag{SG3}$$

A centered version differs only by a translation. In particular $T_r^* T_r = \prod_j F_{n,j}^* F_{n,j}$ is independent of that choice. Fix an integer $p \geq 1$, and put

$$E_{0,r} = (T_r^* T_r)^p H_r, \qquad B_r = R_r E_{0,r}. \tag{SG4}$$

The desired extension is

$$E_r = E_{0,r} B_r^{-1}. \tag{SG5}$$

The following sections prove that B_r^{-1} exists and that this formula has all the claimed bounds.

47.2 2. The coarse normalizer is uniformly invertible

Fix a coarse Fourier momentum θ in its Brillouin zone. Its fine aliases are

$$p_\nu = \theta + 2\pi\nu/r$$

reduced to the fine Brillouin zone. The block-hold multiplier on this fibre is, up to its centered phase,

$$q_\nu = \prod_{j=1}^{d} \frac{\sin(rp_{\nu,j}/2)}{n\sin(ap_{\nu,j}/2)}. \tag{SG6}$$

The value at a zero denominator is its continuous value. The physical normalization in (SG1) gives the exact identity

$$\sum_\nu |q_\nu|^2 = 1. \tag{SG7}$$

The absolute value of the multiplier of T_r is $|q_\nu|$. Hence the multiplier of $E_{0,r}$ into alias ν is $\overline{q}_\nu|q_\nu|^{2p}$, with the harmless phase adjusted to the hold convention, and

$$b_r(\theta) = \sum_\nu |q_\nu|^{2p+2} \tag{SG8}$$

is the multiplier of B_r.

Every factor of the principal alias satisfies

$$\left|\frac{\sin(r\theta_j/2)}{n\sin(a\theta_j/2)}\right| \geq \frac{2}{\pi}, \qquad |\theta_j| \leq \pi/r.$$

Indeed, the numerator is bounded below by $2/\pi$ times its argument in absolute value, while the denominator is bounded above by that argument. Therefore

$$b_{p,d} := (2/\pi)^{(2p+2)d} \leq b_r(\theta) \leq 1. \tag{SG9}$$

The upper bound uses $|q_\nu| \leq 1$ and (SG7). These inequalities hold for every n, including on an arbitrary compatible finite torus. They prove the uniform L^2 invertibility of B_r.

The physical-support calculation also gives a useful fact not supplied by the Fourier lower bound alone. One application of $T_r^*T_r$ shifts a fine coordinate by at most $(n-1)a$. After p applications, holding and then averaging can connect only coarse cells whose integer indices differ by at most p in each coordinate. Thus B_r is a coarse convolution of range at most p. Its coefficients are nonnegative and sum to one, although its inverse need not have nonnegative coefficients.

47.3 3. An exponential convolution bound for the inverse

First construct the same finite-range convolution on the infinite coarse lattice $\mathbb{Z}^d$. The preceding Fourier estimate is valid for every continuous coarse momentum. Set

$$A_r = I - B_r, \qquad \rho = 1 - b_{p,d} < 1.$$

Then

$$\|A_r\|_{2\to 2} \leq \rho, \qquad \operatorname{supp}(A_r^k\delta_0) \subset [-pk, pk]^d. \tag{SG10}$$

For a coefficient sequence, unweighted Cauchy–Schwarz on this support gives

$$\|A_r^k \delta_0\|_{\ell^1} \le (2pk+1)^{d/2} \rho^k. \tag{SG11}$$

This is the appropriate estimate for the Neumann series. It does not assume $\|I - B_r\|_{\ell^1 \to \ell^1} < 1$, which need not hold.

Choose

$$\gamma = \frac{-\log \rho}{2dp} > 0.$$

The weighted form of (SG11) is

$$\sum_{z \in \mathbb{Z}^d} e^{\gamma |z|_1} |(A_r^k \delta_0)(z)| \le (2pk+1)^{d/2} \rho^{k/2}. \tag{SG12}$$

Consequently

$$B_r^{-1} = \sum_{k \ge 0} A_r^k$$

has a convolution kernel $c_r(z)$ obeying

$$\sum_z e^{\gamma |z|_1} |c_r(z)| \le C_{p,d} := \sum_{k \ge 0} (2pk+1)^{d/2} \rho^{k/2} < \infty. \tag{SG13}$$

The constants depend on p, d, and are independent of the number of fine sites in a block, the block spacing, and the torus volume.

On a finite torus its inverse is the periodization of this kernel. The absolute sum of a periodization is bounded by the original absolute sum, and the torus distance of a periodized point is no greater than the distance of its lift. Thus (SG13) holds there with the torus distance as well.

In particular B_r^{-1} is bounded on every coarse L^q, $1 \le q \le \infty$, with the same mesh-independent bound. It commutes with all coarse differences. Since $B_r 1 = 1$, it preserves constants.

It now follows directly from (SG4)–(SG5) that

$$R_r E_r = I. \tag{SG14}$$

This is an exact equality, not an approximate reconstruction.

47.4 4. Uniform fractional Sobolev regularity

At spacing b, use the physical discrete Bessel norm

$$\|u\|_{H_b^t}^2 = \sum_\xi (1 + \widehat{\Delta}_b(\xi))^t |\widehat{u}(\xi)|^2, \qquad \widehat{\Delta}_b(\xi) = 4b^{-2} \sum_j \sin^2(b\xi_j/2). \tag{SG15}$$

Smooth lift theorem. For every fixed

$$0 \le t < 2p + \tfrac{1}{2},$$

there is a constant $C_{p,d,t}$ such that

$$\|E_r \lambda\|_{H_a^t} \le C_{p,d,t} \|\lambda\|_{H_r^t}. \tag{SG16}$$

The constant is uniform in the microscopic spacing, the ratio r/a, and the physical torus volume.

To prove this, first consider $E_{0,r}$. The squared norm on one alias fibre is the input coefficient squared times

$$\sum_{\nu} |q_\nu|^{4p+2}(1+\widehat{\Delta}_a(p_\nu))^t. \tag{SG17}$$

We show that this quantity is bounded by $C_{p,d,t}(1+\widehat{\Delta}_r(\theta))^t$.

In one dimension write $z = r\theta \in [-\pi,\pi]$. For a nonprincipal alias,

$$|q_k| \le C\,\frac{|\sin(z/2)|}{|z+2\pi k|}. \tag{SG18}$$

This follows from $|\sin(ap_k/2)| \ge (2/\pi)|ap_k/2|$ on the fine Brillouin zone. Thus

$$\begin{aligned}\sum_{k\neq 0} |q_k|^{4p+2}|p_k|^{2t} &\le C_{p,t} r^{-2t}|\sin(z/2)|^{4p+2}\sum_{k\neq 0}|k|^{-4p-2+2t}\\ &\le C_{p,t}|\theta|^{2t}.\end{aligned} \tag{SG19}$$

The series converges exactly in the stated strict range. The second inequality also uses $2t < 4p+1 < 4p+2$ and bounded $|z|$, so that $|\sin(z/2)|^{4p+2} \le C_{p,t}|z|^{2t}$. The principal alias contributes at most $|\theta|^{2t}$. At $t=0$, the assertion instead follows immediately from (SG7).

In d dimensions use

$$|p|^{2t} \le C_{d,t}\sum_j |p_j|^{2t}$$

and factor the other coordinate sums, each of which is at most one: $\sum_k |q_k|^{4p+2} \le \sum_k |q_k|^2 = 1$. The lattice symbols on their Brillouin zones are uniformly comparable to the corresponding squared momenta. This proves the required bound for (SG17), including the inhomogeneous term.

Finally B_r^{-1} multiplies each coarse fibre by $b_r(\theta)^{-1} \le b_{p,d}^{-1}$, which is bounded in every H_r^t. This proves (SG16).

The threshold is not an artefact of the estimate. In one dimension, fix a nonzero coarse momentum strictly inside the coarse Brillouin zone and let n tend to infinity. At $t = 2p+1/2$, the aliases below a fixed fraction of the fine cutoff contribute a harmonic sum, so the squared norm grows at least as a constant times $\log n$. Increasing p is therefore the direct way to obtain more regularity.

For the present fractional-curvature application, $p=2$ gives

$$\|E_r\lambda\|_{H_a^{3+s}} \le C_{d,s}\|\lambda\|_{H_r^{3+s}}, \qquad 0 < s < \tfrac{1}{2}, \tag{SG20}$$

with room below the actual $t < 9/2$ threshold.

47.5 5. Exact cancellation of fine differences

The supremum derivative estimate does not follow from Sobolev embedding in four dimensions; it has its own exact filter proof.

Let

$$D_{a,j} = \frac{\tau_{ae_j} - I}{a}, \qquad D_{r,j}^{\mathrm{fine}} = \frac{\tau_{re_j} - I}{r}.$$

The elementary geometric sum gives

$$D_{a,j}F_{n,j} = D_{r,j}^{\mathrm{fine}}, \qquad D_{a,j}F_{n,j}^{*} = \tau_{-(n-1)ae_j}D_{r,j}^{\mathrm{fine}}. \tag{SG21}$$

Also

$$D_{r,j}^{\mathrm{fine}} H_r = H_r D_{r,j}. \tag{SG22}$$

Each fine derivative therefore consumes one line-filter factor and becomes a coarse derivative. There are $2p$ such factors in each coordinate.

For any multiindex α with $\alpha_j \le 2p$, choose $u_j + v_j = \alpha_j$, $0 \le u_j, v_j \le p$. Commuting the fine convolutions gives an exact identity of the form

$$D_a^\alpha E_{0,r} = \tau_{\zeta_\alpha} \left[\prod_j (F_{n,j}^*)^{p-u_j} F_{n,j}^{p-v_j} \right] H_r D_r^\alpha, \tag{SG23}$$

where ζ_α is the sum of the shifts from (SG21). Every remaining filter is a positive normalized average. Holding and translation are L^∞ contractions. Hence

$$\|D_a^\alpha E_{0,r}\lambda\|_\infty \le \|D_r^\alpha \lambda\|_\infty. \tag{SG24}$$

Use (SG13) and commute coarse differences through B_r^{-1} to obtain

$$\boxed{\|D_a^\alpha E_r \lambda\|_\infty \le C_{p,d} \|D_r^\alpha \lambda\|_\infty, \qquad \alpha_j \le 2p.} \tag{SG25}$$

In particular, already $p = 1$ controls all total orders at most two, while $p = 2$ gives

$$\|E_r\lambda\|_{W_a^{2,\infty}} \le C_d \|\lambda\|_{W_r^{2,\infty}}. \tag{SG26}$$

These are physical differences with their displayed spacings. There is no power of a^{-1} left over and no suppressed factor of r^{-1}.

The same identities give L^q derivative bounds, $1 \le q \le \infty$, since all remaining filters and the hold are contractions in the physical normalization and B_r^{-1} has the uniform convolution bound.

47.6 6. Locality, symmetries, and uses of the lemma

The kernel of $E_{0,r}$ has physical range $O_p(r)$. Combining this fact with (SG13) gives an exponential coarse-scale kernel envelope for E_r. In a convention

$$(E_r\lambda)(x) = \sum_y e_r(x,y)\lambda(y),$$

one has, for a smaller $\gamma' > 0$,

$$\sup_x \sum_y e^{\gamma'|x-y|/r} |e_r(x,y)| \le C_{p,d}, \tag{SG27}$$

and the corresponding physical column bound

$$\sup_y r^{-d} a^d \sum_x e^{\gamma'|x-y|/r} |e_r(x,y)| \le C_{p,d}. \tag{SG28}$$

On a torus use its shortest periodic distance. Thus local data give an extension with tails that can be counted in coarse units. A theorem requiring literal compact support must still account for these tails; exponential decay is not compact support.

The centered construction commutes with lattice reflections, coordinate permutations, and coarse translations. It preserves constants and real Lie-algebra values, and commutes with constant adjoint

transformations. No covariance under an arbitrary position-dependent background gauge transformation is asserted.

For $p = 2$, the combined linear statement is

$$R_r E_r = I, \qquad \|E_r\lambda\|_{W_a^{2,\infty}\cap H_a^{3+s}} \le C_{d,s}\|\lambda\|_{W_r^{2,\infty}\cap H_r^{3+s}}, \quad 0 < s < \tfrac{1}{2}. \tag{SG29}$$

Consequently exponentiating a small $E_r\lambda$ produces a smooth fine gauge parameter in the chosen potential chart. It avoids the discontinuities of exponentiating a sharp block hold. It exactly realizes the desired parameter under the linear site average, and also under the specifically defined logarithmic operation $\exp[R_r \log v]$ within one common logarithm chart. This last operation is mentioned only to identify what is exact; it is not being substituted for the original nonlinear gauge cocycle.

47.7 7. What a nonlinear gauge-cocycle inverse still requires

There are three maps that must not be conflated:

1. the site box average R_r, inverted by (SG5);
2. point restriction to coarse sites, which appears in ordinary physical gauge covariance of a raw block average;
3. the field-dependent gauge cocycle after the explicit coarse straightening used in the centered-mean chapter.

For example, if the actual coarse field is $M(U)$, the analyzed representative is $M(U)^{g(U)}$, and a fine gauge transformation is v, the induced coarse transformation is, in the convention $V^h = hVh^{-1}$,

$$\Gamma_U(v) = g(U^v)\,(v|_{\text{coarse}})\,g(U)^{-1}. \tag{SG30}$$

This is an exact identity from physical gauge covariance. At the flat point, the straightening identity $Q_{\text{raw}} = Q_{\text{bl}} - d_r r_{\text{conn}}$ gives

$$D\big[\log\Gamma_1(e^{\omega})\big]_{\omega=0} = R_r\omega. \tag{SG31}$$

Here the connector sign is the one fixed in the centered-mean chapter. Equation (SG31) is a derivative statement; it does not identify the nonlinear map with R_r.

The lift proves that this flat derivative has a uniformly regular right inverse. It does not yet establish that the derivative at a nonzero background remains invertible in the same regularity norm, or that the nonlinear remainder is uniformly tame on the required complex domain.

For clarity, a sufficient quantitative inverse criterion can be stated without hiding those requirements. Let Y_r be the actual coarse gauge-parameter Banach space, X_a the fine space, and suppose the bounds proved above apply to $E_r : Y_r \to X_a$. Define

$$\mathcal{G}_{U,r}(z) = \log\Gamma_U(\exp(E_r z)).$$

If the actual map is analytic on a common ball, satisfies $\mathcal{G}_{U,r}(0) = 0$, and obeys

$$\sup_{\|z\|_{Y_r}<R} \|D\mathcal{G}_{U,r}(z) - I\|_{Y_r\to Y_r} \le \kappa < 1, \tag{SG32}$$

then every target ϑ with $\|\vartheta\|_{Y_r} < (1-\kappa)R$ has a unique solution in that ball of

$$z = \vartheta - \big(\mathcal{G}_{U,r}(z) - z\big). \tag{SG33}$$

The solution is analytic and satisfies $\|z\|_{Y_r} \le (1-\kappa)^{-1}\|\vartheta\|_{Y_r}$. The resulting $\omega = E_r z$ obeys the fine norm bound and realizes the actual desired coarse gauge transformation exactly.

This is the Banach contraction proof, and its hypotheses are explicit. In the Yang–Mills application, (SG32) must be proved for the actual hierarchical straightening cocycle, including its background dependence, all retained boundary data, and the specified fine gauge. A low-norm pointwise implicit-function theorem does not provide its higher-norm version. A global Sobolev algebra estimate may also introduce an unintended volume dependence; one should use the required local/tame norms or prove the stated operator bound directly.

The new linear lemma removes the sharp-extension obstruction. It leaves a well-defined nonlinear operator estimate, rather than assuming that gauge-orbit equivalence already preserves a fine $W^{2,\infty}$ or H^{3+s} representative.

47.8 Source and dependency note

Equations (SG1)–(SG29) are proved here from finite convolution identities and Fourier alias estimates. They use no unverified propagator theorem. The actual raw averaging map and its gauge covariance are the maps in T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, CMP **109** (1987), 249–301, equations (0.4), (0.11), (0.12). The explicit straightened map and its flat connector are proved in the companion centered-mean chapter, (CM15)–(CM18). The nonlinear criterion (SG32) is a sufficient condition, not an assertion already verified for those maps.

48.1 1. Statement, pairings, and zero modes

Let $\mathbb{T}_T^4$ be a periodic physical torus of side T. All lattices are nested, with fine spacing a, retained spacing $r = L^k a \le 1$, and one fixed odd L. Use physical-volume pairings at every lattice spacing. Fine links are

$$U_\mu(x) = e^{a u_\mu(x)}, \qquad U_\mu^A(x) = e^{a A_\mu(x)}.$$

Bond-midpoint shifts can be inserted throughout without changing the estimates.

Write $\mathcal{M}_r$ for the genuine composition of the actual raw averages in the original construction. Let

$$V = \mathcal{M}_r(U) = \mathcal{M}_r(U^A) \tag{NK1}$$

be an equality of actual coarse link matrices in the original fixed representatives. Assume that the physical constraint qualification, or the original variational construction with its gauge-coordinate identities retained, supplies a multiplier for the stationary point u. This multiplier existence is an explicit premise; stationarity alone is not used to infer the rank of the whole nonlinear constraint map. Normalize the multiplier by dividing out the overall scalar Wilson coefficient. The required physical KKT equation is

$$J_a(u) = (D\mathcal{M}_r(U))^{\mathtt{t}}\lambda, \qquad d_V^{\mathtt{t}}\lambda = 0. \tag{NK2}$$

The superscript $\mathtt{t}$ denotes the transpose for the invariant bilinear color pairing. On real compact fields it agrees with the usual physical adjoint. Complex estimates below use absolute Hermitian norms after the algebraic identities have been established. The Ward equation in (NK2) follows from gauge invariance of the action and surjectivity of restriction of fine gauge transformations to coarse sites.

The geometric Euler force is

$$J_a(u) = \mathcal{C}_U \Phi_a(F_a(u)), \qquad F_a(u) = a^{-2}\log P(U), \qquad \Phi_a(F) = a^{-2}\pi_{\mathfrak{g}}\sinh(a^2 F). \tag{NK3}$$

A fixed trace normalization only multiplies both sides by one constant and is absorbed in the definition of λ.

Fix

$$0 < s < \tfrac{1}{2}, \qquad \|u\|_{W^{1,\infty}} + \|A\|_{W^{1,\infty}} \le M, \qquad K_u = \|u\|_{H_a^{1+s}}, \quad K_A = \|A\|_{H_a^{1+s}}. \tag{NK4}$$

All logarithms and actual average factors lie on the common regular branch. The constant M is sufficiently small depending on s, L, T, but not on a, r, k. The assumptions admit complex fields in that same common branch. In particular they do not require a real minimizer to be analytically continued by a new gauge choice.

Define

$$\mathcal{D}_s(A,u) = \|F_a(A)\|_{H_a^{1+s}} + \|A\|_2 + (M + M^2)(K_A + K_u). \tag{NK5}$$

Theorem NK. Under (NK1)–(NK4),

$$\boxed{\|\lambda\|_{H_r^s} + \|J_a(u)\|_{H_a^s} + \|F_a(u)\|_{H_a^{1+s}} \le C_{s,L,T}\,\mathcal{D}_s(A,u).} \tag{NK6}$$

The constant is independent of both lattice spacings and the number of averages. No high regularity of the fine longitudinal part of u is asserted.

The low norm K_u on the right is not the claimed conclusion. If the admitted regular-branch theorem gives

$$\|u\|_{C^{1,\beta}} \le C_{\mathrm{low}}\mathcal{B}(A), \qquad 0 < s < \beta, \tag{NK7}$$

uniformly in the microscopic mesh and retained spacing, then the translation-difference characterization gives

$$K_u \le C_{s,\beta,T} C_{\mathrm{low}}\mathcal{B}(A). \tag{NK8}$$

Thus (NK6) closes the new fractional curvature norm from an existing low bound and the corresponding input norm.

48.2 2. An exact lattice curvature expansion with a uniform strong remainder

The following elementary lattice estimate supplies the nonlinear Euler error without assuming second derivatives of u.

For a plaquette, write its four exponentials in their actual path order. Its logarithm has a convergent BCH series on $\|au\|_\infty < c$. The linear term divided by a^2 is exactly $d_a u$. The quadratic term is a fixed sum of commutators of values on the four bonds. At a constant field its value is $[u_\mu, u_\nu]$.

Choose one nearby actual bond coordinate for each component. Subtract these constants in the quadratic term before estimating it. Every difference is an exact fine nearest-neighbor difference times a. All terms of degree at least three in the logarithm have, after division by a^2, coefficient a^{n-2}. Consequently

$$F_a(u) = d_a u + \mathcal{B}(u,u) + a\mathcal{R}_a(u, \nabla_a u), \tag{NK9}$$

where $\mathcal{B}$ is a fixed local bracket expression, equal to the usual constant-curvature bracket on constant data. The particular nearby coordinates used in $\mathcal{B}$ only change $\mathcal{R}_a$. The latter is analytic on the common small branch and starts with terms of types $u\nabla u$ and u^3.

Uniform discrete fractional product estimates, with $0 < s < 1$, give

$$\|\mathcal{R}_a(u, \nabla_a u)\|_{H^s_a} \le C_s(M + M^2)K_u. \tag{NK10}$$

The H^{1+s} product estimate gives

$$\|\mathcal{B}(u,u)\|_{H^{1+s}_a} \le C_s M K_u.$$

For the remaining term use the literal fine-lattice inverse inequality

$$a\|\mathcal{R}_a\|_{H^{1+s}_a} \le C_s \|\mathcal{R}_a\|_{H^s_a} \qquad (a \le 1).$$

It follows that the complete nonlinear curvature correction

$$G_a(u) = F_a(u) - d_a u$$

satisfies

$$\boxed{\|G_a(u)\|_{H^{1+s}_a} \le C_s(M + M^2)K_u, \quad \|F_a(u)\|_{H^s_a} \le C_s(1 + M + M^2)K_u, \quad \|F_a(u)\|_\infty \le C(M + M^2).} \tag{NK11}$$

This estimate is uniform in a. It uses a fine-lattice inverse inequality only on a term that already carries the compensating factor a. It never assumes an H^{2+s} or H^2 bound for the full potential.

Let d_a, δ_a be the flat exterior incidence and its physical adjoint. The exact Wilson incidence in (NK3) has the form

$$\mathcal{C}_U = \delta_a + \mathcal{T}_U,$$

where $\mathcal{T}_U$ is a finite sum of shifts and zero-order coefficient multipliers. The coefficients are of the form

$$a^{-1}(\mathrm{Ad}_{U^{\pm 1}} - I).$$

Their discrete $W^{1,\infty}$ norms are bounded by $C(M + M^2)$. Moreover

$$\Phi_a(F) = F + a^4 \mathcal{P}_a(F), \qquad \|\mathcal{P}_a(F)\|_{H^s_a} \le C\|F\|^2_\infty \|F\|_{H^s_a}.$$

Thus the exact force equation (NK3) is equivalently

$$\delta_a d_a u = J_a(u) + f_a(u),$$

where

$$f_a(u) = -\mathcal{T}_U F_a(u) - a^4 \mathcal{C}_U \mathcal{P}_a(F_a(u)) - \delta_a G_a(u). \tag{NK12}$$

The crude difference bound $\|\mathcal{C}_U H\|_{H^s} \le C(a^{-1} + M)\|H\|_{H^s}$ costs only a^3 on the flux correction. Equations (NK10)–(NK11) therefore prove

$$\boxed{\|f_a(u)\|_{H^s_a} \le C_s(M + M^2)K_u.} \tag{NK13}$$

This is an identity for the actual Wilson force, not for a continuum surrogate.

48.3 3. Two actual-map estimates and the flat Schur bounds

Let Q_r be the direct flat box-and-line average; R_r is the corresponding site box average. Their exact chain relation is

$$Q_r d_a = d_r R_r, \qquad \delta_a Q_r^* = R_r^* \delta_r. \tag{NK14}$$

Let Π_a^0, Π_r^0 be the flat projections onto mean-zero co-closed one-forms. Harmonic constants are kept separately. The actual transverse constraint theorem GQ26 gives

$$e_c := \Pi_r^0 Q_r(u - A), \qquad \|e_c\|_{H^s_r} \le C_{s,L} r^2 (M + M^2)(K_u + K_A). \tag{NK15}$$

That theorem uses the exact accumulated node gauges, including their nonlinear remainder, to prove (NK15) in the original fine representatives. No fine gauge is changed here.

The actual-current comparison FH32, applied to the complete multiplier satisfying its actual Ward equation, gives

$$J_a(u) = Q_r^* \lambda + e_Q,$$
$$\|e_Q\|_{H^s_a} \le C_{s,L}\big(M r^{1-s} + (M + M^2) r^{2-s}\big)\|\lambda\|_2 \le C_{s,L}(M + M^2)\|\lambda\|_{H^s_r}. \tag{NK16}$$

Its raw intermediate potential and curvature hypotheses follow from GQ25 and (NK11), applied at each retained scale. This is the comparison along the genuine raw averaging chain. It uses the Ward equation on each complete intermediate current before unrolling against flat tails; it never applies a Ward-restricted bound to an isolated error.

The flat direct prolongation also satisfies, for $s < 1/2$,

$$\|Q_r^* \xi\|_{H^s_a} \le C_{s,L}\|\xi\|_{H^s_r}. \tag{NK17}$$

All-depth flat filter composition is exact.

Let $L_a = -\Delta_a$ on mean-zero co-closed fine one-forms. Define

$$C_r = \Pi_r^0 Q_r L_a^{-1} Q_r^* \Pi_r^0\Big|_{\operatorname{ran}\Pi_r^0}, \qquad S_r = C_r^{-1}. \tag{NK18}$$

At a nonzero coarse Fourier fibre, let λ_0 denote the principal fine Laplacian symbol. The actual box-and-line symbols have principal alias bounded below by a fixed constant, all aliases have $\lambda_\nu \ge \lambda_0$, and their squared coefficients sum to at most one. Hence

$$c_L \lambda_0^{-1} I \le C_r(\theta) \le \lambda_0^{-1} I, \qquad \|S_r(\theta)\| \le C_L \lambda_0. \tag{NK19}$$

The principal symbol is uniformly comparable to the coarse symbol. These facts prove three bounds:

$$\begin{aligned} \|S_r e\|_{H^s_r} &\le C_{s,L} r^{-2}\|e\|_{H^s_r}, \\ \|S_r \Pi_r^0 Q_r L_a^{-1} \Pi_a^0 f\|_{H^s_r} &\le C_{s,L}\|f\|_{H^s_a}, \\ \|S_r \Pi_r^0 Q_r A\|_{H^s_r} &\le C_{s,L}\|A_T\|_{H^{2+s}_a}, \qquad A_T = \Pi_a^0 A. \end{aligned} \tag{NK20}$$

For the middle bound, weighted Cauchy–Schwarz on an alias fibre gives the squared multiplier bound

$$(1+\lambda_c)^s\|S_r\|^2\sum_\nu |q_\nu|^2\lambda_\nu^{-2}(1+\lambda_\nu)^{-s}\le C_{s,L}. \tag{NK21}$$

In the vector case apply the same bound componentwise and use the orthogonal projections, whose norms are one. The last bound in (NK20) follows in the same way with the two derivatives placed on the input.

The chain identity (NK14) removes a fine gradient from $\Pi_r^0 Q_r A$. The low frequency of the transverse input is paid by $\|A\|_2$, and the high frequencies by its curl. Therefore

$$\|A_T\|_{H_a^{2+s}}\le C_s(\|d_aA\|_{H_a^{1+s}}+\|A\|_2)\le C_s(\|F_a(A)\|_{H_a^{1+s}}+\|A\|_2+(M+M^2)K_A). \tag{NK22}$$

No massless inverse is used at the harmonic mode.

48.4 4. Ward control of the longitudinal multiplier

Decompose the actual coarse multiplier by the flat Hodge decomposition:

$$\lambda=\lambda_T+\lambda_L+\lambda_H,\qquad \lambda_T=\Pi_r^0\lambda,\quad \lambda_H=P_0\lambda,\quad \lambda_L=d_r(-\Delta_r)^{-1}\delta_r\lambda. \tag{NK23}$$

The inverse in the last expression acts only on mean-zero site fields.

The actual coarse output V has a physical logarithm B with

$$\|B\|_{W^{1,\infty}}\le C_L M \tag{NK24}$$

by GQ25. Consequently

$$\delta_r\lambda=(\delta_r-d_V^{\mathsf t})\lambda=\mathcal R_B\lambda,\qquad \|\mathcal R_B\lambda\|_{H_r^s}\le C_{s,L}(M+M^2)\|\lambda\|_{H_r^s}. \tag{NK25}$$

The operator $\mathcal R_B$ is zero order, with uniformly $W^{1,\infty}$ coefficients. On the fixed physical torus,

$$\|d_r(-\Delta_r)^{-1}\|_{H_r^s\to H_r^s}\le C_T.$$

Thus

$$\boxed{\|\lambda_L\|_{H_r^s}\le C_{s,L,T}(M+M^2)\|\lambda\|_{H_r^s}.} \tag{NK26}$$

This is the only place where a long-wavelength gauge inverse is needed. The required small radius can depend on T; no volume-uniform conclusion is being asserted.

48.5 5. The harmonic multiplier is fixed by the mean Euler equation

Take the physical mean of the exact equation $\delta_a d_a u=J_a(u)+f_a(u)$. The left side has zero mean. Flat Q_r^* preserves constants and the physical mean. Thus (NK16) gives

$$\lambda_H=-P_0(e_Q+f_a(u)). \tag{NK27}$$

It follows that

$$\boxed{\|\lambda_H\|_2\le C_{s,L}(M+M^2)(K_u+\|\lambda\|_{H_r^s}).} \tag{NK28}$$

The H^s and L^2 norms of a constant agree. This argument does not solve a singular Schur equation at zero momentum or discard a possibly nonzero harmonic Wilson force.

48.6 6. Transverse KKT inversion and absorption

Project (NK12) onto the mean-zero co-closed fine subspace. Let $u_T = \Pi_a^0 u$. Equations (NK16), (NK23) and preservation of constants by Q_r^* give

$$L_a u_T = Q_r^* \lambda_T + f_T,$$

$$f_T = \Pi_a^0 \{ f_a(u) + e_Q + Q_r^* \lambda_L \}. \tag{NK29}$$

The harmonic flat lift disappears under Π_a^0. Equations (NK13), (NK16)–(NK17), and (NK26) yield

$$\|f_T\|_{H_a^s} \le C_{s,L,T}(M + M^2)(K_u + \|\lambda\|_{H_r^s}). \tag{NK30}$$

The flat chain identity implies $\Pi_r^0 Q_r u = \Pi_r^0 Q_r u_T$. Apply $\Pi_r^0 Q_r L_a^{-1}$ to (NK29), and use (NK15):

$$C_r \lambda_T = \Pi_r^0 Q_r A + e_c - \Pi_r^0 Q_r L_a^{-1} f_T. \tag{NK31}$$

The first line of (NK20) costs r^{-2}, exactly canceled by (NK15). The other lines of (NK20), (NK22), and (NK30) prove

$$\|\lambda_T\|_{H_r^s} \le C_{s,L} \mathcal{D}_s(A, u) + C_{s,L,T}(M + M^2)\|\lambda\|_{H_r^s}. \tag{NK32}$$

Adding (NK26), (NK28), and (NK32), and choosing the one fixed small radius so that the total final coefficient is at most $1/2$, gives

$$\|\lambda\|_{H_r^s} \le C_{s,L,T} \mathcal{D}_s(A, u). \tag{NK33}$$

There is no depth-dependent smallness condition.

Equation (NK16) together with (NK17) now bounds the physical current in H_a^s. Finally, invert L_a in (NK29) on the mean-zero subspace, apply a curl, and use (NK11):

$$\begin{aligned} \|F_a(u)\|_{H_a^{1+s}} &\le \|d_a u_T\|_{H_a^{1+s}} + \|G_a(u)\|_{H_a^{1+s}} \\ &\le C_{s,L,T}\{\|\lambda_T\|_{H_r^s} + \|f_T\|_{H_a^s} + (M + M^2)K_u\}. \end{aligned} \tag{NK34}$$

The torus Poincare constant is finite and independent of the mesh. This proves (NK6).

48.7 7. A smaller regularity class already closes scalar interpolation

A full H^2 norm of an arbitrary fine potential need not be inferred from curvature regularity. The following class is sufficient for the scalar interpolation operations used in the fractional hull:

$$\mathcal{Y}_s(A) = \|A\|_{W^{1,\infty}} + \|A\|_{H^{1+s}} + \|F_a(A)\|_{H^{1+s}}. \tag{NK35}$$

Use the actual lattice curvature and all the mesh conventions above. By (NK11), on the fixed small chart this is equivalent, with a fixed polynomial change of bound, to

$$\|A\|_{W^{1,\infty}} + \|A\|_{H^{1+s}} + \|d_a A\|_{H^{1+s}}. \tag{NK35a}$$

Thus the third part is a curl norm; it is not an asserted high-regularity norm of the arbitrary fine gauge direction.

Let χ_i be fixed smooth scalar cutoffs with uniformly bounded derivatives through order three and let $A = \sum_{i=1}^n \chi_i A_i$, with n uniformly finite and all fields in the common small chart. It also suffices to have a uniform C^{2,β_χ} bound with $\beta_\chi > s$. A merely bounded second derivative is not the sufficient

hypothesis used here: even for constant A_i, the term $(d\chi_i)A_i$ needs the fractional regularity of the second derivative of the cutoff.

For one common base-site sample of χ, the exact lattice rule is

$$\begin{aligned}(d_a(\chi A))_{\mu\nu}(x) = \chi(x)(d_a A)_{\mu\nu}(x) \\ + (D_{a,\mu}\chi)(x)A_\nu(x+ae_\mu) - (D_{a,\nu}\chi)(x)A_\mu(x+ae_\nu).\end{aligned} \tag{NK35b}$$

Component-midpoint sampling changes this formula by terms with a factor a, one cutoff derivative and one first difference of A; the compensated inverse argument in (NK9)–(NK11) bounds those terms in H^{1+s}. Uniform $W^{3,\infty}$ control makes $D_a\chi$ a uniform $W^{2,\infty}$ multiplier, hence a bounded multiplier on H^{1+s}, for every $s<1$. Under the alternative C^{2,β_χ} premise the fractional translation estimate gives the same bound for $s<\beta_\chi$.

Substitute $d_aA_i = F_a(A_i) - G_a(A_i)$ into (NK35b), then add $G_a(A)$. Every term is bounded in H^{1+s} by (NK11), these scalar multiplier bounds and the corresponding $\mathcal{Y}_s(A_i)$. Thus

$$\mathcal{Y}_s\left(\sum_i \chi_i A_i\right) \le C_{\chi,n,s}\,\mathcal{P}\left(\sum_i \mathcal{Y}_s(A_i)\right), \tag{NK36}$$

with a fixed polynomial on the small chart and no inverse power of a.

This statement is about potentials expressed in one admitted common gauge. It does not manufacture such a gauge for independent backgrounds. It shows that a hidden fine H^2 gauge estimate is unnecessary for this particular interpolation closure once the actual low chart is supplied.

49 The flat Landau gauge and a uniform fractional longitudinal estimate

This note corrects the flat gauge projection and proves the longitudinal estimate in the actual raw contour coordinates. The site average appearing in the gauge condition and the node restriction appearing in the raw gauge transformation are different maps. Their mismatch is resolved here by a positive mixed biharmonic Schur operator. No nonlinear change of fine gauge is made.

The proved result is a uniform flat theorem. The background-dependent projection, the nonlinear adapted constraints, and mixed determining boundaries of the non-Abelian construction remain separate matters, described precisely in Section 6.

49.1 1. The gauge projection in the primary source

Dimock's primary treatment of the multiscale Abelian construction, Section 3.6.2, equations (267)–(271), uses the orthogonal projection onto

$$\Delta \ker S, \tag{LF1}$$

where S is the scalar site-block average. Equation (271), which explicitly cites equation (217) of Balaban's 1984 propagator paper II, is

$$R_{\mathrm{L}} = I - GS^*(SG^2S^*)^{-1}SG, \qquad G = (\Delta + S^*aS)^{-1}. \tag{LF2}$$

Here a is a positive coarse diagonal weight. This is not the sharp projection $I - S^*S$.

On a homogeneous periodic lattice, remove the constant scalar mode. Since Δ is self-adjoint and invertible on that space,

$$(\Delta \ker S)^{\perp} = \Delta^{-1} \operatorname{ran} S^*. \tag{LF3}$$

Consequently the flat Landau condition on a one-form a is

$$R_{\mathrm{L}}\delta a = 0 \quad \Longrightarrow \quad \delta a = \Delta^{-1}S^*\nu. \tag{LF4}$$

The parameter a in (LF2) changes the representation of this subspace, not the subspace. This follows either from orthogonal complementation or from $\Delta GS^* = S^*(I - aSGS^*)$.

Write the fine Hodge decomposition as

$$a = a_{\mathrm{T}} + d\phi, \qquad \delta a_{\mathrm{T}} = 0, \qquad \phi = \Delta^{-1}\delta a.$$

Then (LF4) says

$$\phi = \Delta^{-2}S^*\nu. \tag{LF5}$$

Thus the longitudinal potential is a biharmonic lift of coarse scalar data. Replacing (LF2) by a sharp block projection would give a different potential and a different regularity problem.

The non-Abelian variational paper CMP **102**, 277–309, states its original gauge condition as $R(U_0)D^*_{U_0}A = 0$ in (21), referring for the definition to the background-field propagator and regular-configuration papers. Its equations (134)–(140) use a bounded operator DPD^*, with $P = I - R$; this is consistent with the smoothed projection above. That citation does not by itself establish a fractional perturbation bound for $R(U_0)$.

49.2 2. The mixed node/box biharmonic operator

Let $a > 0$ be the fine spacing, $r = na$, and let n be odd. All tori have compatible periods. Scalar products have their physical volume factors. The centered coarse block at y contains the fine sites

$$y + ak, \qquad k_j = -(n-1)/2, \dots, (n-1)/2.$$

Let $S = S_r$ be their normalized average, $H = S^*$ the block hold, and $N = N_r$ the restriction to the center sites y. The two scalar maps S and N are not identified.

For a coarse Fourier frequency $\theta \in [-\pi/r, \pi/r]^d$, use the fine aliases

$$p_\nu = \theta + \frac{2\pi\nu}{r}, \qquad \nu_j = -(n-1)/2, \dots, (n-1)/2.$$

Up to the harmless choice at a Brillouin boundary, these are all the fine aliases. Centered blocks make the scalar average multiplier real:

$$q_\nu(\theta) = \prod_{j=1}^{d} \frac{\sin((r\theta_j + 2\pi\nu_j)/2)}{n \sin((r\theta_j + 2\pi\nu_j)/(2n))}. \tag{LF6}$$

The value of a removable quotient at zero is one. In this physical Fourier normalization, S has alias row q_ν, H has alias column q_ν, and N has alias row 1. In particular,

$$\sum_\nu |q_\nu|^2 = 1, \qquad q_0 \geq (2/\pi)^d. \tag{LF7}$$

Write

$$\lambda_\nu = \frac{4}{a^2} \sum_j \sin^2(ap_{\nu,j}/2), \qquad \lambda_r = \frac{4}{r^2} \sum_j \sin^2(r\theta_j/2).$$

Then

$$\lambda_\nu \geq \lambda_0, \qquad \lambda_r \leq \lambda_0 \leq (\pi^2/4)\lambda_r. \tag{LF8}$$

Define the mixed scalar operator on the mean-zero sector by

$$C_{\mathrm{mix}} = N\Delta^{-2}H. \tag{LF9}$$

Its symbol is

$$c_{\mathrm{mix}}(\theta) = \sum_\nu q_\nu \lambda_\nu^{-2}. \tag{LF10}$$

The q_ν have signs, so a positive denominator cannot be concluded by replacing them with $|q_\nu|^2$. The following exact heat-kernel estimate supplies the required positivity:

$$\boxed{q_0\lambda_0^{-2} \leq c_{\mathrm{mix}}(\theta) \leq 2^d q_0 \lambda_0^{-2}, \qquad \theta \neq 0.} \tag{LF11}$$

49.2.1 Proof of (LF11)

First work in one coordinate and put $z = r\theta$. By evenness it suffices to take $0 < z \le \pi$. For $t > 0$, set

$$A_t(x) = \frac{\exp[-4ta^{-2}\sin^2(x/(2n))]}{\sin(x/(2n))}, \qquad 0 < x \le n\pi.$$

The principal term in the one-dimensional heat sum is

$$\frac{\sin(z/2)}{n} A_t(z).$$

Pair the aliases k and $-k$, $1 \le k \le (n-1)/2$. Their combined contribution is

$$(-1)^{k+1}\frac{\sin(z/2)}{n}\{A_t(2\pi k - z) - A_t(2\pi k + z)\}. \tag{LF12}$$

The bracket is nonnegative and decreases with k. To verify this without an infinite-series or endpoint approximation, put

$$b_T(y) = e^{-T\sin^2 y}/\sin y, \qquad T = 4ta^{-2}, \quad 0 < y \le \pi/2.$$

It decreases, and direct differentiation gives

$$\frac{b_T''(y)}{b_T(y)} = 4T^2\sin^2 y\cos^2 y + 2T + 1 + 2\cot^2 y > 0. \tag{LF13}$$

Therefore the integral of $-b_T'$ over an interval of fixed length decreases when the interval moves to the right. This proves the asserted monotonicity of the brackets in (LF12).

The finite alternating sum of the paired terms starts positive, hence lies between zero and its first term. The first bracket is at most $A_t(z)$, since $2\pi - z \ge z$. Thus the full one-dimensional heat sum is between its principal term and twice that term:

$$q_0 e^{-t\lambda_0} \le \sum_k q_k e^{-t\lambda_k} \le 2q_0 e^{-t\lambda_0}. \tag{LF14}$$

At $z = 0$, all nonprincipal box multipliers vanish, so the same statement follows directly. For $n = 1$, there are no nonprincipal aliases.

The d-dimensional heat sum factors into these one-dimensional sums because the Laplacian eigenvalue is a sum of its coordinate eigenvalues. Integrating

$$\lambda^{-2} = \int_0^\infty te^{-t\lambda}\,dt$$

proves (LF11). All sums are finite before the integral. No independence, continuum sampling theorem, or limiting sign argument is used.

49.3 3. A uniform lift with prescribed node values

Equation (LF11) makes the following exact right inverse well-defined:

$$H_{\mathrm{node}} = \Delta^{-2} H C_{\mathrm{mix}}^{-1}, \qquad N H_{\mathrm{node}} = I. \tag{LF15}$$

This maps a coarse scalar field to the subspace in (LF5) and gives it exactly the prescribed node values. It is the appropriate lift for a raw contour constraint, whose gauge variation uses node restriction.

For $0 \le t \le 4$,

$$\|H_{\mathrm{node}} c\|_{H_a^t} \le C_{d,t} \|c\|_{H_r^t}. \tag{LF16}$$

The constants are independent of the fine cutoff, n, the number of blocking steps, and the compatible torus volume. Constants may be added separately and are mapped to constants.

Indeed, on a nonzero coarse fibre the left-hand norm squared has multiplier

$$c_{\mathrm{mix}}^{-2} \sum_{\nu} |q_\nu|^2 \lambda_\nu^{-4} (1+\lambda_\nu)^t. \tag{LF17}$$

For $t \le 4$, the function $\lambda^{-4}(1+\lambda)^t$ is decreasing. By (LF7), (LF8), and (LF11), (LF17) is bounded by

$$C(1+\lambda_0)^t \le C_t (1+\lambda_r)^t.$$

This proves (LF16).

The gradient version avoids any volume-dependent Poincare constant. For $0 \le s \le 2$,

$$\|dH_{\mathrm{node}} c\|_{H_a^{1+s}} \le C_{d,s} \|d_r c\|_{H_r^{1+s}}. \tag{LF18}$$

Here the fine multiplier in place of (LF17) contains $\lambda_\nu^{-3}(1+\lambda_\nu)^{1+s}$, which is decreasing for $s \le 2$. The same proof gives the bound by $\lambda_r(1+\lambda_r)^{1+s}$.

For comparison, prescribing block data instead of node data gives the usual constrained biharmonic lift

$$H_{\mathrm{box}} = \Delta^{-2} H (S \Delta^{-2} H)^{-1}.$$

It obeys the same estimates even more directly because its Schur symbol is $\sum |q_\nu|^2 \lambda_\nu^{-2}$. It is not substituted for (LF15) in the raw-coordinate argument.

49.4 4. The raw linear average is uniformly bounded in the needed norm

Let the total blocking ratio be $n = L^k$ with one fixed odd L. Write $Q_{\mathrm{raw}}^{(k)}$ for the linearization at the identity of the actual iterated centered contour average. Let $Q_{\mathrm{bl}}^{(k)}$ be the direct box-and-line average.

Their exact gradient identities differ:

$$Q_{\mathrm{raw}}^{(k)} d = d_r N, \qquad Q_{\mathrm{bl}}^{(k)} d = d_r S. \tag{LF19}$$

At one step the difference is a coarse exact gradient. If r_j is the linear endpoint connector,

$$Q_{\mathrm{raw},j} = Q_{\mathrm{bl},j} - d_{j+1} r_j.$$

Iteration gives the exact identity

$$Q_{\mathrm{raw}}^{(k)} = Q_{\mathrm{bl}}^{(k)} - d_r r_{\mathrm{tot}}, \qquad r_{\mathrm{tot}} = \sum_{j=0}^{k-1} N_{j+1\to k}\, r_j\, Q_{\mathrm{bl}}^{(j)}. \tag{LF20}$$

This is obtained by applying the first identity in (LF19) to the already accumulated gradient at each step. It is not an associativity assumption about the nonlinear mean.

The centered connector r_j has coefficient sum $O_L(\ell)$, support of diameter $O_L(\ell)$, and zero constant moment, where $\ell = L^{j+1}a$. Its Fourier multiplier consequently satisfies

$$|\rho_j(p)| \le C_L \ell^2 |p|.$$

For $0 < s < 1/2$, the sparse-node estimate in four dimensions is

$$\|N_{j+1\to k} r_j f\|_{H^s_r} \le C_{s,L} r^2 (\ell/r)^s \|f\|_{H^{1+s}_{\ell/L}}. \tag{LF21}$$

One proof applies alias Cauchy–Schwarz to the final sampling. After the zero-moment factor $\ell^2|p|$, the squared alias sum is at most

$$C\ell^4 \sum_{|\nu| \lesssim r/\ell} (1+|\nu|)^{-2s} \le C_s r^4 (\ell/r)^{2s}.$$

The principal alias obeys the same bound. This is a finite-lattice estimate; it does not assert that point restriction is bounded on continuum $H^{1+s}(\mathbb{R}^4)$.

The complete flat box/line map is a contraction, up to fixed midpoint phases, in the physical discrete Bessel norms. This follows from its normalized alias row and $1 + \lambda_r \le 1 + \lambda_\nu$. Thus summing (LF21) over the geometric scales gives

$$\|r_{\mathrm{tot}} f\|_{H^s_r} \le C_{s,L} r^2 \|f\|_{H^{1+s}_a}. \tag{LF22}$$

For $r \le 1$, two coarse inverse powers give

$$\|d_r v\|_{H^{1+s}_r} \le C_d r^{-2} \|v\|_{H^s_r}.$$

Combining this with (LF20)–(LF22) proves

$$\boxed{\|Q^{(k)}_{\mathrm{raw}} f\|_{H^{1+s}_r} \le C_{s,L} \|f\|_{H^{1+s}_a}, \qquad 0 < s < 1/2.} \tag{LF23}$$

Other fixed upper bounds for r only change the harmless inhomogeneous-norm constant. The crucial statement is uniformity as a/r tends to zero.

49.5 5. The exact flat longitudinal closure

Theorem. Let $0 < s < 1/2$. Suppose a fine one-form a on a compatible periodic four-torus satisfies the original flat Landau condition

$$R_{\mathrm{L}} \delta a = 0$$

and the raw iterated linear constraint

$$Q^{(k)}_{\mathrm{raw}} a = B.$$

Then

$$\boxed{\|a\|_{H^{1+s}_a} \le C_{s,L} \{ \|a_{\mathrm{T}}\|_{H^{1+s}_a} + \|B\|_{H^{1+s}_r} \}.} \tag{LF24}$$

All harmonic one-form modes are included in a_{T}. The constant is independent of the fine cutoff and blocking depth.

Proof. Use the Hodge decomposition in Section 1. By (LF5), the zero-mean scalar potential belongs to the biharmonic range. If $c = N\phi$, the invertibility in (LF11) gives $\phi = H_{\mathrm{node}} c$. The raw constraint and (LF19) imply

$$d_r c = B - Q^{(k)}_{\mathrm{raw}} a_{\mathrm{T}}. \tag{LF25}$$

Apply (LF18) to $d\phi$, then (LF23) to a_{T}, and finally add the transverse part. This gives (LF24). No sharp hold, discontinuous fine gauge transformation, or replacement of N by S occurs.

The first-order Hodge estimate can express the transverse norm through a curvature norm and harmonic data if desired. It is useful to keep (LF24) in the displayed form so no volume-dependent infrared bound is hidden.

There is also a precise inhomogeneous version. Given an arbitrary a, set

$$f = R_{\mathrm{L}}\delta a, \qquad a_f = d\Delta^{-1}f.$$

Then $a - a_f$ satisfies the homogeneous flat Landau condition. The same proof, using (LF23), yields

$$\|a\|_{H^{1+s}_a} \le C_{s,L}\{\|a_{\mathrm{T}}\|_{H^{1+s}_a} + \|Q^{(k)}_{\mathrm{raw}}a\|_{H^{1+s}_r} + \|d\Delta^{-1}R_{\mathrm{L}}\delta a\|_{H^{1+s}_a}\}. \tag{LF26}$$

Equation (LF26) identifies the exact gauge-error norm that a nonlinear perturbation argument must control.

49.6 6. What this proves for the current construction, and what it does not

The primary variational theorem already gives a local gauge for an actual minimizer. In CMP **102**, printed page 279, Theorem 1 equations (9)–(10), the first-derivative Holder bound on a retained layer of spacing r is proportional to

$$\epsilon_1 r^{-2-\beta}. \tag{LF27}$$

The constant is uniform in the fine cutoff, but the layer spacing is part of the bound. An old inner layer cannot be renamed as the current outer scale.

Balaban I (1987), (3.31)–(3.32), provides current-scale $C^{1,\beta}$ control of the outer field used in the local Taylor argument. Equations (4.16)–(4.18) transfer it to the averaged coarse field B. Equation (3.36) states the relevant pair-domain inclusion for the inner reconstructed field and its own current. These are different assertions. In particular, the cited pair-domain inclusion is not by itself a current-scale H^{1+s} estimate for the inner minimizer at an arbitrarily old retained layer.

The flat theorem above closes the exact linear longitudinal issue, including the raw-node/averaged-gauge mismatch. It makes the remaining nonlinear task narrower. To use (LF26) for the actual small-background representative, one needs uniform estimates of the following two kinds on the same original domain:

$$\begin{aligned} \|d\Delta^{-1}R_{\mathrm{L}}\delta a\|_{H^{1+s}} &\le \varepsilon\|a\|_{H^{1+s}} + F_{\mathrm{gauge}}, \\ \|q^{(k)}_{\mathrm{raw}}(a) - Q^{(k)}_{\mathrm{raw}}a\|_{H^{1+s}_r} &\le \varepsilon\|a\|_{H^{1+s}_a} + F_{\mathrm{constraint}}. \end{aligned} \tag{LF28}$$

Here the first line must follow from the actual $R(U_0)D^*_{U_0}a = 0$, including the background, boundary conditions, and any known inhomogeneity. The second line is now proved on the common centered small-potential chart by CM34 in the centered-mean note: its right-hand side is

$$C_{s,L}r(M + M^2)\|a\|_{H^{1+s}_a},$$

where M controls the actual $W^{1,\infty}$ chart. Thus it has the additional factor r. This estimate applies to the original raw averaging map in that chart; no adapted coordinate is silently substituted. If the first line is also established with ε below the fixed constant in (LF26), the two errors can be absorbed. The first line remains unproved for a general original background projection.

There is a useful exact alternative in the primary source. CMP **102**, Section F, printed pages 301–302, (148)–(158), constructs local nested determining cubes for an actual minimizer and gauges it relative to the identity. Equation (153) then gives precisely a flat $R\delta A = 0$; the text after (158) explicitly identifies the external background as the identity. However, the gauge changes V to V', and the actual constraints become the adapted B in (154)–(156). The displayed bound (155) controls its sup norm. A current-scale H^{1+s} bound for that transformed B does not follow merely from (155). This alternative replaces the background-projection question with a concrete source-coordinate and boundary question.

In particular:

- Formula (LF2) is verified here for the flat Abelian/multiscale construction. The background paper's precise $R(U_0)$, its covariant scalar averaging map, and their fractional operator bounds have not been reconstructed merely by citing Dimock.
- CMP **102** (20), (42), and (44) use adapted nonlinear coordinates $Q_j(U_0, \eta A)$, with B defined through the preceding regular-configuration theorem. The raw and adapted constraint maps must be related explicitly before transferring a nonlinear estimate between them.
- Mixed determining sets and local boundary conditions are not Fourier diagonal. Their actual collars and a localized analogue of the above inverse must be retained. A periodic flat theorem alone is not that boundary theorem.
- A separately wall-weakened propagator need not preserve the original gauge equation. The protected-seed telescope applies only after matching its full typed factors and any projection operators to the actual construction.

These restrictions leave a concrete operator/domain problem rather than an unexamined claim that every auxiliary background is a smooth critical point.

49.7 7. The mixed lift is also uniformly bounded in $W^{2,p}$

The Fourier proof gives a useful stronger statement. For every $1 \le p \le \infty$,

$$\|H_{\mathrm{node}} c\|_{W_a^{2,p}} \le C_d \|c\|_{W_r^{2,p}}. \tag{LF29}$$

Here $W^{2,p}$ uses all nearest-neighbor physical differences through order two. The estimate includes the constant mode. It is uniform in the odd ratio n and the compatible periodic volume. The following kernel proof avoids assuming that $\lambda_r^2 c_{\mathrm{mix}}$ is analytic at zero.

Scale r to one for this proof, so $a = 1/n$. For a fine offset ξ in one coarse block, the symbol giving the fine output above a coarse plane wave is

$$b_\xi(\theta) = \frac{\sum_\nu e^{ip_\nu \cdot \xi} q_\nu \lambda_\nu^{-2}}{c_{\mathrm{mix}}(\theta)}. \tag{LF30}$$

For any multi-index $|\alpha| \le 2$, the corresponding fine-difference symbol is

$$b_\xi^\alpha(\theta) = \frac{\sum_\nu e^{ip_\nu \cdot \xi} d_a(p_\nu)^\alpha q_\nu \lambda_\nu^{-2}}{c_{\mathrm{mix}}(\theta)}, \qquad d_{a,j}(p) = a^{-1}(e^{iap_j} - 1). \tag{LF31}$$

Offsets crossed by these fine differences remain in a fixed enlargement of the unit block. Since $n\xi$ has integer coordinates, reindexing the aliases at a coarse Brillouin face gives exact periodicity in θ.

For $\nu \neq 0$, all denominators in the sums are analytic and uniformly separated from zero in a fixed complex neighborhood of the real coarse Brillouin cube. In that neighborhood,

$$|q_\nu| \le C \prod_j (1 + |\nu_j|)^{-1}, \qquad |\lambda_\nu| \ge c|\nu|^2, \qquad |d_a(p_\nu)| \le C(1 + |\nu|).$$

Consequently the nonprincipal sums in (LF30)–(LF31) are uniformly bounded and analytic. Their absolute tails are bounded by

$$C_\alpha \sum_{m \ge 1} m^{|\alpha|-5} (1 + \log m)^{d-1} < \infty, \qquad |\alpha| \le 2. \tag{LF32}$$

To obtain the displayed tail, choose a coordinate attaining $\max_j |\nu_j| = m$; its box factor contributes m^{-1}, and the sums of the remaining factors are $O((1 + \log m)^{d-1})$. This count is independent of the finite alias cutoff.

Let

$$R(\theta) = \sum_{\nu \neq 0} q_\nu \lambda_\nu^{-2}, \qquad D(\theta) = q_0 + \lambda_0^2 R(\theta).$$

Both are analytic in the same fixed neighborhood. Equation (LF11) gives

$$D(\theta) \ge q_0 \ge (2/\pi)^d$$

on the real Brillouin cube, including the continuous value at zero. Uniform analytic bounds therefore give a fixed smaller complex neighborhood on which $|D| \ge c_d > 0$. Multiplying numerator and denominator in (LF31) by λ_0^2 gives

$$b_\xi^\alpha(\theta) = \frac{e^{i\theta\cdot\xi} d_a(\theta)^\alpha q_0 + \lambda_0^2 \sum_{\nu \neq 0} e^{ip_\nu \cdot \xi} d_a(p_\nu)^\alpha q_\nu \lambda_\nu^{-2}}{D(\theta)}. \tag{LF33}$$

This explicitly removes the principal pole, including on the nearby complex characteristic set $\lambda_0 = 0$. Equations (LF32)–(LF33) prove a uniform analytic strip bound for b_ξ^α, for all offsets and all odd n. The analytic continuations agree at Brillouin faces by their exact periodic identity.

Thus the coarse convolution kernels $k_\xi^\alpha(z)$ satisfy

$$|k_\xi^\alpha(z)| \le C_\alpha e^{-\gamma|z|_1}, \qquad |\alpha| \le 2, \tag{LF34}$$

with fixed constants. For a finite torus, periodize these infinite-volume kernels; the same weighted unperiodized estimate controls every resulting convolution.

At zero frequency, (LF33) gives

$$b_\xi(0) = 1, \qquad \nabla_\theta b_\xi(0) = i\xi.$$

It follows that H_{node} reproduces constants and affine scalar functions. Equivalently, its fine derivative kernels satisfy

$$\sum_z k_\xi^\alpha(z) = 0 \quad (|\alpha| \ge 1), \qquad \sum_z z_j k_\xi^\alpha(z) = 0 \quad (|\alpha| = 2). \tag{LF35}$$

For the first fine derivative, subtract $c(y)$ under the convolution and telescope the coarse differences along a lattice path. For the second derivative, subtract the affine discrete Taylor polynomial at y; the remainder is bounded in ℓ^p by $C|z|_1^2 \sum_{|\beta|=2} \|D^\beta c\|_{\ell^p}$. The cancellations in (LF35) remove the subtracted terms. Sum (LF34), using its first or second moment. This proves, after restoring r,

$$\|D_a^\alpha H_{\mathrm{node}} c\|_{\ell_a^p} \le C_\alpha \sum_{|\beta|=|\alpha|} \|D_r^\beta c\|_{\ell_r^p}, \qquad |\alpha| \le 2. \tag{LF36}$$

The physical fine norm is the average over the n^d fine offsets of the corresponding coarse norms; the bounds are uniform over those offsets. This proves (LF29), including $p = 1, \infty$. For periodic inputs the discrete Taylor argument uses their periodic extension on the infinite coarse lattice, so no moment is taken modulo the torus period.

The separate gauge-section construction can now combine (LF29) with the nodal restriction estimate $N : W_a^{2,p} \to W_r^{2,p}$, $p > 4$, and the mean-zero Poisson estimate to study

$$\mathcal{J} = (I - H_{\mathrm{node}} N)\Delta^{-1}\delta.$$

For a scalar ϕ with $N\phi = 0$, the identity $\mathcal{J}(d\phi) = \phi$ follows exactly, including removal and restoration of its constant mode. Also $\mathcal{J}a = 0$ is equivalent to $R_{\mathrm{L}}\delta a = 0$ by (LF3)–(LF5). These identities identify the correct linear gauge correction. Its nodal Sobolev bound and the nonlinear fixed point belong to the separate gauge-section construction; they are not inferred only from (LF29).

50.1 1. The norms and the two product estimates

Fix

$$p > 8, \qquad 0 < s < \tfrac{1}{2}, \qquad \alpha = 1 - \frac{4}{p} > s. \tag{WP1}$$

At mesh b, use physical volume in every L^p norm and write

$$M_p(A) = \|A\|_{L^p} + \|\nabla_b A\|_{L^p}, \qquad K_s(A) = \|A\|_{H^{1+s}_b}. \tag{WP2}$$

For these iteration estimates the spatial components use the componentwise norm $\|A\|_{L^p} = (\sum_\mu \|A_\mu\|_{L^p}^p)^{1/p}$, with the analogous sum over μ, ν for first differences. The color norm within each component is fixed. Each flat component then has its own positive box-and-line contraction with coefficient one. The usual vector-valued norms are equivalent by a fixed finite-dimensional constant, applied once to the complete bound rather than once per blocking step.

The discrete Sobolev and Morrey inequalities, with constants uniform in b, give

$$\|A\|_\infty + [A]_{C^\alpha} \le C_{p,T} M_p(A). \tag{WP3}$$

One direct justification is to extend a site field by piecewise multilinear interpolation on each mesh cube. Its continuum $W^{1,p}$ norm is bounded by the discrete norm, and its values at the mesh sites are unchanged. Apply the continuum Morrey inequality to that extension. Bond-midpoint component lattices are translates of the site lattice, so the same argument applies to each component.

Put $q = 2p/(p-2)$; then $2 < q < 4$. Let $\mathcal{B}^s_{q,2}$ denote the norm obtained by taking the L^q norm of translation differences, followed by the square integral with the usual fractional weight. The Fourier comparison for discrete differences and the uniform H^1-to-L^q Sobolev inequality imply

$$\|A\|_{\mathcal{B}^s_{q,2}} \le C_{s,p,T} \|A\|_{H^{1+s}}. \tag{WP4}$$

Indeed, apply the H^1-to-L^q inequality to each translation difference of A, then integrate its square with the fractional weight. The resulting Fourier multiplier is bounded by a constant times $(1 + \widehat{\Delta})^{1+s}$. The same argument gives the undifferenced term. No continuum point trace is used in (WP4).

The exact translation identity for a product gives

$$\|fg\|_{H^s} \le C_{s,p,T} \left(\|f\|_\infty \|g\|_{H^s} + \|g\|_{L^p} \|f\|_{\mathcal{B}^s_{q,2}} \right) \tag{WP5}$$

whenever the norms on the right are finite. For the second term use Hölder with $1/p + 1/q = 1/2$ on the factor containing the translation difference of f; for the first term use its supremum norm on the other product difference. The zero-order term is bounded in the same way. Matrix products obey this estimate with a fixed dimension-dependent constant.

In particular, applying (WP5) to each term of the exact shifted derivative of AB gives

$$\|AB\|_{H^{1+s}} \le C_{s,p,T}\left(M_p(A)K_s(B) + M_p(B)K_s(A)\right). \tag{WP6}$$

The derivative terms use $\nabla A \in H^s \cap L^p$, (WP3), and (WP4). This proves the estimate without an assumed bounded first derivative.

A second useful estimate is the multiplier bound

$$\|bf\|_{H^s} \le C_{s,p,T}\|b\|_{W^{1,p}}\|f\|_{H^s}. \tag{WP7}$$

To see it directly, use (WP3) in the product difference. The additional integral at short distances is bounded by $\int_0^1 t^{2\alpha-2s-1}\,dt$, which is finite by (WP1). Long distances and the undifferenced term are bounded by the supremum of b. The discrete difference sum satisfies the same bound uniformly as its lower cutoff tends to zero.

50.3 3. The accumulated coarse gauge in two different norms

The connector r_ℓ has zero constant moment. Its one-step L^p estimate is

$$\|r_\ell(A)\|_{L^p_\ell} \le C_L \ell^2 \|\nabla_a A\|_{L^p_a}. \tag{WP11}$$

Restriction from the ℓ-site lattice to the retained r-site lattice has L^p norm at most $(r/\ell)^{4/p}$. Thus the intermediate node factors satisfy

$$\|b_{j,r}\|_{L^p_r} \le C_L r^{4/p} \ell_j^{2-4/p} M. \tag{WP12}$$

The Fourier estimate GQ10 is unchanged, since it uses the H^{1+s} norm, not a supremum derivative. It gives

$$\|b_{j,r}\|_{H^s_r} \le C_{s,L} r^2 (\ell_j/r)^s K. \tag{WP13}$$

Sum these geometric series. At a retained site, $\|f\|_\infty \le r^{-4/p}\|f\|_{L^p_r}$. The exact ordered-product difference identity, and then the local logarithm, consequently give

$$\|\kappa\|_{L^p_r} \le Cr^2 M, \qquad \|\kappa\|_{H^s_r} \le Cr^2 K, \qquad \|\kappa\|_\infty \le Cr^{2-4/p} M. \tag{WP14}$$

The constants include bounded analytic exponentials, controlled by the sum of the intermediate supremum norms. The exponents $2 - 4/p$ and s are positive, so no microscopic-depth factor appears.

The weaker supremum power in (WP14) must not be used in place of the L^p bound when estimating a mixed gauge term. In the exact gauge action GQ19–22, terms of the form κB are estimated by (WP5):

$$\|\kappa B\|_{H^s} \le C\left(\|B\|_\infty \|\kappa\|_{H^s} + \|\kappa\|_{L^p}\|B\|_{\mathcal{B}^s_{q,2}}\right) \le Cr^2 MK. \tag{WP15}$$

Terms containing two gauge factors and the physical division by r are bounded by

$$r^{-1}\|\kappa\|_\infty \|\kappa\|_{H^s} \le Cr^{3-4/p} MK \le Cr^2 MK. \tag{WP16}$$

The difference $\kappa(y + re_\mu) - \kappa(y)$ obeys the same two norms. Higher analytic words are summed by applying the exact product-difference identity to one factor at a time: for a difference on a B factor use one gauge factor in L^p, and for a difference on a gauge factor use the other factors in supremum norm. This proves the complete nonlinear gauge remainder estimate without discarding noncommuting terms.

It follows that the actual raw map obeys

$$\|\Pi_r(q^{(k)}_{r,\mathrm{raw}}(A) - Q_r A)\|_{H^s_r} \le Cr^2(M + M^2)K, \tag{WP17}$$

and its raw intermediate charts have uniform $W^{1,p}$ and H^{1+s} bounds. For the former apply two coarse inverse differences to the L^p bound on κ, and one to the L^p bound on the nonlinear remainder. For the latter use the corresponding H^s estimates, exactly as in GQ25.

The stronger CM34 comparison also persists:

$$\|q^{(k)}_{r,\mathrm{raw}}(A) - Q^{(k)}_{r,\mathrm{raw}} A\|_{H^{1+s}_r} \le Cr(M + M^2)K. \tag{WP18}$$

For clarity, the one additional exponent in that proof can be checked explicitly. Intermediate nonlinear straightened defects in H^{1+s} have size $O(\ell_j(M+M^2)K)$. Applying GQ10 to the connector differences therefore sums to $O(r^3(M+M^2)K)$. The difference between the logarithm of the ordered gauge product and its linear sum has size at most $Cr^{4-4/p}MK$ in H^s, which is bounded by Cr^3MK because $p>4$ and $r\le 1$. Two inverse coarse derivatives on these gauge terms cost r^{-2}. The remaining nonlinear coarse gauge remainder costs only one inverse derivative. Together these give (WP18).

50.4 4. The KKT theorem in the finite-integrability chart

Let the genuine raw constraint and physical KKT equation be NK1–3. Assume

$$M_p(u)+M_p(A)\le M, \qquad \|F_a(u)\|_\infty+\|F_a(A)\|_\infty\le M, \tag{WP19}$$

with M sufficiently small, and use $K_u=K_s(u)$, $K_A=K_s(A)$. The fields are on one common logarithmic branch. The curvature supremum in (WP19) is an additional assumption, not a Sobolev embedding consequence.

The exact BCH remainder NK9 has

$$\|\mathcal{R}_a(u,\nabla u)\|_{H^s}\le C(M+M^2)K_u. \tag{WP20}$$

This follows from (WP5) on its quadratic terms, which are finite sums of shifted $u\nabla u$. The terms of field degree at least three form a convergent analytic series in small shifted values of u itself. No bounded supremum of ∇u is used. The bracket part is bounded in H^{1+s} by (WP6), and the factor a on $\mathcal{R}_a$ compensates for the one inverse fine derivative, just as in NK11. The Wilson incidence coefficients have $W^{1,p}$ norm $O(M+M^2)$, so (WP7) supplies their H^s multiplier bounds. The flux correction uses the separately assumed curvature supremum and the same factor a^3 as in NK12–13.

The actual-current comparison uses only the bounded physical potential, bounded curvature, the exact Ward equation, and the all-depth flat H^s tails. These premises follow from (WP3), (WP17), and the actual curvature propagation in PG19–27. The coarse Ward correction is multiplication by coefficients of $W^{1,p}$ norm $O(M)$, so (WP7) also replaces the only remaining $W^{1,\infty}$ multiplier step in NK25.

All Fourier Schur estimates and the harmonic multiplier argument are unchanged. Repeating NK23–34 therefore proves

$$\boxed{\|\lambda\|_{H^s_r}+\|J_a(u)\|_{H^s_a}+\|F_a(u)\|_{H^{1+s}_a}\le C_{s,p,L,T}\left[\|F_a(A)\|_{H^{1+s}_a}+\|A\|_2+(M+M^2)(K_A+K_u)\right].} \tag{WP21}$$

The still-unknown low norm K_u remains explicit. At finite cutoff it is finite; absorbing it requires an additional gauge/constraint estimate. This chapter supplies the norm-compatible KKT implication, not that absorption by assertion.

50.5 5. Transfer through a controlled gauge transformation

Suppose a fine site gauge is $g=e^{\phi}$, with a uniform small $W^{2,p}$ bound on ϕ. Then g, g^{-1} are bounded in $C^{1,\alpha}$. Their multiplication and conjugation operators are uniformly bounded on H^{1+s}, since $s<\alpha$: differentiate the product once and use the same fractional difference estimate as in (WP7). They are also bounded on H^s.

The exact plaquette and Euler-force covariance are

$$F_a(U^g)(x) = \mathrm{Ad}_{g(x)}\, F_a(U)(x), \qquad J_a(U^g)(x) = \mathrm{Ad}_{g(x)}\, J_a(U)(x). \tag{WP22}$$

On the common logarithmic branch the first identity follows by conjugating the plaquette product before taking its logarithm. The second follows by differentiating the gauge-invariant action in physical left-trivialized link coordinates. Thus the curvature and current estimates transfer between such gauges without a fine-mesh loss.

If g is exactly the identity at every final coarse center, covariance of the actual raw averaging chain also gives

$$\mathcal{M}_r(U^g) = \mathcal{M}_r(U). \tag{WP23}$$

This identifies the precise sufficient gauge construction for removing the remaining K_u premise: a uniformly controlled $W^{2,p}$ site gauge, fixed at the final centers, placing the field in the flat smoothed Landau slice. Establishing that construction and the slice estimate are separate arguments. Equations (WP21)–(WP23) show why, if they are established, no change of the prescribed raw coarse representative and no $W^{2,\infty}$ estimate are needed.

51.1 1. The flat projected Landau condition and its exact scalar consequence

Work on a fixed physical periodic torus $\mathbb{T}^4_T$, with nested centered odd lattices $r = na$, $n = L^k$, $r \le 1$. All scalar products use physical volume. Let

$$R = R_{a,r}$$

be normalized site block averaging, R^* its block hold, and $S = S_{a,r}$ restriction to the actual block centers. The latter is not the adjoint of the former. Let

$$G = (-\Delta_a)^{-1}$$

on mean-zero fine site fields. Inverses on coarse sites below also act on their mean-zero subspace.

Define

$$\mathscr{R}_0 = I - GR^*(RG^2R^*)^{-1}RG. \tag{PL1}$$

This is the orthogonal projection onto the complement of $\operatorname{ran}(GR^*)$. Equivalently, it projects onto $\Delta_a \ker R$. Indeed, GR^* is the orthogonal complement of that image; the displayed Gram formula gives its orthogonal projection.

For a fine one-form u, impose the exact flat projected Landau equation

$$\mathscr{R}_0 \delta_a u = 0. \tag{PL2}$$

Decompose the one-form by the flat physical Hodge decomposition:

$$u = u_T + d_a\phi + u_H, \qquad u_T = \Pi_a^0 u, \quad u_H = P_0 u, \quad \phi = G\delta_a u, \quad P_0\phi = 0. \tag{PL3}$$

Here u_T is mean-zero co-closed. Equation (PL2) is equivalent to

$$\delta_a u = GR^*\sigma, \qquad \phi = G^2R^*\sigma \tag{PL4}$$

for a unique mean-zero coarse site field σ. In particular this is a biharmonic scalar lift. An ordinary harmonic lift $GR^*\sigma$ for ϕ would correspond to a different projected gauge.

The coarse scalar datum fixed by the raw gauge covariance is $S\phi$, not $R\phi$. The required normalizer is therefore

$$C_{\mathrm{mix}} = SG^2R^*. \tag{PL5}$$

The following section imports its proved uniform invertibility on nonzero coarse momenta.

51.3 3. The full raw remainder, including its first-order gradient

Let

$$Q_{\mathrm{raw}} = Dq^{(k)}_{r,\mathrm{raw}}(0)$$

be the true linearization of the composed raw average, in physical potentials. The centered-mean chapter CM31–36 proves

$$Q_{\mathrm{raw}} d_a = d_r S, \qquad \|Q_{\mathrm{raw}} w\|_{H^{1+s}_r} \le C_{s,L} \|w\|_{H^{1+s}_a}, \tag{PL19}$$

and

$$\begin{aligned} q^{(k)}_{r,\mathrm{raw}}(w) &= Q_{\mathrm{raw}} w + N_{\mathrm{raw}}(w), \\ \|N_{\mathrm{raw}}(w)\|_{H^s_r} &\le C r^2 (M + M^2) \|w\|_{H^{1+s}_a}, \\ \|N_{\mathrm{raw}}(w)\|_{H^{1+s}_r} &\le C r (M + M^2) \|w\|_{H^{1+s}_a}. \end{aligned} \tag{PL20}$$

The estimates hold on the actual common small chart, without changing the fine field. They subtract the entire true raw linearization, which contains the linear endpoint gradient; subtracting only Q_{bl} would not give (PL20).

The raw derivative preserves the physical mean and constant one-forms:

$$P_0 Q_{\mathrm{raw}} w = P_0 w. \tag{PL21}$$

This follows from $Q_{\mathrm{raw}} = Q_{\mathrm{bl}} - d_r \kappa^{(1)}$, with the periodic gradient having zero mean.

51.4 4. The slice theorem and all harmonic modes

Theorem PL. Fix $0 < s < 1/2$, a fixed physical torus and a fixed odd L. There is $M_*(s, L, T) > 0$, independent of a, r, k, with the following property. Let A, u be finite-lattice physical potentials with

$$\|A\|_{W^{1,\infty}} + \|u\|_{W^{1,\infty}} \le M \le M_*,$$

on the common regular branch. Suppose:

1. the genuine raw constraints agree, $\mathcal{M}_r(e^{au}) = \mathcal{M}_r(e^{aA})$;
2. the physical multiplier and Ward equations (NK2) hold, with their stated constraint-qualification premise;
3. u satisfies the exact identity-background projected Landau equation (PL2).

Then, with

$$\mathcal{I}_s(A) = \|A\|_{H^{1+s}_a} + \|F_a(A)\|_{H^{1+s}_a},$$

one has

$$\boxed{\|u\|_{H^{1+s}_a} + \|\lambda\|_{H^s_r} + \|J_a(u)\|_{H^s_a} + \|F_a(u)\|_{H^{1+s}_a} \le C_{s,L,T} \mathcal{I}_s(A).} \tag{PL22}$$

The estimate assumes no uniform H^{1+s} bound for u in advance. At each finite lattice that norm is finite, so it can be carried through the identities and then absorbed. No unknown high norm enters the choice of M_*.

Proof. Put $K_u = \|u\|_{H^{1+s}_a}$, $K_A = \|A\|_{H^{1+s}_a}$, and $m = M + M^2$. The conditional KKT estimate NK6 is applicable with this finite K_u. It gives

$$\|\lambda\|_{H^s_r} + \|J_a(u)\|_{H^s_a} + \|F_a(u)\|_{H^{1+s}_a} \le C_{s,L,T}\{\mathcal{I}_s(A) + m K_u\}. \tag{PL23}$$

The projected fine Euler equation NK29, inverted on the mean-zero co-closed subspace, also gives

$$\|u_T\|_{H_a^{1+s}} \le C_{s,L,T}\{\mathcal{I}_s(A) + mK_u\}. \tag{PL24}$$

The fixed torus constant here is legitimate and independent of the microscopic mesh.

Let $B = q^{(k)}_{r,\mathrm{raw}}(A) = q^{(k)}_{r,\mathrm{raw}}(u)$. Equation (PL20) and the raw linear bound give

$$\|B\|_{H_r^{1+s}} \le CK_A. \tag{PL25}$$

Use the Hodge decomposition (PL3), actual gradient covariance (PL19), and constant preservation:

$$d_r S\phi = B - Q_{\mathrm{raw}}u_T - u_H - N_{\mathrm{raw}}(u).$$

The left side is a coarse exact gradient. Applying the flat coarse gradient projection removes u_H and has norm one in the Bessel norm. Equations (PL17)–(PL20) therefore imply

$$\|d_a\phi\|_{H_a^{1+s}} \le C_{s,L}\{K_A + \|u_T\|_{H_a^{1+s}} + rmK_u\}. \tag{PL26}$$

The harmonic one-form is not discarded. Take the mean of the equality of raw outputs, use (PL21), and obtain

$$u_H = A_H + P_0N_{\mathrm{raw}}(A) - P_0N_{\mathrm{raw}}(u).$$

Thus

$$\|u_H\|_2 \le \|A_H\|_2 + Cr^2m(K_A + K_u). \tag{PL27}$$

The H^{1+s} norm of this constant one-form equals its L^2 norm. The constant scalar gauge parameter in ϕ was already fixed to zero and is not the same mode as the harmonic one-form u_H.

Add (PL24), (PL26), and (PL27). Since $r \le 1$,

$$K_u \le C_{s,L,T}\mathcal{I}_s(A) + C_{s,L,T}mK_u. \tag{PL28}$$

Choose M_* so that the last coefficient is at most $1/2$. This closes K_u. Substitute the resulting bound into (PL23) to prove (PL22).

The harmonic multiplier λ_H, separate from u_H, was controlled in NK27–28 by the mean physical Euler equation. No inverse Laplacian or mixed normalizer was applied to that mode.

52 A uniform periodic gauge section preserving the actual raw constraint

This chapter constructs a small gauge section on a fixed physical periodic torus. It does not identify that section with Bałaban's original background-dependent Landau representative. The constructed fine gauge equals the identity at every retained site, so it preserves the actual nonlinear raw averaging constraint exactly.

The construction uses the mixed node/box biharmonic lift proved in the flat-Landau chapter. A uniform $W^{2,p}$ bound suffices. No $W^{2,\infty}$ estimate for the gauge transformation is asserted.

52.1 1. Flat operators and the mixed lift

Work on a periodic physical four-torus of side T. Let a be its fine spacing, $r = na \le 1$ its retained spacing, and n an odd integer. Use physical volume factors in all norms and adjoints. Let R_r be normalized averaging over centered scalar site blocks and N_r restriction to their centers. Let $G_a = \Delta_a^{-1}$ act on mean-zero scalars and annihilate constants.

The flat Landau projection R_0 is the orthogonal projection onto $\Delta_a \ker R_r$. Its complement on the mean-zero sector is

$$P_0 = G_a R_r^* (R_r G_a^2 R_r^*)^{-1} R_r G_a, \qquad R_0 = I - P_0. \tag{GS1}$$

On constants set $P_0 = I$, $R_0 = 0$. This is the smoothed projection verified in the flat-Landau chapter, not the sharp block projection.

The mixed biharmonic lift is

$$C_{\mathrm{m}} = N_r G_a^2 R_r^*, \qquad H_{\mathrm{m}} = G_a^2 R_r^* C_{\mathrm{m}}^{-1}, \qquad N_r H_{\mathrm{m}} = I. \tag{GS2}$$

Initially these formulas act on coarse mean-zero fields. Extend H_{m} by mapping a coarse constant to that same fine constant. Equations LF11–18 prove positivity of C_{m} and its uniform fractional Sobolev estimates. The additional offset-kernel argument LF29–36 proves

$$\boxed{\|H_{\mathrm{m}} c\|_{W_a^{2,p}} \le C_{p,T} \|c\|_{W_r^{2,p}}, \qquad 1 \le p \le \infty.} \tag{GS3}$$

The constants are independent of a/r and the blocking depth. To specify the imported input precisely: at a fixed fine offset ξ in a rescaled unit coarse cell, the multiplier is

$$E_\xi(\theta) = \frac{\sum_\nu e^{ip_\nu \cdot \xi} q_\nu \lambda_\nu^{-2}}{\sum_\nu q_\nu \lambda_\nu^{-2}}. \tag{GS4}$$

Multiplication of numerator and denominator by the principal λ_0^2 cancels the principal pole. The denominator is $q_0 + \lambda_0^2 A_0$, uniformly nonzero on the real cube and a fixed complex strip. Nonprincipal

aliases after at most two fine derivatives have the convergent absolute bound $\sum_{m\ge1} m^{j-5}(1+\log m)^3$, $j \le 2$. The actual offsets satisfy $n\xi \in \mathbb{Z}^4$, so coarse periodicity follows from exact alias permutation.

These facts give uniformly exponential coarse kernels for the fine differences through order two. Moreover $E_\xi(\theta) = e^{i\theta\cdot\xi} + O(|\theta|^4)$: constants and affine fields are reproduced. Subtracting an affine coarse polynomial before estimating a second fine difference leaves a discrete Taylor remainder, whose polynomial path count is paid by that exponential kernel. This is the mechanism of (GS3). It does not assert arbitrarily high regularity of a sharp conditional mean.

52.2 2. Uniform restriction to retained nodes

For $2 < p < \infty$,

$$\boxed{\|N_r\psi\|_{W_r^{2,p}} \le C_{p,T}\|\psi\|_{W_a^{2,p}}.} \tag{GS5}$$

This is a point restriction only after two derivatives of regularity. It makes no assertion about point restriction of a general H^{1+s} field.

Here is a local proof. On a patch B of diameter Cr, choose an affine polynomial ℓ_B by the discrete first- and second-order Poincaré conditions. On a smaller patch, the scale-one discrete Sobolev and Poincaré inequalities imply

$$\|\psi - \ell_B\|_\infty \le C_p r^{2-4/p}\|\nabla_a^2\psi\|_{L^p(B)}. \tag{GS6}$$

The constants are uniform in a/r. Here is one direct derivation of the point estimate. On the rescaled unit lattice with spacing $\eta = a/r$, the massive scalar Green kernel of $1+\Delta_\eta$, obtained by integrating the product heat kernel against e^{-t}, obeys

$$|G_{1,\eta}(z)| \le C(|z|+\eta)^{-2}(1+|z|)^{-6}. \tag{GS6a}$$

The elementary heat-kernel estimate used here follows either from its product Fourier formula by fixed-order summation by parts or from the nearest-neighbor walk expansion for times below η^2. The physical $\ell_\eta^{p'}$ norm of (GS6a) is uniform for $p' < 2$: the near-origin radial sum has exponent $3-2p' > -1$, and the single origin cell costs $\eta^{4-2p'}$. Convolution Hölder applied to $v = G_{1,\eta}(1+\Delta_\eta)v$ therefore gives

$$\|v\|_\infty \le C_p\big(\|v\|_p + \|\nabla_\eta^2 v\|_p\big)$$

for compactly supported v. Insert a fixed smooth cutoff supported in a slightly larger unit cube; its exact difference product rule adds only the first- and zeroth-order norms. Choose the affine polynomial with slopes equal to the mean first differences and its constant fixed by the mean value. Two ordinary discrete Poincaré inequalities, proved by coordinate-path telescoping, bound those lower norms by the second differences on the enlarged patch. Scaling back gives (GS6), including the factor $r^{2-4/p}$. This proves the required local point estimate strictly above $p = 4/2$, without using a piecewise multilinear interpolant as a $W^{2,p}$ function.

Every coarse second difference annihilates the affine polynomial, so

$$|D_{r,\mu}D_{r,\nu}N_r\psi(y)| \le C_p r^{-4/p}\|\nabla_a^2\psi\|_{L^p(B(y,Cr))}. \tag{GS7}$$

Raise to the p-th power and sum with physical weights r^4. These cancel r^{-4}, and the enlarged patches overlap a bounded number of times. The zeroth-order point estimate is

$$|\psi(y)| \le C_p r^{-4/p}\big(\|\psi\|_{L^p(B)} + r\|\nabla_a\psi\|_{L^p(B)} + r^2\|\nabla_a^2\psi\|_{L^p(B)}\big). \tag{GS8}$$

It gives the L^p part after summation. The first-difference estimate follows by the same affine subtraction, retaining its linear coefficient and applying the first Poincaré estimate. Since $r \le 1$, these prove (GS5). The smallest compatible periodic torus is handled by the ordinary periodic version, with constants depending only on the fixed T.

52.3 3. A bounded gauge-solving operator

Define on fine one-forms

$$\mathcal{J}_r = (I - H_{\mathrm{m}}N_r)G_a\delta_a, \qquad \delta_a = d_a^*. \tag{GS9}$$

For $2 < p < \infty$,

$$\boxed{\|\mathcal{J}_r w\|_{W_a^{2,p}} \le C_{p,T}\|w\|_{W_a^{1,p}}.} \tag{GS10}$$

Indeed, the periodic scalar elliptic estimate gives

$$\|G_a\delta_a w\|_{W_a^{2,p}} \le C_{p,T}\|\delta_a w\|_{L_a^p} \le C_{p,T}\|w\|_{W_a^{1,p}}. \tag{GS11}$$

For the flat nearest-neighbor Laplacian, its discrete Calderón–Zygmund constant is uniform in the mesh. The second-difference Fourier multipliers have uniform L^2 bounds and the mean-zero kernels have uniform size and difference estimates; the usual singular-integral proof gives every finite $1 < p < \infty$. The low-frequency constant in (GS11) depends only on the fixed physical T. Apply (GS5) and then (GS3) to the second term in (GS9).

The exact algebra is

$$N_r\mathcal{J}_r = 0, \qquad \mathcal{J}_r d_a\phi = \phi \quad (N_r\phi = 0), \qquad \mathcal{J}_r w = 0 \iff R_0\delta_a w = 0. \tag{GS12}$$

For the second identity, $G_a\Delta_a\phi = \phi - \bar{\phi}$, and the extension of H_{m} on constants cancels $\bar{\phi}$ exactly. For the third, put $\psi = G_a\delta_a w$. The equation $(I - H_{\mathrm{m}}N_r)\psi = 0$ means precisely that ψ belongs to the biharmonic range $\mathrm{ran}(G_a^2R_r^*)$. This is equivalent to $\delta_a w \in \mathrm{ran}(G_aR_r^*) = \ker R_0$. Conversely every such ψ is reproduced from its node values by (GS2).

These identities avoid a separate L^p bound for the orthogonal projection R_0. They do not use a discontinuous block hold as a smooth fine gauge.

52.4 4. The actual nonlinear gauge map

Choose a finite $p > 4$. Suppose the original connection has a small physical potential

$$U_\mu(x) = e^{au_\mu(x)}, \qquad \|u\|_{W_a^{1,p}} \le \varepsilon. \tag{GS13}$$

The embeddings $W^{1,p} \subset L^\infty$ and $W^{2,p} \subset C^{1,\alpha}$, for every fixed $\alpha < 1 - 4/p$, have uniform fine-mesh constants on the fixed torus. Set

$$X_r^p = \{\phi : N_r\phi = 0\},$$

with its $W_a^{2,p}$ norm. For $\phi \in X_r^p$, define

$$\mathcal{U}_a(u,\phi)_\mu(x) = a^{-1}\log\left(e^{\phi(x)}e^{au_\mu(x)}e^{-\phi(x+ae_\mu)}\right). \tag{GS14}$$

This is the actual gauge-transformed potential. On one common small product ball, it is holomorphic and has the form

$$\mathcal{U}_a(u,\phi) = u - d_a\phi + \mathcal{N}_a(u,\phi). \tag{GS15}$$

Its remainder satisfies

$$\|\mathcal{N}_a(u,\phi)\|_{W_a^{1,p}} \le C_{p,T}(\|u\|_{W_a^{1,p}} + \|\phi\|_{W_a^{2,p}})\|\phi\|_{W_a^{2,p}}, \tag{GS16}$$

and the corresponding Lipschitz bound in ϕ, with the same small coefficient on a small ball.

To check the apparent inverse mesh, first write $\phi(x + ae_\mu) - \phi(x) = aD_{a,\mu}\phi(x)$. The pure gauge factor in (GS14), after division by a, is an analytic bounded coefficient times $D_a\phi$. The other factor conjugates u. Once the linear terms are removed, each monomial has at least two factors and contains a gauge factor. Terms containing two physical first-order quantities but no undifferentiated ϕ carry an explicit a. Thus no bare a^{-1} remains. The exact lattice difference product rule, the Banach algebra property of $W^{1,p}$, and $W^{2,p} \subset W^{1,\infty}$ prove (GS16) and its derivative estimate on a fixed smaller ball. All finite shifts and matrix orders are retained.

52.5 5. A gauge section fixing the raw target exactly

The desired slice is

$$R_0\delta_a\widetilde{u} = 0, \qquad \widetilde{u} = \mathcal{U}_a(u,\phi). \tag{GS17}$$

By (GS12), (GS15), this is equivalent to

$$\phi = \mathcal{J}_r(u + \mathcal{N}_a(u,\phi)). \tag{GS18}$$

Its right side lies in X_r^p. Equations (GS10), (GS16) give a contraction on $\|\phi\|_{W^{2,p}} \le C_{p,T}\varepsilon$, after one fixed small choice of ε. This choice depends on p, T, not on a, r, n. The unique solution satisfies

$$\boxed{\|\phi\|_{W_a^{2,p}} \le C_{p,T}\|u\|_{W_a^{1,p}}, \qquad \|\widetilde{u}\|_{W_a^{1,p}} \le C_{p,T}\|u\|_{W_a^{1,p}}, \qquad R_0\delta_a\widetilde{u} = 0.} \tag{GS19}$$

It is real on the real compact-group chart, by real iteration or uniqueness, and holomorphic on the common complexified small ball.

Since $N_r\phi = 0$, $e^{\phi(y)} = 1$ at every retained site. Exact covariance of the actual raw iterated average gives

$$M^k(U^{e^\phi}) = M^k(U). \tag{GS20}$$

The action, physical critical orbit, and target representative of the raw constraint are unchanged. This uses actual retained node values, not block site means. It is stronger than an orbit-only comparison.

At each finite cutoff all lattice norms are finite. A subsequent estimate may therefore absorb the constructed field's H^{1+s} norm with a uniform small coefficient before establishing its uniform bound. This does not assert that $W^{1,p}$ alone implies H^{1+s}.

52.6 6. Curvature and current in the original gauge

For a plaquette based at x, the principal logarithmic curvature transforms exactly:

$$F_a(\widetilde{u})(x) = \mathrm{Ad}_{e^{\phi(x)}} F_a(u)(x). \tag{GS21}$$

The geometric current has the same covariance at its base site. For real fields the curvature sup norm is unchanged. For complex fields, the common small $W^{2,p}$ ball bounds the adjoint norm.

Choose $0 < s < 1 - 4/p$, and then $s < \alpha < 1 - 4/p$. The gauge and its inverse have uniform $C^{1,\alpha}$ norms. Multiplication by such a function is bounded on H^{1+s}: differentiate once and use the fractional translation-difference product bound. Hence

$$\|F_a(u)\|_{H_a^{1+s}} \le C_{p,s,T}\|F_a(\widetilde{u})\|_{H_a^{1+s}}, \qquad \|J_a(u)\|_{H_a^s} \le C_{p,s,T}\|J_a(\widetilde{u})\|_{H_a^s}. \tag{GS22}$$

The reverse bounds hold as well. With $p > 8$, every fixed $s < 1/2$ is allowed.

The input defining the target should remain unchanged. If $M^k(U) = M^k(U^A)$, (GS20) gives $M^k(U^{e^\phi}) = M^k(U^A)$ with the original A. Gauging A is unnecessary and could introduce an uncontrolled inhomogeneous gauge term into its high potential norm. No H^{2+s} claim about ϕ is used in the curvature transfer.

52.7 7. Use conditions and limits

This is a periodic small-potential theorem. Its combination with the reconstructed slice requires the $W^{1,p}$ versions of coarsening, the full raw remainder, and the KKT estimates. Those are separate estimates in the companion weak-chart and projected-Landau chapters. The curvature sup bound used in the actual-current comparison is retained from the original physical representative; it is not inferred from $W^{1,p}$ in four dimensions.

The theorem reaches the flat projected slice without an unverified identification of the original non-Abelian projector $R(U_0)$. It does not identify an independently localized auxiliary background with a genuine constrained stationary point. Mixed determining domains, boundary gauges, and original wall interpolations still require their actual collars and a matching local construction or extension. A global periodic Fourier theorem is not that boundary theorem.

The analytic statement applies to physical families already on the common small $W^{1,p}$ chart. It does not turn arbitrary independent-current pairs into such families or replace original domain matching.

In particular, separate small $W^{1,\infty}$ gauges on blocks measured in their own retained-scale units do not by themselves imply (GS13) on one fixed physical torus or buffer. Rescaling changes the physical

derivative norms, and compatibility of the local transition gauges is another requirement. To apply this chapter uniformly to original inner readouts, one must establish a single admissible physical $W^{1,p}$ chart for the whole readout family, with a bound independent of its earlier retained scale. Removing the unknown output's H^{1+s} norm from the reconstruction estimate does not prove this low-chart hypothesis.

53.1 1. Statement

Fix a compact matrix gauge group, a physical four-torus of side T, a fixed odd blocking factor L, and

$$0 < s < \tfrac{1}{2}, \qquad 8 < p < \infty. \tag{PR1}$$

Let a be the microscopic spacing and $r = L^k a \le 1$ a retained spacing. Use physical volume pairings. Let $\mathcal{M}_r$ be the genuine composition of the centered raw reference-log averages in the preceding chapters, including the admitted endpoint contour means. For fine physical potentials A, u, put

$$U_\mu^A(x) = e^{aA_\mu(x)}, \qquad U_\mu(x) = e^{au_\mu(x)}, \qquad F_a(u) = a^{-2}\log P(U). \tag{PR2}$$

All logarithms are on the common small branch. Let $J_a(u)$ be the geometric Wilson Euler force with the fixed conventions of NK3; the overall scalar coupling is divided out of its multiplier.

Theorem PR. There are $\varepsilon_* > 0$ and $C < \infty$, depending on s, p, L, T and the fixed group, but independent of a, r, k, with the following property. Suppose

$$\begin{aligned}
&\|A\|_{W_a^{1,p}} + \|u\|_{W_a^{1,p}} + \|F_a(A)\|_\infty + \|F_a(u)\|_\infty \le \varepsilon_*,\\
&\mathcal{M}_r(U) = \mathcal{M}_r(U^A) = V,\\
&J_a(u) = (D\mathcal{M}_r(U))^{\mathsf{t}}\lambda, \qquad d_V^{\mathsf{t}}\lambda = 0.
\end{aligned} \tag{PR3}$$

The multiplier equation is an explicit hypothesis, including its constraint qualification when inferred from stationarity. The transpose uses the invariant bilinear color pairing. For real fields this is the physical adjoint; complex bounds use an absolute Hermitian norm after the algebraic identities.

Then

$$\boxed{\|\lambda\|_{H_r^s} + \|J_a(u)\|_{H_a^s} + \|F_a(u)\|_{H_a^{1+s}} \le C\left(\|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}}\right).} \tag{PR4}$$

There is also a gauge $g = e^\phi$, equal to the identity at every retained center, such that the transformed physical potential $\widetilde{u}$ obeys

$$\begin{aligned}
&\|\phi\|_{W_a^{2,p}} + \|\widetilde{u}\|_{W_a^{1,p}} \le C\|u\|_{W_a^{1,p}},\\
&R_0\delta_a\widetilde{u} = 0, \qquad \mathcal{M}_r(U^g) = V,\\
&\|\widetilde{u}\|_{H_a^{1+s}} \le C\left(\|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}}\right).
\end{aligned} \tag{PR5}$$

Here R_0 is the precisely defined flat smoothed Landau projection in GS1, with scalar constants treated there. In particular, the theorem does not assume $R_0\delta_a u = 0$ in the original gauge.

A uniformly small original $W^{1,\infty}$ chart is a sufficient special case of the low potential and curvature conditions in (PR3), on the fixed physical volume. The more general $W^{1,p}$ statement retains bounded curvature separately. No Sobolev embedding from $W^{1,p}$ to bounded curvature is asserted.

53.2 2. Proof with the original input and target fixed

Apply the gauge-section theorem GS19 to u. It constructs a holomorphic $W^{2,p}$ gauge parameter ϕ with zero values at the retained centers, and a transformed field $\widetilde{u}$ in the exact flat slice. Its small $W^{1,p}$ norm follows from that same theorem. Gauge covariance and the uniform adjoint norm give

$$\|F_a(\widetilde{u})\|_\infty \le C\|F_a(u)\|_\infty. \tag{PR6}$$

For real compact gauges the constant in (PR6) is one. Choose the initial radius ε_* below the fixed gauge-section and slice radii, including these finitely many comparison constants.

Because g is exactly the identity at every retained node, the nonlinear raw constraint is unchanged:

$$\mathcal{M}_r(U^g) = \mathcal{M}_r(U^A). \tag{PR7}$$

The input A has not been transformed. This matters: a $W^{2,p}$ gauge does not automatically give an H^{2+s} bound for its parameter, so transforming A would require an additional estimate for the inhomogeneous gauge term of its potential. The exact nodal condition makes that operation unnecessary.

The physical KKT equation transforms with the same coarse multiplier. To check it without differentiating the selected gauge as a function of u, fix its value g at this field. If $\mathcal{A}_g$ denotes adjoint transport of a fine physical tangent, exact raw covariance gives

$$D\mathcal{M}_r(U^g)\mathcal{A}_g = D\mathcal{M}_r(U). \tag{PR8}$$

The invariant bilinear pairing has $\mathcal{A}_g^{\mathsf{t}} = \mathcal{A}_g^{-1}$. Taking transposes in (PR8), and using covariance of the action derivative, yields

$$(D\mathcal{M}_r(U^g))^{\mathsf{t}}\lambda = \mathcal{A}_g J_a(u) = J_a(\widetilde{u}). \tag{PR9}$$

The coarse field is the same V, so its Ward equation is also unchanged. Thus every hypothesis of the finite-p slice theorem PL29–33 holds for $(\widetilde{u}, A, \lambda)$.

At each finite lattice, $K_{\widetilde{u}} = \|\widetilde{u}\|_{H^{1+s}}$ is finite. The slice argument first proves

$$K_{\widetilde{u}} \le C\left(\|A\|_{H^{1+s}} + \|F_a(A)\|_{H^{1+s}}\right) + C(M + M^2)K_{\widetilde{u}}. \tag{PR10}$$

The constants are independent of the finite dimension and retained spacing. The chosen small radius makes the last coefficient at most $1/2$. Absorbing it gives the last line of (PR5), and PL30 supplies the multiplier, transformed current and transformed curvature estimates.

Finally choose $s < \alpha < 1 - 4/p$. The constructed g, g^{-1} have uniformly bounded $C^{1,\alpha}$ norms. Equation GS22 transfers the H^{1+s} curvature and H^s current bounds back to the original field. The multiplier itself was unchanged. This proves (PR4) and completes the proof.

The scalar constant in the gauge parameter is fixed by its retained node values. Harmonic one-forms are controlled by the mean of the exact raw constraint in PL27/PL32. Harmonic multipliers are controlled by the mean Euler equation in NK27–28. None of these modes is inverted by a singular zero-momentum Green function.

53.3 3. Common complex parameters

Suppose that, for every cutoff, $u(z), A(z), \lambda(z)$ are holomorphic physical KKT families on the same polydisc $|z_i| < \rho_i$, and (PR3) holds uniformly there. Assume also

$$\sup_z \left(\|A(z)\|_{H^{1+s}} + \|F_a(A(z))\|_{H^{1+s}}\right) \le K. \tag{PR11}$$

The gauge fixed point is holomorphic on the same smaller uniform weak chart. The proof above supplies strong bounds there without an initial strong bound for $u(z)$. Consequently Cauchy's formula gives, for example,

$$\|\partial_z^\gamma F_a(u)(0)\|_{H^{1+s}} + \|\partial_z^\gamma J_a(u)(0)\|_{H^s} \le CK\,\gamma! \prod_i \rho_i^{-\gamma_i}. \tag{PR12}$$

Use smaller radii and take their limit if the input supremum is stated only on the open polydisc. This is a uniform strong-norm consequence for actual physical families. It does not assert that a weak independent pair (U, J), or an independently wall-modified background, belongs to such a family.

53.4 4. Consequence for the integrated dimension-six remainder

For actual reconstructed fields satisfying (PR3), the output curvature now has the fractional norm required by the integrated quadratic Taylor argument D6.10–13. In any common original chart where that covariant translation estimate applies, its quadratic remainder is bounded by

$$C|v|^{2+s}\|f\|_{C^{2+s}}\left(\|A\|_{H^{1+s}}+\|F_a(A)\|_{H^{1+s}}\right)^2. \tag{PR13}$$

The dimension-six rescaling removes two powers of the retained displacement and leaves the strict factor $|v|^s$. This conclusion concerns the completed integrated quadratic expression with the exact descendants and commutator terms identified in that chapter. It does not turn a pointwise second derivative of curvature into a bounded function, and it does not normalize the whole dimension-six mixing matrix.

54.1 1. Statement with both Euler and constraint errors

Use the physical periodic lattices, actual raw averaging map, and conventions of PR. Fix

$$0 < s < \tfrac{1}{2}, \qquad 8 < p < \infty, \qquad r = L^k a \le 1. \tag{RS1}$$

Write the actual coarse physical logarithm as

$$q_r(u) = r^{-1}\log \mathcal{M}_r(e^{au}), \qquad V_u = \mathcal{M}_r(e^{au}).$$

The common logarithm branch is part of the small chart. Let $\Pi_r^0, P_{L,r}, P_{H,r}$ be, respectively, the flat mean-zero co-closed, mean-zero gradient, and harmonic projections on coarse one-forms. Define the constraint-error norm

$$\mathfrak{B}_{s,r}(b) = \|\Pi_r^0 b\|_{H_r^{2+s}} + \|P_{L,r}b\|_{H_r^{1+s}} + \|P_{H,r}b\|_2. \tag{RS2}$$

It keeps the different roles of the constraint components visible. In particular,

$$\mathfrak{B}_{s,r}(b) \le C\|b\|_{H_r^{2+s}} \le C_s r^{-2}\|b\|_{H_r^s}. \tag{RS3}$$

Theorem RS. There are a small radius and a constant, with the same parameter dependence and cutoff independence as PR, such that the following holds. Assume

$$\|u\|_{W_a^{1,p}} + \|A\|_{W_a^{1,p}} + \|F_a(u)\|_\infty + \|F_a(A)\|_\infty \le \varepsilon_*, \tag{RS4}$$

and suppose the actual equations are

$$\begin{aligned} q_r(u) - q_r(A) &= b, \\ J_a(u) &= (D\mathcal{M}_r(e^{au}))^{\mathsf{t}}\lambda + E, \\ d^{\mathsf{t}}_{V_u}\lambda &= 0. \end{aligned} \tag{RS5}$$

The last equation is exact. The Euler residual E is a fine physical one-form, with the same tangent convention as J_a. Set

$$\mathcal{D}_s(A,E,b) = \|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}} + \|E\|_{H_a^s} + \mathfrak{B}_{s,r}(b). \tag{RS6}$$

Then

$$\boxed{\|\lambda\|_{H_r^s} + \|J_a(u)\|_{H_a^s} + \|F_a(u)\|_{H_a^{1+s}} \le C\mathcal{D}_s(A,E,b).} \tag{RS7}$$

The gauge of GS, equal to the identity at every retained center, also gives a representative $\widetilde{u}$ in the exact projected Landau slice satisfying

$$\|\widetilde{u}\|_{H_a^{1+s}} \le C\mathcal{D}_s(A,E,b). \tag{RS8}$$

The Wilson gauge identity and (RS5) imply $d_U^{\mathsf{t}}E = 0$, with the exact fine covariant differential. Thus the exact-Ward version concerns a physical, gauge-compatible residual. It is not a theorem for an arbitrary separately prescribed Ward defect. The latter needs the additional endpoint analysis considered separately.

54.2 2. Proof of the residual multiplier estimate

First suppose u is already in the slice. Put $K_u = \|u\|_{H_a^{1+s}}$, $K_A = \|A\|_{H_a^{1+s}}$, and $m = M + M^2$, where M denotes the small weak radius. At a finite cutoff K_u is finite; no uniform estimate for it is assumed.

The actual raw-remainder estimate, applied to u and A separately, gives

$$\Pi_r^0 Q_r(u - A) = \Pi_r^0 b + e_c, \qquad \|e_c\|_{H_r^s} \le Cr^2 m(K_u + K_A). \tag{RS9}$$

Here Q_r is the flat box-and-line map used in NK. No equality of the two coarse targets was used in this subtraction.

The exact Ward equation on λ permits the complete-current estimate from NK/WP:

$$(D\mathcal{M}_r(e^{au}))^{\mathsf{t}}\lambda = Q_r^*\lambda + e_Q, \qquad \|e_Q\|_{H_a^s} \le Cm\|\lambda\|_{H_r^s}. \tag{RS10}$$

The identity for the geometric Wilson force is unchanged. Hence

$$\delta_a d_a u = Q_r^*\lambda + e_Q + E + f_a(u), \qquad \|f_a(u)\|_{H_a^s} \le CmK_u. \tag{RS11}$$

The longitudinal multiplier bound still follows from the actual coarse Ward equation at V_u:

$$\|\lambda_L\|_{H_r^s} \le C_T m\|\lambda\|_{H_r^s}. \tag{RS12}$$

Its harmonic component is now fixed by

$$\lambda_H = -P_0(e_Q + E + f_a(u)). \tag{RS13}$$

In particular E contributes its actual mean; no singular zero-mode inverse is introduced.

Apply the same transverse Schur inverse S_r as in NK18 to (RS9) and (RS11). The additional terms are exactly

$$S_r\Pi_r^0 b, \qquad -S_r\Pi_r^0 Q_r L_a^{-1}\Pi_a^0 E. \tag{RS14}$$

The second is bounded in H_r^s by $C\|E\|_{H_a^s}$, by NK20. The fibre bound $\|S_r(\theta)\| \le C\lambda_0(\theta)$ also gives

$$\|S_r\Pi_r^0 b\|_{H_r^s} \le C\|\Pi_r^0 b\|_{H_r^{2+s}}. \tag{RS15}$$

All nonlinear constraint terms retain their factor r^2, which cancels the possible r^{-2} of S_r. The argument of NK23–33, with (RS12)–(RS15), therefore yields

$$\|\lambda\|_{H_r^s} + \|J_a(u)\|_{H_a^s} + \|F_a(u)\|_{H_a^{1+s}} + \|u_T\|_{H_a^{1+s}} \le C\{\mathcal{D}_s(A,E,b) + mK_u\}. \tag{RS16}$$

The occurrence of the actual E in J_a is included in this estimate. Coefficients multiplying λ have been absorbed using only the weak radius.

54.3 3. Slice closure and transfer to the original representative

The exact raw linearization obeys

$$Q_{\mathrm{raw}} d_a\phi = d_r N_r\phi$$

and preserves the physical mean. In the projected Landau slice, $\phi = E_{\mathrm{mix}}N_r\phi$, and the mixed lift controls $d_a\phi$ by $d_rN_r\phi$ in H^{1+s}. Apply the coarse gradient projection to

$$Q_{\mathrm{raw}}(u_T + d_a\phi + u_H) = q_r(A) + b - N_{\mathrm{raw}}(u). \tag{RS17}$$

The new contribution to the longitudinal potential is exactly $P_{L,r}b$, so

$$\|d_a\phi\|_{H_a^{1+s}} \le C\{K_A + \|u_T\|_{H_a^{1+s}} + \|P_{L,r}b\|_{H_r^{1+s}} + rmK_u\}. \tag{RS18}$$

The mean of the same raw constraint gives

$$u_H - A_H = P_0 b + P_0\{N_{\mathrm{raw}}(A) - N_{\mathrm{raw}}(u)\}. \tag{RS19}$$

The full strong raw-remainder bound therefore controls the harmonic potential by $C\{K_A + \|P_{H,r}b\|_2 + rmK_u\}$. Combining (RS16), (RS18) and (RS19) gives

$$K_u \le C\mathcal{D}_s(A,E,b) + CmK_u. \tag{RS20}$$

The weak radius is chosen so that the final coefficient is at most $1/2$. This proves (RS7)–(RS8) in the slice.

For a general admitted representative, use the exact retained-center gauge of GS and hold its value fixed when differentiating the raw map. The input A, both coarse targets, b, and λ are unchanged. The transformed Euler residual is

$$\widetilde{E} = \mathcal{A}_g E. \tag{RS21}$$

Uniform $C^{1,\alpha}$ control of g, g^{-1}, with $s < \alpha < 1 - 4/p$, gives $\|\widetilde{E}\|_{H^s_a} \le C\|E\|_{H^s_a}$. The argument of PR8–10 then applies to the transformed field. The same controlled gauge transfers curvature and current back. This proves the theorem.

As in PR, a family satisfying these hypotheses on one common complex polydisc inherits Cauchy derivative bounds from its uniform strong estimate. The theorem does not itself construct that polydisc for an auxiliary formation map.

54.4 4. A concrete summable old-layer error

The flat estimate LR45 is

$$\|dP_\ell \delta\|_{L^2 \to H^s} \le C_s \ell^{-2-s}, \qquad 0 \le s \le 1. \tag{RS22}$$

Suppose, as a specified model for one component of an actual residual, that

$$E_{\mathrm{proj}} = \sum_{j=0}^{J} dP_{\ell_j} \delta f_j, \qquad \ell_j = L^{-j},$$

with a fixed physical separation $R > 0$ and

$$\|f_j\|_2 \le B e^{-cR/\ell_j}. \tag{RS23}$$

The scale ℓ_j is the old layer's physical spacing. It is not replaced by the current spacing. With $m = 2 + s$, the triangle inequality and (RS22) give

$$\|E_{\mathrm{proj}}\|_{H^s} \le C_s B \sum_{j \ge 0} L^{jm} e^{-cRL^j} \le C_{s,L,c} B \begin{cases} R^{-m}, & 0 < R \le 1, \\ e^{-c'R}, & R \ge 1. \end{cases} \tag{RS24}$$

The last bound is independent of the terminal layer and of the fine cutoff. For $R \le 1$, split at the last index with $RL^j \le 1$. The first part is a geometric sum bounded by CR^{-m}; in the remainder factor out R^{-m} and sum $(RL^j)^m e^{-cRL^j}$ on a geometric sequence starting in a fixed compact interval away from zero. For $R \ge 1$, keep half the exponential as $e^{-cR/2}$ and sum the remainder using the same geometric estimate. Any fixed $0 < c' < c/2$ is adequate after changing the constant.

This is an actual cancellation of the old-layer polynomial price by physical separation. A fixed old-layer buffer, in contrast, supplies only e^{-cM}; it leaves $\sum_j L^{j(2+s)} e^{-cM}$, which is not uniformly bounded. The distinction is quantitative and cannot be removed by naming the same buffer in different units.

There is a second qualification. The isolated term $dP_\ell \delta f_j$ need not be covariantly divergence-free. Equation (RS24) estimates a norm of one component. To use Theorem RS, all components of the actual residual must first be assembled so that its exact Ward hypothesis holds. Alternatively, an additional theorem must control the non-Ward endpoint lift. The scalar companions of a projected propagator therefore cannot be dropped just because (RS24) is summable.

55.1 1. The exact flat connector has order minus one

Let $h_j = L^j a$, $0 \le j \le k$, with $h_k = r$, and use base-site coordinates for bond potentials in this section. Write

$$D_{j,\mu}(p) = h_j^{-1}(e^{ih_j p_\mu} - 1), \qquad P(p) = \max_\nu |D_{0,\nu}(p)|. \tag{CR1}$$

All formulas have removable continuous values at zero factors. Let $R_j(p)$ be the centered scalar site-box filter from spacing a to spacing h_j. The whole straight box-and-line filter for component μ is exactly

$$q_{j,\mu}(p) = R_j(p)\frac{D_{j,\mu}(p)}{D_{0,\mu}(p)}. \tag{CR2}$$

This is the Fourier form of the gradient chain identity, with the literal normalized line filter. It includes its phase in base-site coordinates.

At the j-th step define the centered one-dimensional L-box filters

$$b_{j,\nu}(p) = \frac{1}{L}\sum_{m=-(L-1)/2}^{(L-1)/2} e^{imh_j p_\nu}. \tag{CR3}$$

The endpoint connector averages integrals from the block center to a uniformly sampled block point, over the coordinate-permutation paths in the actual map. Its component symbol is

$$\rho_{j,\mu}(p) = \frac{b_{j,\mu}(p) - 1}{D_{j,\mu}(p)}\Phi_{j,\mu}(p), \qquad \Phi_{j,\mu}(p) = \frac{1}{4!}\sum_\pi \prod_{\nu \prec_\pi \mu} b_{j,\nu}(p). \tag{CR4}$$

To verify (CR4), a one-dimensional path ending at mh_j contributes

$$h_j \sum_{\text{oriented path}} e^{ith_j p_\mu} = \frac{e^{imh_j p_\mu} - 1}{D_{j,\mu}(p)}.$$

The orientation includes negative endpoints. Average over the centered endpoints. Each coordinate visited before μ contributes its independent endpoint phase, whose average is $b_{j,\nu}$. Coordinates visited afterward contribute one. The derivative of an admitted inner group mean at the flat diagonal is the arithmetic mean, so the same formula applies to that version of the endpoint paths. In particular $|\Phi_{j,\mu}| \le 1$ on real momenta.

Let $K_{a,r}$ be the full accumulated flat connector:

$$Q_{\text{raw}}^{(k)} = Q_r - d_r K_{a,r}.$$

Linearizing the exact accumulated coarse gauge gives

$$K_{a,r,\mu}(p) = \sum_{j<k} \rho_{j,\mu}(p) q_{j,\mu}(p) = \frac{1}{D_{0,\mu}(p)} \sum_{j<k} R_j(p)(b_{j,\mu}(p) - 1)\Phi_{j,\mu}(p). \tag{CR5}$$

Restriction of each intermediate node factor to a final node introduces no additional Fourier amplitude. The intermediate gradient factor in (CR2) cancels the denominator in (CR4) exactly. At $D_{0,\mu} = 0$, the component in (CR5) is zero by centered symmetry, using its continuous value.

For completeness, the prior filter in (CR5) is the complete flat box-and-line map, not an assumed raw filter. At one step write $Q_{\text{raw},j} = Q_{\text{bl},j} - d_{j+1}\rho_j$. The two chain identities $Q_{\text{raw},j} d_j = d_{j+1} N_j$ and $Q_{\text{bl},j} d_j = d_{j+1} R_{\text{step},j}$ give $\rho_j d_j = R_{\text{step},j} - N_j$, directly also by telescoping endpoint paths. If $Q_{\text{raw}}^{(j)} = Q_{\text{bl}}^{(j)} - d_j K_j$, then exact multiplication gives

$$K_{j+1} = N_j K_j + \rho_j Q_{\text{bl}}^{(j)}, \qquad K_0 = 0.$$

Its iteration is precisely (CR5), with the intermediate N's evaluated at the final retained centers.

Put $q = |D_{0,\mu}(p)|$, with $0 < q \le P$. The centered symmetry gives a second-order zero:

$$|b_{j,\mu}(p) - 1| \le C_L \min\{1, (h_j q)^2\}. \tag{CR6}$$

For example, pair the endpoints to express this difference as a sum of $\sin^2(mh_j p_\mu/2)$, and use $|\sin(nx)| \le n|\sin x|$. The box factor in a coordinate attaining P gives

$$|R_j(p)| \le C \min\{1, (h_j P)^{-1}\}. \tag{CR7}$$

Both estimates hold for unreduced fine momentum, including aliases at intermediate scales.

Therefore each term in (CR5) is bounded by

$$\frac{C_L}{q} \min\{1, (h_j q)^2\} \min\{1, (h_j P)^{-1}\}. \tag{CR8}$$

There is no logarithmic loss. To see this, set $x_j = h_j P$ and $t = q/P$. In the three ranges $x_j \le 1$, $1 < x_j \le t^{-1}$, and $x_j > t^{-1}$, the bounds are respectively

$$\frac{q}{P^2} x_j^2, \qquad \frac{q}{P^2} x_j, \qquad \frac{1}{q x_j}.$$

Each geometric sum is at most C_L/P. The middle sum uses its upper endpoint t^{-1}, while the last sum uses its lower endpoint. Missing ranges only improve the estimate. Hence

$$\boxed{|K_{a,r,\mu}(p)| \le \frac{C_L}{P(p)}, \qquad |K_{a,r}(p)| \le \frac{C_L}{|d_a(p)|} \quad (p \neq 0).} \tag{CR9}$$

The constant is independent of a, r, k. The own-coordinate second-order zero in (CR6) is essential. A bound with only one transverse box factor and $\rho_{j,\mu} = O(h_j)$ would leave a constant contribution at each of many scales.

55.2 2. Flat divergence compatibility controls the endpoint lift

Let $N = N_{a,r}$ restrict a fine scalar to retained centers, and let its adjoint use physical volume. On every coarse Fourier fiber, N^*q has the same coefficient $\widehat{q}(\theta)$ in every fine alias. The raw gauge identity is

$$Q^{(k)}_{\mathrm{raw}} d_a = d_r N. \tag{CR10}$$

Suppose the flat residual equation is

$$\delta_a d_a u = (Q^{(k)}_{\mathrm{raw}})^* \lambda + E. \tag{CR11}$$

Taking the flat divergence gives the necessary compatibility

$$\delta_a E = -N^* q, \qquad q = \delta_r \lambda. \tag{CR12}$$

In particular q has zero coarse mean. At a nonzero fine alias p_ν,

$$|\widehat{q}(\theta)|^2 \le |d_a(p_\nu)|^2 |\widehat{E}(p_\nu)|^2.$$

The minimal gradient lift of the divergence is $-d_a \Delta_a^{-1} N^* q$, and the orthogonal flat gradient projection commutes with every Bessel weight. Thus

$$\|d_a \Delta_a^{-1} N^* q\|_{H^s_a} \le \|E\|_{H^s_a}. \tag{CR13}$$

Together with (CR9), this proves the exact endpoint estimate

$$\boxed{\|K^*_{a,r} q\|_{H^s_a} \le C_L \|E\|_{H^s_a}.} \tag{CR14}$$

The symbol comparison itself works for every fixed real s. The reconstruction applications below use $0 < s < 1/2$, which is the range of the sharp flat prolongation estimate.

The principal fine alias in (CR12) also gives

$$\|d_r \Delta_r^{-1} q\|_{H^s_r} \le C_{s,T} \|E\|_{H^s_a}. \tag{CR15}$$

One may either compare the principal gradient weights directly, or first compare the coarse H^{s-1} norm of q to the fine divergence norm and then apply the coarse inverse. Scalar constants are removed by the flat inverse; in this flat case q already has zero mean.

These estimates show why a direct long, thin contour is not an appropriate counterexample to the actual fixed-L hierarchy. The full connector is a sum of paths whose lengths and smoothing scales are comparable at each step. Formula (CR5) retains that structure.

55.3 3. Atomic compatibility in a nonlinear physical chart

There is a second proof mechanism that extends to the nonlinear map. It uses an explicit atomic norm and the exact full-space one-step endpoint identity.

Work on the periodic weak chart of WP, with $p > 8$, $0 < s < 1/2$, and uniformly small potential $W^{1,p}$ and bounded curvature. Let the genuine geometric Wilson force satisfy

$$J_a(u) = (D\mathcal{M}_r(U))^{\mathsf{t}}\lambda + E, \qquad V = \mathcal{M}_r(U), \qquad q = d_V^{\mathsf{t}}\lambda. \tag{CR16}$$

No separate coarse Ward condition is imposed. Gauge invariance of the actual Wilson action gives $d_U^{\mathsf{t}} J_a(u) = 0$. Exact raw covariance therefore implies

$$d_U^{\mathsf{t}} E = -N^{\mathsf{t}} q. \tag{CR17}$$

The superscript denotes the invariant bilinear transpose. The scalar restriction matrix is real and independent of the field, so its transpose is the physical point embedding used above.

For $a \le r \le 1$, the physical Fourier normalization gives

$$\|N^* q\|^2_{H_a^{s-1}} = \sum_{\theta} |\widehat{q}(\theta)|^2 \sum_{\nu \text{ in fiber}} (1 + \widehat{\Delta}_a(p_\nu))^{s-1}. \tag{CR18}$$

There are exactly $(r/a)^4$ aliases. Since $s < 1$ and $\widehat{\Delta}_a(p_\nu) \le Ca^{-2}$, every weight in the inner sum is at least $c_s a^{2-2s}$. Hence

$$\boxed{r^2 a^{-1-s}\|q\|_{L^2_r} \le C_s \|N^* q\|_{H_a^{s-1}}.} \tag{CR19}$$

This includes a coarse constant q and the case $r = a$. No inverse Laplacian has been used.

The difference $d_U^{\mathsf{t}} - \delta_a$ is a zero-order shifted multiplier with uniformly $W^{1,p}$ coefficients on the stated chart. WP7 therefore gives

$$\|d_U^{\mathsf{t}} E\|_{H_a^{s-1}} \le C(1 + M)\|E\|_{H_a^s}.$$

Equations (CR17)–(CR19) prove

$$r^2 a^{-1-s}\|q\|_2 \le C\|E\|_{H_a^s}. \tag{CR20}$$

The principal alias in (CR18), together with comparability of the principal fine and coarse symbols, gives the additional bound

$$\|q\|_{H_r^{s-1}} \le C\|E\|_{H_a^s}. \tag{CR21}$$

Unlike the flat case, a covariant divergence can have a nonzero ordinary mean. Equations (CR19)–(CR21) retain that mode. A coarse flat inverse below always removes its scalar mean before inversion.

55.4 4. Full-space current propagation with a Ward defect

Let $U_{j+1} = M_j(U_j)$ be the actual raw hierarchy, with spacings h_j and $\ell_j = h_{j+1} = Lh_j$. Let

$$\lambda_k = \lambda, \qquad \lambda_j = (DM_j(U_j))^{\mathsf{t}}\lambda_{j+1}, \qquad q_j = d_{U_j}^{\mathsf{t}}\lambda_j. \tag{CR22}$$

Raw covariance gives the exact defect propagation

$$q_j = N_{j,r}^* q, \qquad \|q_j\|_2 = (r/h_j)^2\|q\|_2. \tag{CR23}$$

The second equality is the dimension-four physical norm of a point embedding: its nonzero site amplitudes are multiplied by $(r/h_j)^4$.

The full endpoint formula PG15d–e, before any Ward restriction, is

$$(DM_j)^{\mathsf{t}} = T_j^0 + B_j - r_j^{\mathsf{t}} d_{U_{j+1}}^{\mathsf{t}}. \tag{CR24}$$

Here T_j^0 is the flat box-and-line prolongation. The coherent raw potential chart and the curvature bound give

$$\|B_j\|_{2\to 2} \le C_L(M\ell_j + F_*\ell_j^2), \qquad \|r_j^{\mathsf{t}}\|_{2\to 2} \le C_L\ell_j. \tag{CR25}$$

The first bound combines the transported-reference difference with the full-space curvature error. The second is a fixed-L physical endpoint-path estimate: the path variation has length $O_L(\ell_j)$, and its finite row and column bounds use the physical site/bond volume factors. Both estimates hold on unrestricted inputs. They are not obtained by interpolating a bound stated only on a Ward subspace.

Consequently

$$\|\lambda_j\|_2 \le [1 + C_L(M\ell_j + F_*\ell_j^2)]\|\lambda_{j+1}\|_2 + C_L r^2 \ell_j^{-1}\|q\|_2.$$

The products of the first coefficients are bounded by $\exp[C_L(Mr + F_* r^2)]$. Summing the second terms geometrically gives

$$\|\lambda_j\|_2 \le C\|\lambda\|_2 + Cr^2 h_j^{-1}\|q\|_2. \tag{CR26}$$

Unroll (CR24) using only the complete flat tails:

$$(D\mathcal{M}_r)^{\mathsf{t}}\lambda - Q_r^*\lambda = \sum_{j<k}(T_0^0 \cdots T_{j-1}^0)[B_j\lambda_{j+1} - r_j^{\mathsf{t}} q_{j+1}]. \tag{CR27}$$

Each flat tail has a uniform H^s norm for $s < 1/2$, regardless of the divergence of its input. The inverse estimate at spacing h_j, (CR23), and (CR25)–(CR26) give

$$\|r_j^{\mathsf{t}} q_{j+1}\|_{H^s} \le Cr^2\ell_j^{-1-s}\|q\|_2,$$

and

$$\begin{aligned}\|B_j\lambda_{j+1}\|_{H^s} \le{}& C(M\ell_j^{1-s} + F_*\ell_j^{2-s})\|\lambda\|_2 \\ &+ Cr^2(M\ell_j^{-s} + F_*\ell_j^{1-s})\|q\|_2.\end{aligned}$$

All three scale sums are geometric for the stated fixed $s > 0$. The endpoint sum is dominated by the smallest scale. The extra defect terms from B_j are bounded by that same smallest-scale quantity times $C(M + F_*)$. Thus

$$\begin{aligned}\|(D\mathcal{M}_r)^{\mathsf{t}}\lambda - Q_r^*\lambda\|_{H_a^s} \le{}& C(Mr^{1-s} + F_* r^{2-s})\|\lambda\|_2 \\ &+ Cr^2 a^{-1-s}\|q\|_2.\end{aligned} \tag{CR28}$$

Apply the actual compatibility estimate (CR20):

$$\boxed{\|(D\mathcal{M}_r)^{\mathsf{t}}\lambda - Q_r^*\lambda\|_{H_a^s} \le C(Mr^{1-s} + F_* r^{2-s})\|\lambda\|_2 + C\|E\|_{H_a^s}.} \tag{CR29}$$

This is the required non-Ward current comparison. It is proved from unrestricted one-step identities and atomic defect propagation. The Ward-restricted FH32 estimate has not been applied to any non-Ward current.

55.5 5. The KKT estimate with a compatible Euler residual

Keep the actual raw constraint

$$\mathcal{M}_r(U) = \mathcal{M}_r(U^A) \tag{CR30}$$

and the weak chart of PR3, but replace its multiplier/Ward lines by (CR16). The source E is measured in the original fine physical H^s norm. In this section M bounds the potential $W^{1,p}$ norms and both curvature suprema, as in WP19; in particular $F_* \le CM$.

First, the longitudinal multiplier obeys

$$\lambda_L = d_r \Delta_r^{-1} \delta_r \lambda, \qquad \delta_r \lambda = q + (\delta_r - d_V^{\mathsf{t}})\lambda.$$

The expression on the right has zero mean because it equals the ordinary divergence on the left. The inverse removes that mean in its argument. By (CR21), the flat inverse estimate $H^{s-1} \to H^s$, and the weak-chart multiplier bound,

$$\|\lambda_L\|_{H^s_r} \le C_{s,p,T}\|E\|_{H^s_a} + C_{s,p,T}(M + M^2)\|\lambda\|_{H^s_r}. \tag{CR31}$$

Write $(D\mathcal{M}_r)^{\mathsf{t}}\lambda = Q_r^*\lambda + e_Q$. Equation (CR29) bounds e_Q by a small multiple of the multiplier norm plus $C\|E\|_{H^s}$. Insert

$$J_a(u) = Q_r^*\lambda + e_Q + E$$

into the exact Wilson force identity NK12. The same mean Euler equation as in NK27 now gives

$$\|\lambda_H\|_2 \le C\|E\|_{H^s_a} + C(M + M^2)\big(K_u + \|\lambda\|_{H^s_r}\big). \tag{CR32}$$

The harmonic coarse multiplier is not discarded.

The mean-zero transverse Schur equation is unchanged except that its right-hand side contains E, e_Q, and the longitudinal multiplier controlled by (CR31). The transverse constraint error still has the proved $r^2(M + M^2)(K_A + K_u)$ bound. Applying NK20 and absorbing the same fixed small multiplier coefficient therefore yields

$$\begin{aligned} &\|\lambda\|_{H^s_r} + \|J_a(u)\|_{H^s_a} + \|F_a(u)\|_{H^{1+s}_a} \\ &\qquad \le C\left[\|F_a(A)\|_{H^{1+s}_a} + \|A\|_2 + (M + M^2)(K_A + K_u) + \|E\|_{H^s_a}\right]. \end{aligned} \tag{CR33}$$

The same estimate controls the transverse potential in H^{1+s}, with its fixed torus constant. This is the residual version of NK/WP, before eliminating the unknown low output norm.

55.6 6. The assembled residual theorem

Theorem CR. Fix the group, physical torus, odd L, $p > 8$, and $0 < s < 1/2$. For a sufficiently small weak radius independent of a, r, k, assume

- the potential $W^{1,p}$ and curvature supremum hypotheses in PR3 for A, u;
- the actual equality of raw constraints (CR30);
- the physical Euler equation (CR16) for some coarse multiplier and $E \in H^s_a$.

There is no separate coarse Ward hypothesis. Then

$$\boxed{\|\lambda\|_{H^s_r} + \|J_a(u)\|_{H^s_a} + \|F_a(u)\|_{H^{1+s}_a} \le C\left[\|A\|_{H^{1+s}_a} + \|F_a(A)\|_{H^{1+s}_a} + \|E\|_{H^s_a}\right].} \tag{CR34}$$

The constructed gauge of GS still fixes every retained center. In that gauge,

$$\|\widetilde{u}\|_{H_a^{1+s}} \le C\left[\|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}} + \|E\|_{H_a^s}\right]. \tag{CR35}$$

Proof. Use GS to put u into the flat smoothed slice. The raw target remains exactly the one determined by the original, untransformed A. Fixing that gauge while differentiating the constraint, as in PR8–9, transforms the residual to $\widetilde{E} = \mathrm{Ad}_g E$ with the same coarse multiplier. The $C^{1,\alpha}$ gauge bound gives $\|\widetilde{E}\|_{H^s} \le C\|E\|_{H^s}$. The transformed equation has the same actual divergence compatibility, and its weak chart and curvature supremum remain small after the fixed comparison constants are included in the initial radius.

Apply (CR33) in this slice. The longitudinal-potential and harmonic-one-form arguments PL25–33 use only the actual constraint, the flat slice, and the transverse estimate; they are unchanged. They give

$$K_{\widetilde{u}} \le C\left[\|A\|_{H^{1+s}} + \|F_a(A)\|_{H^{1+s}} + \|\widetilde{E}\|_{H^s}\right] + C(M+M^2)K_{\widetilde{u}}.$$

At finite cutoff the last norm is finite. Absorb it using one fixed weak-radius choice, then apply (CR33). Finally transfer curvature and current back through GS22. This proves (CR34)–(CR35).

In the flat theory the same proof requires no smallness hypothesis: the nonlinear remainders and chart errors vanish. With the exact slice it gives the corresponding potential estimate, and without the slice it still gives the curvature/current estimate. Equations (CR9)–(CR15) provide an independent direct control of the flat endpoint term.

55.7 7. Allowing a raw constraint residual

The same result includes the constraint residual of RS. Put

$$q_r(u) = r^{-1}\log \mathcal{M}_r(e^{au}), \qquad q_r(u) - q_r(A) = b,$$

and define, with the flat coarse Hodge projections,

$$\mathfrak{B}_{s,r}(b) = \|\Pi_r^0 b\|_{H_r^{2+s}} + \|P_{L,r}b\|_{H_r^{1+s}} + \|P_{H,r}b\|_2. \tag{CR36}$$

Corollary CR. Under the weak chart hypotheses of Theorem CR, replace the exact constraint equality by this residual equation. Keep the actual Euler residual equation (CR16), with $V = \mathcal{M}_r(e^{au})$, and impose no separate coarse Ward condition. Then

$$\begin{aligned}\|\lambda\|_{H_r^s} + \|J_a(u)\|_{H_a^s} + \|F_a(u)\|_{H_a^{1+s}} + \|\widetilde{u}\|_{H_a^{1+s}} \\ \le C\left[\|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}} + \|E\|_{H_a^s} + \mathfrak{B}_{s,r}(b)\right],\end{aligned} \tag{CR37}$$

where $\widetilde{u}$ is the same retained-center-fixed slice representative.

Indeed, (CR17)–(CR29) depend only on the actual Euler equation and raw gauge covariance at U, not on equality with the input target. The transverse raw constraint now has the extra term $\Pi_r^0 b$, exactly as in RS9. Its Schur contribution obeys

$$\|S_r\Pi_r^0 b\|_{H_r^s} \le C\|\Pi_r^0 b\|_{H_r^{2+s}},$$

by the principal-alias bound for S_r. The longitudinal and harmonic potential equations add respectively $P_{L,r}b$ and $P_{H,r}b$, as in RS18–19. Their norms are exactly the other two terms in (CR36). Every

weak-radius absorption is unchanged. The gauge fixes the retained centers, so both original targets, b, and λ stay unchanged; only E receives the controlled adjoint transport.

This corollary supersedes the separate exact-Ward hypothesis of Theorem RS. The actual residual components must still be assembled before using their gauge identity, but their total residual need not be covariantly divergence-free.

56 A real periodic weak chart for the actual regular critical orbit

This chapter closes the low-chart premise of the periodic reconstruction theorem for a genuine **real periodic regular critical orbit** supplied by the original variational theorem. It constructs a representative with the same actual raw coarse field. The initial representative may be arbitrarily oscillatory by a fine gauge transformation; no ordinary high Sobolev norm is asserted in that initial gauge.

The argument does not use energy minimality to infer pointwise curvature. The pointwise input comes from the stronger, explicitly scaled conclusion of the original regular-orbit theorem. It also does not identify a wall-modified auxiliary field with a critical orbit, or construct a common complex-parameter gauge chart.

56.1 1. The precise source input and its units

Use the uniform periodic version of Bałaban's variational problem, on the compatible block lattices admitted there. Let the physical torus have fixed side T, fine spacing a, and retained spacing $r = L^k a \leq 1$, with fixed odd L. In the variational paper, coordinates with retained spacing one have fine spacing

$$\eta = \frac{a}{r}. \tag{WC1}$$

The defining regularity condition (2), the coarse assumption (7), and Theorem 1/(8) of *The variational problem and background fields in renormalization group method for lattice gauge theories*, CMP **102** (1985), imply the following. If the actual coarse field V has

$$\sup_{p_c} |1 - V(\partial p_c)| \le \varepsilon_1,$$

then its admitted regular critical orbit satisfies

$$\sup_{p_f} |1 - U(\partial p_f)| \le B_3 \varepsilon_1 \eta^2. \tag{WC2}$$

Here B_3 is independent of the blocking depth and volume in that theorem. Proposition 8 gives this improvement for every critical configuration in its original sufficiently small regular class. This does not assert that every unrestricted global minimizer belongs to that class.

The map is the actual raw iterated average. Bałaban I, CMP **109** (1987), states after (0.10) that the preceding results apply to the admitted group averages; its (1.1)–(1.2) explicitly imports this critical/minimal orbit and plaquette implication for its averaging map. We retain the outer reference-log average and its admitted endpoint means.

Suppose the target is $V = \mathcal{M}_r(U^A)$, where $U^A = e^{aA}$, and

$$\|A\|_{W_a^{1,p}} \le m, \qquad \|F_a(A)\|_\infty \le \alpha, \qquad 8 < p < \infty. \tag{WC3}$$

All norms use physical volume. The real intermediate-curvature estimate PG19–20, whose leading two-form average has norm one, gives

$$\sup_{p_c} |1 - V(\partial p_c)| \le C r^2 \alpha \tag{WC4}$$

when the fixed physical weak radius is sufficiently small. The fine plaquette branch in (WC2) is small. Since its principal logarithm is comparable to its distance from the identity, (WC1)–(WC4) give

$$\boxed{\|F_a(U)\|_\infty \le C\alpha.} \tag{WC5}$$

This is the scale cancellation being used. In contrast, the companion source bound for the physical current is only $O(\alpha/r)$; it is not a uniform current estimate and is not used below.

56.2 2. Fine holonomies from the actual coarse target

The small raw target also controls the harmonic obstruction to a small periodic potential. All fields in this section are real compact-group fields, so adjoint transport is isometric.

Let $U_j = \mathcal{M}^j(U)$, with spacings $a_j = L^j a$. For each coarse bond compare U_j with the original fine straight transport along that bond, using the same lifted path on the periodic universal cover. If E_j is the supremum of these group distances, the literal one-step reference-log formula and its endpoint contour means give

$$E_{j+1} \le L E_j + C_L a_{j+1}^2 \|F_{a_j}(U_j)\|_\infty, \qquad E_0 = 0. \tag{WC6}$$

Indeed, every contour entering one mean has the same lifted endpoints as its reference bond. Its relative loop fills an area $O_L(a_{j+1}^2)$. Plaquette telescoping bounds the relative holonomy by that area times the physical curvature. The analytic mean preserves this estimate. Replacing the L old bonds

of the reference by their fine straight transports costs at most LE_j, by telescoping products of unitary matrices.

PG20 gives $\|F_{a_j}(U_j)\|_\infty \le C\alpha$. Division of (WC6) by a_{j+1}^2 leaves coefficient L^{-1} on E_j/a_j^2, hence

$$E_j \le C_L a_j^2 \alpha. \tag{WC7}$$

The same reasoning applies on the periodic cover: the relative comparison loops have zero winding, even when the reference bond itself follows a wrapping path. The local tree-gauge proof of PG20 extends to this cover for the same reason; it needs the contractible plaquette patch, not injectivity of that patch's projection onto the torus.

Let $H_\mu(U)$, $\mu = 1, \dots, 4$, be the fine fundamental coordinate holonomies based at one retained center. Their coarse counterparts are products of T/r bonds of V. The actual raw weak-chart estimate WP supplies $\|r^{-1} \log V\|_\infty \le C_{p,T} m$. Product telescoping and (WC7) therefore give

$$\delta := \max_\mu \|H_\mu(U) - 1\| \le C_{p,L,T}(m + r\alpha). \tag{WC8}$$

There is no choice of a logarithm for a large torus holonomy here: (WC8) first puts those holonomies in a fixed small neighborhood of the identity.

56.3 3. A bounded periodic potential from curvature and holonomy

We prove a lattice gauge lemma independently of stationarity.

Lemma WC-A. Fix a compact matrix group and a physical four-torus of side T. Suppose a real lattice connection has

$$\sup_p a^{-2}\|1 - U_p\| \le b, \qquad \max_\mu \|H_\mu(U) - 1\| \le \delta. \tag{WC9}$$

If $T^2 b + \delta$ is sufficiently small, there is a periodic real gauge q such that

$$U^q = e^{au_0}, \qquad \|u_0\|_\infty \le C_T(b + \delta). \tag{WC10}$$

The constant is independent of a. More explicitly its geometric scale is $C(Tb + \delta/T)$.

56.3.1 Proof: local tree gauges

Choose a fixed finite cover of the torus by coordinate cubes of diameter less than $T/2$, with larger coordinate cubes giving fixed buffers. Choose a partition of unity χ_i subordinate to the smaller cubes, with $\sum_i |\nabla \chi_i| \le C/T$. For sufficiently fine meshes the supports and their neighboring lattice sites lie inside the larger cubes.

Fix a common base point o, paths from o to the patch anchors z_i, and coordinate-order paths inside each lifted patch from z_i to x. Let $q_i(x)$ be their transport from o to x. The local gauge-transformed bond is a conjugate of the loop between two coordinate-order paths ending at neighboring sites. That loop fills at most CT/a fine plaquettes. Consequently

$$B_i(x, x + ae_\mu) := q_i(x) U_\mu(x) q_i(x + ae_\mu)^{-1}, \qquad \|B_i - 1\| \le CaTb. \tag{WC11}$$

This uses only the exact product-of-plaquettes identity and the unitary telescoping inequality. It differentiates neither the curvature nor the initial gauge.

56.3.2 Proof: overlap alignment

On an overlap, $q_i(x)q_j(x)^{-1}$ is the holonomy of a loop formed from a fixed number of coordinate segments of total length $O(T)$. Coordinate interchanges fill its zero-winding part with $O((T/a)^2)$ plaquettes. Its winding part is a product of a bounded number of the four based fundamental cycles and their inverses. All conjugating transports are unitary. Thus, uniformly on each enlarged overlap,

$$\|q_i(x)q_j(x)^{-1} - 1\| \le C(T^2 b + \delta) =: \tau. \tag{WC12}$$

This gives the needed alignment of the patch values; small derivatives of transition functions alone would not do so. The exact bond identities also give

$$\|D_a(q_i q_j^{-1})\|_\infty \le CTb \tag{WC13}$$

where the overlap contains the relevant bond.

56.3.3 Proof: gluing the gauges

For a sufficiently small relative cluster in a compact group, define the weighted local mean by its logarithmic equation

$$\sum_i \chi_i \log(q_i q^{-1}) = 0. \tag{WC14}$$

The implicit-function theorem at a common group value has derivative minus the identity. It gives a uniformly smooth mean for all nonnegative weights with sum one, including zero weights. The mean is equivariant under common left and right multiplication. Its derivative in the input values is uniformly bounded; on the weight hyperplane $\sum_i d\chi_i = 0$, its weight derivative is $O(\tau)$. These assertions follow directly by differentiating (WC14) in the relative logarithmic chart.

Set $q(x)$ to this mean of the available $q_i(x)$, weighted by $\chi_i(x)$. For a neighboring pair use the union of the two active sets; the buffers ensure that every required q_i is defined at both sites. Right equivariance, (WC11), and the input derivative bound compare $q(x)U_\mu(x)$ with the same-weight mean at $x + ae_\mu$, at cost $CaTb$. Changing the weights costs at most $C(a/T)\tau$. Therefore

$$\|q(x)U_\mu(x)q(x + ae_\mu)^{-1} - 1\| \le Ca\left(Tb + \frac{\delta}{T}\right). \tag{WC15}$$

The principal logarithm gives (WC10). This is a globally defined periodic gauge because the paths, patch values and partition were defined on the periodic lattice.

The finitely coarse meshes excluded by the buffer choice have $a \ge cT$. A global coordinate spanning tree then gives each remaining bond as a loop of uniformly bounded length and area, with the same fundamental-cycle control. Its distance from the identity is $O(T^2 b + \delta)$; division by $a \ge cT$ gives (WC10) with another fixed constant. This completes the proof.

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56.5 5. Restoring the original raw target with a smooth correction

Lemmas WC-A and WC-B yield a real gauge g_1 such that

$$U^{g_1} = e^{av}, \qquad \|v\|_{W^{1,p}_a} \le C_{p,L,T}(m+\alpha). \tag{WC25}$$

They do not initially fix its values at retained centers. Write

$$h = N_r g_1, \qquad W = \mathcal{M}_r(U^{g_1}) = V^h, \qquad W = e^{rB}, \quad V = e^{rC}.$$

The actual raw weak-chart estimates, applied to v and the unchanged A, give

$$\|B\|_{W^{1,p}_r} + \|C\|_{W^{1,p}_r} \le C_{p,L,T}(m+\alpha). \tag{WC26}$$

Choose a retained anchor y_0. A constant left multiplication of g_1, and the corresponding constant conjugation of v, B, arranges $h(y_0) = 1$ without changing these norms. The exact coarse intertwining relation is

$$h(y + re_\mu) = e^{-rB_\mu(y)} h(y) e^{rC_\mu(y)}. \tag{WC27}$$

It first gives $\|D_r h\|_\infty \le C(\|B\|_\infty + \|C\|_\infty)$. Coordinate paths of total physical length at most $4T$ then put h in one small identity chart. Taking one further coarse difference in (WC27) gives

$$\|D_r^2 h\|_{L^p} \le C\big(\|D_r B\|_{L^p} + \|D_r C\|_{L^p} + (\|B\|_\infty + \|C\|_\infty)\|D_r h\|_{L^p}\big). \tag{WC28}$$

The derivative of $r^{-1}(e^{rB} - 1)$ is a uniformly bounded analytic coefficient times $D_r B$; no inverse retained spacing is introduced. Consequently

$$\|\xi\|_{W_r^{2,p}} \le C(m+\alpha), \qquad \xi = -\log h. \tag{WC29}$$

Use the exact mixed node/box lift of LF29–36:

$$\phi_2 = H_{\mathrm{m}}\xi, \qquad N_r\phi_2 = \xi, \qquad \|\phi_2\|_{W_a^{2,p}} \le C(m+\alpha). \tag{WC30}$$

Set $g = e^{\phi_2} g_1$ and $U^g = e^{aw}$. Since $g = 1$ at every retained center, raw covariance is exact:

$$\mathcal{M}_r(U^g) = V, \qquad \|w\|_{W_a^{1,p}} \le C_{p,L,T}(m+\alpha). \tag{WC31}$$

The last bound follows from the actual gauge-map estimate GS14–16, with $v \in W^{1,p}$ and $\phi_2 \in W^{2,p}$. The curvature supremum is unchanged by this real gauge. A constant mode was fixed by $h(y_0) = 1$; no unbounded scalar zero mode is hidden in (WC29).

56.6 6. The resulting weak-chart theorem

Theorem WC. Fix T, the compact matrix gauge group, a finite $p > 8$, and the admitted odd blocking factor L. Let A obey (WC3) with $m + \alpha$ sufficiently small. Let U be a representative of the actual periodic regular critical orbit with raw target $\mathcal{M}_r(U^A)$, in the class to which the original variational theorem and its curvature improvement apply. Then there is a real fine gauge g, equal to the identity at every retained center, such that

$$\boxed{U^g = e^{aw}, \qquad \mathcal{M}_r(U^g) = \mathcal{M}_r(U^A), \qquad \|w\|_{W_a^{1,p}} + \|F_a(w)\|_\infty \le C_{p,L,T}(m+\alpha).} \tag{WC32}$$

The radius and constant are independent of a, r, k. All geometric lattices remain within the uniform periodic setting of the imported variational theorem. The gauge g itself need not have a high norm relative to an arbitrarily chosen original representative.

Choose $p > 8$, and shrink the input radius by the fixed constant in (WC32). Provided the physical KKT multiplier exists with the stated qualification, the pair (w, A) then satisfies the low hypotheses of Theorem PR. Its conclusion provides a further same-target representative with

$$\|F_a(\widetilde{w})\|_{H_a^{1+s}} + \|J_a(\widetilde{w})\|_{H_a^s} \le C\big(\|A\|_{H_a^{1+s}} + \|F_a(A)\|_{H_a^{1+s}}\big), \qquad 0 < s < \tfrac{1}{2}. \tag{WC33}$$

This is a statement in a constructed good representative of the genuine orbit. Gauge-invariant completed expressions may be evaluated in that representative. It is not a bound for the ordinary H^{1+s} norm of $F_a(U)$ in every original fine gauge.

56.8 Primary source locations

- T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Commun. Math. Phys. **102** (1985), 277–309: (2), (7), Theorem 1/(8), and Proposition 8. The curvature improvement and its units are the imported source input.
- T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, Commun. Math. Phys. **109** (1987), 249–301: the paragraph after (0.10), and (1.1)–(1.2), for the averaging-map and regular-orbit identification.

The remaining lemmas in this chapter are proved above and use the previously derived finite-lattice averaging, mixed-lift and uniform flat elliptic estimates.

57 Complex physical charts from the original analytic reconstruction

This chapter proves a uniform complex weak chart for a representative of the genuine periodic critical **orbit**, using the original analytic reconstruction about a fixed real reference. The final conversion to the exact raw coarse representative is reduced to one bounded-anchor estimate for the source's axial/Landau gauge bridge. That last source identification is displayed separately; the existence of an analytic gauge map alone is not used as a uniform bound.

The key analytic ingredient is stronger than bounded real values. The source reconstruction is holomorphic on a physical input ball whose radius grows as r^{-1}, with a crude curvature bound $O(r^{-2})$. This controls its Taylor remainder above degree two. A finite real interpolation then bounds the remaining quadratic polynomial in a fixed complex neighborhood.

The later CK construction gives a different same-target holomorphic $W^{1,p}$ branch even at the flat reference. It does not give that representative a uniform complex curvature supremum. The anchor issue in this chapter concerns simultaneous control of both norms in the same-target representative.

57.1 1. Physical input spaces and the reference field

Fix a physical four-torus of side T, a finite $p > 8$, a fixed odd blocking factor L, and nested spacings $a \le r = L^k a \le 1$. Define the linear weak norm

$$\|A\|_{\mathcal{Y}_a} := \|A\|_{W^{1,p}_a} + \|d_a A\|_\infty. \tag{CX1}$$

On a small $W^{1,p}$ potential ball this is equivalent, up to the usual quadratic correction, to the weak potential-plus-curvature conditions used earlier. Indeed the exact plaquette BCH identity gives

$$\|F_a(A) - d_a A\|_\infty \le C\|A\|_\infty^2 \le C_{p,T}\|A\|_{W^{1,p}}^2. \tag{CX2}$$

For strong estimates one may use the linear norm

$$\|A\|_{\mathcal{X}_a} := \|A\|_{\mathcal{Y}_a} + \|A\|_{H^{1+s}_a} + \|d_a A\|_{H^{1+s}_a}, \qquad 0 < s < \tfrac{1}{2}. \tag{CX3}$$

The already proved compensated BCH estimate makes the last curl norm equivalent to using $F_a(A)$ on the common weak ball.

Fix a real A_0 with $\|A_0\|_{\mathcal{Y}_a} \le \varepsilon$, and let $V_0 = \mathcal{M}_r(e^{aA_0})$. The real weak-chart theorem WC supplies a real representative

$$U_0 = e^{aw_0}, \qquad \mathcal{M}_r(U_0) = V_0, \qquad \|w_0\|_{W^{1,p}} + \|F_a(U_0)\|_\infty \le C\varepsilon. \tag{CX4}$$

This representative is fixed while its nearby complex inputs are considered. It is not chosen anew at each complex parameter.

57.2 2. A growing amplitude domain for the actual raw average

There is a constant $c_L > 0$ such that the genuine raw map

$$A \longmapsto \mathcal{M}_r(e^{aA})$$

is holomorphic whenever $r\|A\|_\infty < c_L$, and

$$\|\log \mathcal{M}_r(e^{aA})\|_\infty \le C_L r\|A\|_\infty. \tag{CX5}$$

This statement uses only amplitude, not a spatial derivative bound.

To prove it, apply the same actual internal-path straightening as in CM/GQ. On one fixed-L step of output spacing ℓ_j, its finite analytic expansion has positive box-and-line linear part and a remainder bounded by $C_L\ell_j S_j^2$, where S_j is the input physical potential supremum. Thus

$$S_{j+1} \le S_j + C_L\ell_j S_j^2. \tag{CX6}$$

Since $\sum_j \ell_j \le C_L r$, a bootstrap gives $S_j \le 2S_0$ when rS_0 is small. At this amplitude level, the internal connector needs no zero-moment gain:

$$\|r_{\ell_j}(A_j)\|_\infty \le C_L\ell_j S_j.$$

The exact accumulated node gauge is a product of the corresponding exponentials. Its total logarithmic size is at most $C_L r S_0$. Restoring the raw field by that node gauge therefore keeps every final link within $C_L r S_0$ of the identity. This proves (CX5), including the raw representative and all noncommuting factors.

All finite steps are holomorphic on the same relative branch. The geometric bootstrap keeps them there for the whole growing input ball. Banach-valued Cauchy estimates on a fixed smaller ball consequently give

$$\left\|\log\left[\mathcal{M}_r(e^{aA})\mathcal{M}_r(e^{aA_0})^{-1}\right]\right\|_\infty \le C_L r\|A-A_0\|_\infty \tag{CX7}$$

when $r\|A\|_\infty$ and $r\|A_0\|_\infty$ lie in that smaller ball. This is also obtained by integrating the uniformly bounded derivative along the straight segment. It does not invoke an increasing direct block factor.

57.3 3. What the original analytic theorem supplies

Choose one fixed admissible source parameter $\varepsilon_{\mathrm{ref}} > 0$, below the constants in CMP102, rather than setting that parameter equal to the sharper r^2 curvature of the actual input. The coarse field V_0 satisfies the source hypothesis with this larger bound.

Section G, (172)–(175) and Proposition 9 of CMP **102** supplies an analytic relative Landau reconstruction

$$U_{\mathrm{L}}(B) = e^{ab(B)}U_0, \qquad b(0)=0, \tag{CX8}$$

on a dimensionless coarse ball $\|B\|_\infty < c\varepsilon_{\mathrm{ref}}$. Its own-unit bound (173)/(19), converted by $\eta = a/r$, gives

$$\|b(B)\|_\infty \le C\varepsilon_{\mathrm{ref}}/r, \qquad \|F_a(U_{\mathrm{L}}(B))\|_\infty \le C/r^2. \tag{CX9}$$

For the second bound, the relative covariant first-difference estimate in (19) and the real reference curvature control the actual plaquette logarithm. Its dimensionless fine plaquette remains in the small branch, since its size is $O(\varepsilon_{\mathrm{ref}}(a/r)^2)$.

The construction is an analytic representative of the critical orbit corresponding to $e^B V_0$. The source also gives an analytic inverse gauge back to its axial representative. We do not yet assign a uniform norm to that gauge.

The first estimate in (CX9), holomorphy, and $b(0)=0$ give the sharper response

$$\|b(B)\|_\infty \le Cr^{-1}\|B\|_\infty \tag{CX10}$$

on a smaller coarse ball. This is the Banach-valued Schwarz estimate applied to $z \mapsto b(zB)$. It avoids extracting more regularity than the source's derivative bound (190) states.

Set

$$B(A) = \log\left[\mathcal{M}_r(e^{aA})V_0^{-1}\right]. \tag{CX11}$$

Equations (CX5)–(CX7) and the fixed source radius show that $U_{\mathrm{L}}(A) := U_{\mathrm{L}}(B(A))$ is holomorphic on an L^∞ input ball about A_0 of radius c/r. There

$$\|F_a(U_{\mathrm{L}}(A))\|_\infty \le C/r^2. \tag{CX12}$$

On a fixed small physical input ball, (CX7), (CX10), and the small real reference give a global physical potential

$$U_{\mathrm{L}}(A) = e^{a u_{\mathrm{L}}(A)}, \qquad \|u_{\mathrm{L}}(A)\|_\infty \le C(\varepsilon + \|A - A_0\|_\infty). \tag{CX13}$$

The logarithm here is the actual matrix logarithm of $e^{ab}e^{aw_0}$. Its analytic BCH bound requires only the two displayed amplitude bounds. No first derivative of b has been assumed uniform in current physical units.

57.4 4. A finite interpolation lemma

Let E be any complex Banach space, and f an E-valued holomorphic function on $|z| < R$, with $R \ge 4$ and $\sup \|f(z)\| \le M_R$. Let

$$P_2(z) = f(0) + f'(0)z + \tfrac{1}{2} f''(0) z^2.$$

Cauchy's formula gives, for $|z| \le 1$,

$$\|f(z) - P_2(z)\| \le 2 M_R R^{-3}. \tag{CX14}$$

The degree-two polynomial is determined by its values at $-1, 0, 1$. Its three Lagrange polynomials are bounded by an absolute constant on $|z| \le 1$. Therefore

$$\sup_{|z|\le 1} \|f(z)\| \le C \max_{\xi \in \{-1,0,1\}} \|f(\xi)\| + C M_R R^{-3}. \tag{CX15}$$

The estimate uses only three real evaluations and has no dimension-dependent polarization constant.

Apply it to

$$f(z) = F_a(U_{\mathrm{L}}(A_0 + H_1 + zH_2)), \tag{CX16}$$

where H_1, H_2 are real and $\|H_1\|_{\mathcal{Y}_a} + \|H_2\|_{\mathcal{Y}_a} \le \rho$. If $H_2 = 0$ there is nothing to interpolate. Otherwise let

$$t = \|H_2\|_\infty, \qquad R = \frac{\kappa}{rt}, \tag{CX17}$$

where $\kappa > 0$ is a fixed small fraction of the amplitude and source radii. By reducing ρ once, $R \ge 4$, and the offset H_1 leaves the full disc inside the source domain. Equation (CX12) gives $M_R \le C/r^2$, hence

$$M_R R^{-3} \le C r t^3. \tag{CX18}$$

At the three real nodes in (CX16), the source field is real and belongs to the genuine regular critical orbit. Its real gauge can be different at each node, but compact gauge conjugation preserves the curvature norm. WC2–5 therefore imply

$$\|f(-1)\|_\infty + \|f(0)\|_\infty + \|f(1)\|_\infty \le C(\varepsilon + \rho).$$

Here the input curvature follows from its $\mathcal{Y}_a$ bound and (CX2). Using (CX15) at $z = i$ proves

$$\boxed{\sup_{\|A - A_0\|_{\mathcal{Y}_a} < \rho} \|F_a(U_{\mathrm{L}}(A))\|_\infty \le C(\varepsilon + \rho).} \tag{CX19}$$

The complexified norm may be defined by the sum of the norms of real and imaginary parts; any fixed equivalent complex norm only changes ρ by a constant. All bounds are independent of a, r, k.

This argument would fail with only a fixed source radius and the bound r^{-2}. The growing radius $R \asymp (rt)^{-1}$ is exactly what pays the high Taylor coefficients.

57.5 5. Holomorphic weak representatives of the correct orbit

The small-L^p gauge construction WC17–24 extends to complex fields without a change in its contraction constants. Its algebraic gauge map is holomorphic; the norms in its proof are absolute norms. A complex gauge parameter in the small $W^{1,p}$ ball has a uniformly small supremum, so its adjoint action and inverse are uniformly bounded.

Use that fixed point on the small potential (CX13). It produces a holomorphic $\phi(A) \in W^{1,p}$ and

$$U_{\mathrm{L}}(A)^{e^{\phi(A)}} = e^{av(A)}, \qquad \delta_a v(A) = 0.$$

Before the derivative bootstrap, $\|v(A)\|_{L^p} + \|\phi(A)\|_{W^{1,p}} \le C(\varepsilon + \rho)$. Equation (CX19) and bounded adjoint transport supply the same small curvature bound after the gauge change. The exact curvature/Hodge absorption of WC22–24 therefore gives

$$\boxed{\sup_{\|A - A_0\|_{\mathcal{Y}_a} < \rho} \left(\|v(A)\|_{W^{1,p}} + \|F_a(v(A))\|_\infty\right) \le C(\varepsilon + \rho).} \tag{CX20}$$

All maps are finite-dimensional holomorphic at each cutoff. Their uniform bounds on this one common complex ball make the Banach-norm Cauchy estimates uniform as well. The family is real on real inputs.

This proves a cutoff- and retained-scale-independent complex weak representative of the actual critical **orbit**. It has not yet proved that this representative has the originally prescribed raw coarse field.

57.6 6. The exact bounded-anchor requirement

Let $h_{\mathrm{src}}(A)$ be the coarse value of the source's analytic gauge between its Landau representative and its raw axial representative, with the direction fixed so that

$$\mathcal{M}_r(U_{\mathrm{L}}(A)) = \mathcal{M}_r(e^{aA})^{h_{\mathrm{src}}(A)}. \tag{CX21}$$

The source construction supplies this analytic bridge, with identity at the reference parameter, but the preceding estimates have not bounded its value at a retained anchor. The required additional bound is only

$$\sup_{\|A - A_0\|_{\mathcal{Y}_a} < \rho} \Big(\|h_{\mathrm{src}}(A)(y_0)\| + \|h_{\mathrm{src}}(A)(y_0)^{-1}\|\Big) \le C. \tag{CX22}$$

The gauge from Section 5 already has a bounded anchor. Once (CX22) is known, normalize the product bridge at y_0 by a bounded holomorphic constant conjugation. Its coarse fields on both sides have small $W^{1,p}$ potentials, by (CX20) and the actual raw coarsening estimates. The exact intertwining equation (WC27) now yields

$$\|\log h_{\mathrm{norm}}(A)\|_{W_r^{2,p}} \le C(\varepsilon+\rho). \tag{CX23}$$

For complex matrices, the short path estimate uses $\|h(y)\| + \|h(y)^{-1}\| \le C\exp[CT(\|B\|_\infty + \|C\|_\infty)]$ after the anchor is fixed; it does not assume unitary transport. The rest of WC28–31 is holomorphic and unchanged. The mixed lift then gives a holomorphic same-target representative $w(A)$ satisfying

$$\mathcal{M}_r(e^{aw(A)}) = \mathcal{M}_r(e^{aA}), \qquad \|w(A)\|_{W^{1,p}} + \|F_a(w(A))\|_\infty \le C(\varepsilon+\rho). \tag{CX24}$$

Thus (CX22), rather than a whole new high-regularity inverse theorem, is the exact remaining source-gauge estimate for this route.

57.8 8. Strong-norm consequence once the bridge is matched

Assume (CX22), and consider any physical parameter family $A(z)$ in the common weak ball whose $\mathcal{X}_a$ norm is uniformly at most K. The same-target family (CX24) is a genuine analytic critical family. Once the physical multiplier/constraint qualification is supplied, PR and its exact gauge section give, on one smaller common parameter polydisc,

$$\|F_a(\widetilde{w}(z))\|_{H_a^{1+s}} + \|J_a(\widetilde{w}(z))\|_{H_a^s} + \|\widetilde{w}(z)\|_{H_a^{1+s}} \le CK. \tag{CX29}$$

All gauges selected after the bounded bridge are holomorphic fixed points or fixed linear lifts on common norm balls. Cauchy's formula then gives the corresponding mixed parameter derivatives with the ordinary factorial and inverse-radius factors.

This is a periodic physical-family result. It does not identify separately weakened wall operators with this family, establish mixed-boundary analogues, or transfer an ordinary curvature norm to an arbitrary initial gauge.

57.9 9. An independent normalization on a quantitatively irreducible patch

In this section the compact group is $SU(N)$, with complexification $SL(N,\mathbb{C})$. The bounded-anchor condition can be replaced on a precisely specified open subdomain. The following argument uses the existence of the source bridge only as a finite-cutoff analytic germ. It does not use a bound for that bridge, or an inverse for arbitrary hidden forces.

First make the Coulomb representative in Section 5 based at one fixed retained site y_0. Its gauge parameter has uniformly small supremum. Multiplying it by the inverse of its value at y_0 is therefore a uniformly bounded holomorphic constant gauge transformation. This operation preserves its weak norm bounds and makes its contribution to the coarse bridge equal to the identity at the anchor. In particular, at $A = A_0$, the based holonomies of the input coarse field and of this representative coincide. Denote these two coarse fields by $V(A)$ and $W(A)$.

Choose a fixed finite set of coarse loops $\gamma_1, \dots, \gamma_m$, based at y_0, whose physical lengths are bounded by C_T. Their reference holonomies are

$$H_i^0 = V(A_0)(\gamma_i) = W(A_0)(\gamma_i), \qquad \mathcal{T}\xi = (H_i^0\xi - \xi H_i^0)_{i=1}^m, \quad \xi \in \mathfrak{sl}(N,\mathbb{C}). \tag{CX30}$$

Use the Hilbert–Schmidt inner product on matrices and the corresponding Hilbert direct-sum norm when defining $\mathcal{T}^*$. Assume the quantitative irreducibility condition

$$\|\mathcal{T}\xi\| \ge \gamma\|\xi\|, \qquad \gamma > 0. \tag{CX31}$$

Thus the common complex centralizer of the reference tuple consists only of scalar matrices. The number of loops, their length bound, and the positive number γ are fixed independently of the fine cutoff and retained spacing. No conclusion below is uniform when γ tends to zero.

Set $V_i(A) = V(A)(\gamma_i)$ and $W_i(A) = W(A)(\gamma_i)$. The actual raw coarsening bounds, (CX20), and a fixed smaller common input ball give

$$\|(V_i(A) - H_i^0)_i\| + \|(W_i(A) - H_i^0)_i\| \le C\|A - A_0\|_{\mathcal{Y}_a}. \tag{CX32}$$

To justify the uniform constant, each loop contains $O(T/r)$ links, while a coarse link variation is r times a bounded variation of its physical potential. Product telescoping gives the factor $O(T)$, and complex path norms cost only $\exp(C_T\varepsilon)$. The weak holomorphic bound on a larger fixed ball supplies the variation bound by Cauchy's formula. There is no factor from the number of fine links.

Let

$$\mathcal{T}^\dagger = (\mathcal{T}^*\mathcal{T})^{-1}\mathcal{T}^*.$$

Here $\mathcal{T}^*$ is the fixed adjoint of the reference linear map, extended complex linearly. No conjugation of the variable A or of ξ is taken. Consider the finite-dimensional holomorphic equation

$$\mathscr{F}(\xi, A) := \mathcal{T}^\dagger\big(W_i(A)e^{\xi} - e^{\xi}V_i(A)\big)_{i=1}^m = 0. \tag{CX33}$$

At $(\xi, A) = (0, A_0)$, this equation vanishes and $D_\xi\mathscr{F} = I$. Since $\|\mathcal{T}^\dagger\| \le \gamma^{-1}$, the exponential series and (CX32) give, on a small fixed matrix ball,

$$\|D_\xi\mathscr{F}(\xi, A) - I\| \le C\gamma^{-1}(\|A - A_0\|_{\mathcal{Y}_a} + \|\xi\|). \tag{CX34}$$

Also $\|\mathscr{F}(0, A)\| \le \gamma^{-1}\|(W_i(A) - V_i(A))_i\|$. The contraction form of the holomorphic implicit-function theorem therefore gives a unique solution $\xi(A)$ on a ball of radius $\rho_\gamma > 0$, with

$$\rho_\gamma \le c\min\{\rho, \gamma^2\}, \qquad \|\xi(A)\| \le 2\gamma^{-1}\|(W_i(A) - V_i(A))_i\|. \tag{CX35}$$

The constants may depend on the fixed loop tuple and γ, but not on a, r, k. The displayed upper restriction on ρ_γ means that a sufficiently small fixed multiple of that minimum is an admissible radius; it is not a claim about an optimal radius.

Equation (CX33) by itself is only a projected normal equation. The source's existing analytic bridge makes it an exact gauge intertwiner. Indeed, at each finite cutoff that bridge is analytic in some neighborhood of A_0, equals the identity there at the anchor when $A = A_0$, and satisfies all holonomy intertwining identities. Its anchor logarithm consequently solves (CX33). Uniqueness of the small solution implies that it equals $\xi(A)$ on a possibly cutoff-dependent open neighborhood. Thus

$$W_i(A)e^{\xi(A)} = e^{\xi(A)}V_i(A) \tag{CX36}$$

holds on that neighborhood. Every matrix entry of the difference is holomorphic on the common ball of radius ρ_γ, so the identity theorem extends the equality to that entire ball. This use of an analytic germ requires no lower bound on the size of the original bridge's domain of smallness.

To construct the full coarse gauge, choose fixed anchor-to-site paths p_y of physical length at most C_T, and define

$$h(A,y) = W(A)(p_y)^{-1} e^{\xi(A)} V(A)(p_y). \tag{CX37}$$

It agrees with the actual source bridge on the same small open neighborhood. For each coarse bond, the difference in the required bond intertwining equation is holomorphic on the common ball and vanishes on that neighborhood. Applying the identity theorem again proves

$$W(A) = V(A)^{h(A)} \tag{CX38}$$

for all bonds on the whole common ball. The selected loops need only provide the gap (CX31); they need not generate all lattice cycles. The analytic continuation of the bond identities supplies the remaining consistency conditions.

The path formula and (CX35) give uniform bounds for h and its inverse. Normalize its anchor by the bounded constant $e^{-\xi(A)}$, acting also on the representative. The exact bond recurrence used in WC then proves the small $W_r^{2,p}$ bound for the normalized gauge. The mixed node lift and the final gauge correction consequently give (CX24), now without the extra assumption (CX22), on this γ-dependent ball.

For real input, the fields in (CX38) are compact-group valued. The selected holonomies remain irreducible after reducing the ball. The matrix $h(y_0)^* h(y_0)$ commutes with their tuple and hence is scalar; determinant one makes that positive scalar equal to one. Thus the constructed anchor, and then the entire coarse gauge, is unitary on real input. The logarithm and lifted final correction respect the same real form.

This proves a genuine uniform holomorphic same-target family on quantitatively irreducible reference patches. It does not cover the identity flat reference: there all $H_i^0 = I$, so $\mathcal{T} = 0$. More generally, a flat periodic connection may have nontrivial fundamental holonomies, but they commute and have a nontrivial common centralizer; (CX31) still fails. Uniformity through these reducible loci requires the source normalization estimate or another argument that remains effective when $\gamma = 0$.

57.10 10. Why small holomorphic orbit data alone do not bound a bridge

A bounded anchor is a sufficient device for the construction above, not a logically necessary condition for the existence of every good same-target representative. It cannot, however, be replaced by the general assertion that two small holomorphic orbit representatives have a uniformly bounded holomorphic intertwiner.

Here is an explicit finite-dimensional example. Put

$$H = i\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad X = i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad g_n(z) = e^{nzH}. \tag{CX39}$$

Fix $R > 1$, set $q_n = e^{-8nR}$, and on $|z| < R$ define

$$V_n(z) = e^{q_n(1+z)X}, \qquad W_n(z) = g_n(z) V_n(z) g_n(z)^{-1}. \tag{CX40}$$

Use the matrix operator norm in the following numerical bounds and in (CX41). Both families are holomorphic, are $SU(2)$-valued on the real interval, coincide at $z = 0$, and tend uniformly to the

identity on the full complex disc. Their logarithms satisfy

$$\|\log V_n(z)\| \le (1+R)e^{-8nR}, \qquad \|\log W_n(z)\| \le (1+R)e^{-6nR}.$$

The actual bridge g_n is also holomorphic and equals the identity at the reference. Nevertheless, at $z = i$,

$$\frac{\|\log W_n(i)\|}{\|\log V_n(i)\|} = e^{2n}. \tag{CX41}$$

For any matrix c_n satisfying $W_n(i) = c_n V_n(i) c_n^{-1}$, the common small logarithm branch gives

$$\|c_n\|\,\|c_n^{-1}\| \ge e^{2n}. \tag{CX42}$$

Thus no alternate intertwiner can have uniformly bounded condition number at that point. Assigning the displayed matrix to every link in one direction and the identity to the other directions embeds the example into a flat periodic lattice connection. The phenomenon is already a color-frame degeneracy; it is unrelated to a rough forcing term.

This example addresses only a generic inference from holomorphic small orbit representatives and their reality properties. It is not the normalized source reconstruction and does not satisfy the full target-variable response estimates asserted for that particular map. In particular it is not a counterexample to a possible proof of (CX22) from the source's actual averaged-gauge normalization. It shows why that extra structure, or an argument of comparable strength, cannot simply be omitted.

57.11 Primary source inputs

- T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, CMP **102** (1985): Section G, (172)–(175), (179)–(190), Proposition 9, together with the regularity bounds (19).
- T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, CMP **109** (1987), pp265–266: the axial/Landau transformation and its reference to formula (106) of CMP **98**.

The latter identification is explicit in the 1987 bibliography: its reference [12] is *Averaging operations*, CMP **98**. The citation to (106)[5] on CMP102 p297 uses a different reference number and appears inconsistent with that paper's bibliography. The argument here does not resolve that bibliographic inconsistency or infer the missing formula from it.

The growing-amplitude estimate, finite interpolation, complex small-L^p bootstrap, conditional bounded-anchor completion and quantitatively irreducible patch construction are proved above. The final literal source bridge at flat and other reducible references remains explicitly distinguished from these estimates.

58 The raw-slice hidden inverse is not bounded on arbitrary negative Sobolev forcing

The flat constrained reconstruction with prescribed coarse data and its hidden Euler inverse are different operator questions. The latter cannot have the unrestricted mapping property $W^{-1,p} \to W^{1,p}$ on the exact raw constraint kernel and the smoothed Landau slice. This chapter gives a counterexample for every fixed odd blocking factor $L \ge 3$, in physical dimension four.

The obstruction concerns an arbitrary, nonphysical longitudinal forcing. It does not refute an inverse on a separately specified range satisfying the fine gauge identity. It identifies a restriction required of the Banach-space target in an implicit-function argument.

58.1 1. The operator being tested

Fix a physical torus of side $T = 4$, retained spacing $r = 1$, and fine spacings

$$a = L^{-k}. \tag{FI1}$$

Use the actual composed raw linear map

$$Q_{\mathrm{raw}} = Q_{\mathrm{bl}} - d_r K_{a,1}, \qquad Q_{\mathrm{raw}} d_a = d_r N. \tag{FI2}$$

Let E_{mix} be the mixed biharmonic scalar lift from LF/PL, extended on constants, so $NE_{\mathrm{mix}} = I$. The exact projected Landau slice says that the gradient scalar in the fine Hodge decomposition belongs to the range of E_{mix}.

Define the finite-dimensional space

$$X_a = \{u : Q_{\mathrm{raw}} u = 0,\ R_0 \delta_a u = 0\}. \tag{FI3}$$

The constrained hidden inverse, interpreted variationally, maps a forcing f to the unique $u = \Gamma_a f \in X_a$ satisfying

$$\langle d_a u, d_a z\rangle = \langle f, z\rangle \quad (z \in X_a). \tag{FI4}$$

The physical volume pairings are used. This inverse exists at each finite mesh. Indeed, a zero-energy element is a gradient plus a harmonic constant. The raw mean identity removes the harmonic constant; the raw gauge identity and $NE_{\mathrm{mix}} = I$ then remove the gradient. Thus the quadratic form is positive definite on X_a.

There is a fixed smooth real forcing f_a, with uniformly bounded L^p and $W_a^{-1,p}$ norms for every finite p, for which

$$\|\Gamma_a f_a\|_{H_a^1} \geq c\sqrt{k} - C. \tag{FI5}$$

Consequently no estimate

$$\|\Gamma_a f\|_{W_a^{1,p}} \leq C \|f\|_{W_a^{-1,p}} \tag{FI6}$$

can hold uniformly in a for any $p \geq 2$, on this unrestricted force space. A fixed torus volume compares $W^{1,p}$ from below with H^1 for $p \geq 2$.

58.2 2. A nonzero transverse part of the actual composed connector

We use the exact connector formula CR5, including its derivation by the raw and box chain identities. For a real vector v with all four components nonzero, put

$$R_t(v) = \prod_{\nu=1}^{4} \operatorname{sinc}\left(\frac{L^{-t} v_\nu}{2}\right), \quad b_{t,\nu}(v) = \frac{1}{L} \sum_{j=-(L-1)/2}^{(L-1)/2} e^{ijL^{-t} v_\nu}, \tag{FI7}$$

where $\operatorname{sinc} x = \sin(x)/x$. Define

$$\Phi_{t,\mu}(v) = \frac{1}{4!} \sum_{\pi} \prod_{\nu \prec_\pi \mu} b_{t,\nu}(v), \qquad A_\mu(v) = \sum_{t \in \mathbb{Z}} R_t(v)(b_{t,\mu}(v) - 1)\Phi_{t,\mu}(v). \tag{FI8}$$

On a compact set with all coordinates bounded away from zero, the negative-t tail is $O(L^{4t})$, because each sinc contributes one inverse large argument. The positive-t tail is $O(L^{-2t})$, by the centered quadratic zero. Both tails are uniformly summable. Thus A is continuous there. Set

$$\mathcal{K}_\mu(v) = \frac{A_\mu(v)}{iv_\mu}. \tag{FI9}$$

The permutation identity and telescoping give

$$\sum_\mu (b_{t,\mu} - 1)\Phi_{t,\mu} = \prod_\mu b_{t,\mu} - 1, \qquad R_t \prod_\mu b_{t,\mu} = R_{t-1}.$$

Consequently

$$\sum_\mu A_\mu(v) = -1. \tag{FI10}$$

This is the limiting gradient chain identity. It does not imply that $\mathcal{K}$ is a pure gradient vector.

Take

$$v_* = 2\pi(1, 1, 1, 2). \tag{FI11}$$

At this point $R_t = 0$ for $t \le 0$. At $t = 1$, all four $b_{t,\nu}$ vanish and each $\Phi_{t,\mu} = 1/4$. For $t \ge 2$, all the $b_{t,\nu}$ lie in $(0, 1)$, and $R_t > 0$. Every term in every component of (FI8) is therefore nonpositive.

Symmetry and (FI10) give

$$A(v_*) = (x, x, x, -1 - 3x). \tag{FI12}$$

A pure gradient vector in (FI9) would require $x = -1/7$.

For every $L \ge 5$, the first term alone gives

$$-x \ge \frac{1}{4}\operatorname{sinc}(\pi/L)^3 \operatorname{sinc}(2\pi/L) > \frac{1}{7}. \tag{FI13}$$

For example, monotonicity of sinc on $(0, \pi)$, its lower bound $\operatorname{sinc} t \ge 1 - t^2/6$, and $\pi^2 < 10$ give at $L = 5$

$$\operatorname{sinc}(\pi/5)^3 \operatorname{sinc}(2\pi/5) > (14/15)^3(11/15) > 4/7.$$

The inequality improves when L increases.

For $L = 3$, the first three terms suffice. The first term has magnitude

$$\frac{729}{128\pi^4} > 0.0569.$$

At $t = 2$, write $b = b_{2,1} = b_{2,2} = b_{2,3}$ and $c = b_{2,4}$. The elementary bounds

$$R_2 > 0.86, \qquad 0.84 < b < 0.85, \qquad c > 0.44$$

give

$$\Phi_{2,1} = \frac{1}{4} + \frac{2b + c}{12} + \frac{b^2 + 2bc}{12} + \frac{b^2 c}{4} > 0.624.$$

Thus the second term has magnitude greater than $0.86 \cdot 0.15 \cdot 0.624 > 0.0804$. At $t = 3$, the bounds

$$R_3 > 0.98, \qquad b_{3,1} < 0.983, \qquad \min_\nu b_{3,\nu} > 0.92$$

give $\Phi_{3,1} > 0.88$, hence a third magnitude greater than $0.98 \cdot 0.017 \cdot 0.88 > 0.0146$. These displayed trigonometric bounds follow directly from the alternating sine/cosine Taylor bounds and $3.14159 < \pi < 3.14160$; their deliberately loose margins avoid any limiting numerical assumption. Therefore

$$-x > 0.1519 > 1/7. \tag{FI14}$$

Let $\Pi(v) = I - vv^{\mathsf{t}}/|v|^2$. Equations (FI12)–(FI14) show

$$|\Pi(v_*)\mathcal{K}(v_*)| > 0. \tag{FI15}$$

Continuity gives an open ball B around v_*, with all coordinates bounded away from zero, on which this norm has a positive lower bound. Shrink the ball so the geometric shells $L^m B$ are disjoint.

58.3 3. Transfer to finite aliases

For $p = L^m v$, $v \in B$, set $\epsilon = L^{m-k}$. Reindexing the actual finite connector CR5 gives exactly

$$L^m K_{a,1,\mu}(L^m v) = \frac{1}{\epsilon^{-1}(e^{i\epsilon v_\mu} - 1)} \frac{1}{R_{k-m}(v)} \sum_{t=1-m}^{k-m} R_t(v)(b_{t,\mu}(v) - 1)\Phi_{t,\mu}(v). \tag{FI16}$$

The denominator R_{k-m} is the correction between the finite centered box and its continuum sinc numerator. It tends uniformly to one. The two omitted tails have bounds CL^{-4m} and $CL^{-2(k-m)}$, while the physical difference symbol tends uniformly to iv_μ. The fine Hodge projector tends to $\Pi(v)$ on the same compact set.

It follows from (FI15)–(FI16) that there are fixed integers m_0, n_0 and $c > 0$, independent of k, such that

$$|\Pi_a(p)K_{a,1}(p)^*| \geq cL^{-m}, \quad p \in L^m B, \quad m_0 \leq m \leq k - n_0. \tag{FI17}$$

Here n_0 can also be chosen so all these momenta stay strictly inside the fine Brillouin zone. The star changes only conjugation and the row/column convention; the same transverse lower bound holds.

58.4 4. A fixed longitudinal force detects those aliases

Choose the retained coarse frequency

$$\theta = (\pi/2, 0, 0, 0) \tag{FI18}$$

and initially use the complex scalar mode $\psi(x) = e^{i\theta \cdot x}$. Set $f_a = d_a \psi$. The real part will give a real counterexample. Its L^p and $W^{-1,p}$ norms are bounded uniformly in a.

Put

$$q_a = E^*_{\mathrm{mix}} \Delta_a \psi. \tag{FI19}$$

It is a single coarse scalar mode at θ. If q_0 is the principal centered box multiplier, λ_0 the principal fine Laplacian symbol, and c_{mix} the mixed normalizer, its coefficient is

$$\widehat{q}_a(\theta) = q_0 \lambda_0^{-1} / c_{\mathrm{mix}}. \tag{FI20}$$

LF11 gives

$$\lambda_0/16 \leq |\widehat{q}_a(\theta)| \leq \lambda_0. \tag{FI21}$$

In particular the coefficient stays bounded away from zero.

Consider a fine alias

$$p = \theta + 2\pi\nu \in L^m B. \tag{FI22}$$

For all sufficiently large m, its last three alias indices are nonzero. Because the last three coarse components of θ are zero, the centered final box multiplier is then exactly zero. Consequently every component of $Q_{\mathrm{bl}}(p)$ is zero. The scalar E_{mix} multiplier at that alias is also zero.

Let w be an arbitrary fine transverse one-form supported at this one alias, so $\delta_a w = 0$. Then

$$z = w + d_a E_{\mathrm{mix}} K_{a,1} w \tag{FI23}$$

belongs to X_a. Indeed, its gradient part is in the exact slice, and

$$Q_{\mathrm{raw}} w = -d_r K_{a,1} w, \qquad Q_{\mathrm{raw}} d_a E_{\mathrm{mix}} K_{a,1} w = d_r K_{a,1} w.$$

No box Schur correction is needed for this test vector.

Apply (FI4) with this z. The gradient part drops out of the curl energy. The forcing pairs trivially with w and pairs with its gradient correction as

$$\langle d_a \psi, d_a E_{\mathrm{mix}} K_{a,1} w \rangle = \langle q_a, K_{a,1} w \rangle.$$

Therefore the fine transverse component of the solution is forced to be

$$\widehat{u}_T(p) = \widehat{\Delta}_a(p)^{-1} \Pi_a(p) K_{a,1}(p)^* \, \widehat{q}_a(\theta). \tag{FI24}$$

This identity is exact for every alias in (FI22). It cannot be canceled by a coarse multiplier: the box multiplier is zero there. The slice gradient also has zero coefficient at that alias.

58.5 5. The logarithmic lower bound

The ball $L^m B$ contains at least cL^{4m} points of the alias lattice $\theta + 2\pi\mathbb{Z}^4$ once $m \geq m_0$. On the admitted shell the physical Laplacian is comparable to L^{2m}. By (FI17), (FI21), and (FI24), each such alias contributes at least

$$cL^{-4m}$$

to the squared homogeneous H^1 norm. Thus each disjoint shell contributes at least a fixed positive constant. Summing $m_0 \leq m \leq k - n_0$ gives

$$\|u_T\|^2_{\dot{H}^1_a} \geq ck - C. \tag{FI25}$$

Fine Hodge components are orthogonal for every scalar Bessel weight, so the full u has at least this norm. Taking the real forcing $\mathrm{Re}\, f_a$ gives the same bound, with a fixed factor, on the two conjugate Fourier fibers. This proves (FI5) and disproves (FI6).

58.6 6. Consequence for the next analytic step

The failure is already flat and linear. A small nonlinear perturbation cannot justify the missing unrestricted inverse. The issue is specific to feeding an arbitrary longitudinal source into a slice tied to retained point values through the actual raw constraint.

The physical Yang–Mills Euler force satisfies a gauge identity. Its admissible residual and multiplier have the compatibility analyzed in CR, which prevents prescribing this fixed longitudinal force

independently of its endpoint charges. An analytic implicit-function theorem must therefore use a compatible codomain, or a different controlled parametrization in which the forbidden longitudinal forcing is not an independent source.

The separate question whether the constrained minimizer with prescribed coarse data has a uniform $W^{1,p}$ right inverse remains meaningful. The exact candidate

$$H_{\mathrm{raw}} = H_{\mathrm{bl}} + d_a E_{\mathrm{mix}} K_{a,r} H_{\mathrm{bl}}$$

does preserve the raw target and the slice whenever H_{bl} is the corresponding box-constrained minimizer in that same slice. Its norm question does not require the false unrestricted estimate (FI6). The following chapter FW proves that positive prescribed-data bound and also proves a hidden inverse bound on co-closed forces. Neither statement reinstates the unrestricted estimate disproved here.

59 Uniform finite-p stability of the actual flat raw reconstruction

The hidden inverse on arbitrary negative-Sobolev forcing fails, as shown in FI. The prescribed-data reconstruction nevertheless has a uniform $W^{1,p}$ bound. The distinction matters for an analytic implicit-function theorem: a bounded right inverse for coarse data does not supply the missing unrestricted Euler inverse.

This chapter proves the positive statement on the exact smoothed Landau slice and for the actual composed raw constraint.

59.1 1. Statement and conventions

Fix the physical torus, a fixed odd L, and

$$2 < p < \infty, \qquad r = L^k a \le 1. \tag{FW1}$$

Let $Q = Q_{\mathrm{bl},a,r}$ be the direct centered box-and-line average, R the scalar site average, and N restriction to retained centers. Write

$$Q_{\mathrm{raw}} = Q - d_r K_{a,r}, \qquad Q d_a = d_r R, \qquad Q_{\mathrm{raw}} d_a = d_r N. \tag{FW2}$$

The scalar lifts are

$$E_{\mathrm{box}} = G_a^2 R^* (R G_a^2 R^*)^{-1}, \qquad E_{\mathrm{mix}} = G_a^2 R^* (N G_a^2 R^*)^{-1}, \tag{FW3}$$

on mean-zero modes, with constants preserved separately. The exact slice is

$$R_0 \delta_a u = 0. \tag{FW4}$$

Theorem FW. There is a unique minimizer of $\frac{1}{2}\|d_a u\|_2^2$ with prescribed actual raw datum $Q_{\mathrm{raw}} u = B$ and slice (FW4). Its linear reconstruction operator H_{raw} satisfies

$$\boxed{Q_{\mathrm{raw}} H_{\mathrm{raw}} = I, \qquad \|H_{\mathrm{raw}} B\|_{W_a^{1,p}} \le C_{p,L,T} \|B\|_{W_r^{1,p}}.} \tag{FW5}$$

The constant is independent of the two spacings and the blocking depth. The proof uses no bound for the false unrestricted hidden inverse.

59.2 2. A componentwise full-gradient reconstruction

First consider

$$C = QG_aQ^*, \qquad H_0 = G_aQ^*C^{-1}. \tag{FW6}$$

These formulas act on nonzero coarse modes. On a coarse constant, define H_0 to return the same fine constant. This H_0 minimizes the full gradient energy, component by component. It is an intermediate operator, not yet the curl-energy minimizer.

We prove

$$\|H_0B\|_{W_a^{1,p}} \le C_p\|B\|_{W_r^{1,p}} \tag{FW7}$$

by its offset kernels. In this section rescale $r = 1$, put $n = r/a$, and use the coarse Fourier variable $z \in [-\pi,\pi]^4$. For a fine alias $p_\nu = z + 2\pi\nu$, let $q_{\mu,\nu}(z)$ be the component box-and-line multiplier and

$$\lambda_\nu(z) = 4n^2\sum_j \sin^2\left(\frac{z_j + 2\pi\nu_j}{2n}\right).$$

For a fine offset $\xi \in n^{-1}\mathbb{Z}^4 \cap [-1/2,1/2]^4$, the value multiplier of H_0 is

$$h_{\mu,\xi}(z) = \frac{\sum_\nu e^{i(z+2\pi\nu)\cdot\xi}\overline{q_{\mu,\nu}(z)}\,\lambda_\nu(z)^{-1}}{c_\mu(z)}, \qquad c_\mu(z) = \sum_\nu |q_{\mu,\nu}(z)|^2\lambda_\nu(z)^{-1}. \tag{FW8}$$

The bar denotes its analytic conjugate extension when z is complex. Formula (FW8) also describes the apparent pole at zero by its removable extension.

Here are uniform bounds for that extension. On a fixed sufficiently narrow complex strip, all finite-order coarse derivatives of the alias filters obey

$$|\partial_z^\beta q_{\mu,\nu}(z)| \le C_\beta\prod_{j=1}^4(1+|\nu_j|)^{-1}. \tag{FW9}$$

There is an additional own-direction line factor, which can be discarded for this bound. The finite geometric-sum formula proves (FW9): a nonprincipal denominator is comparable to $1+|\nu_j|$, including aliases near the fine Brillouin boundary, and the principal quotient is removable.

For $\nu \neq 0$, the same strip has

$$|\lambda_\nu(z)| \ge c(1+|\nu|)^2. \tag{FW10}$$

The inverse and its fixed-order derivatives have the corresponding uniform bounds. The physical fine difference symbol, in retained units, is at most $C(1+|\nu|)$.

Consequently the nonprincipal numerator in (FW8), and its first fine differences, have uniformly bounded analytic sums. The only possibly limiting majorant for a first fine difference is

$$\sum_{\nu\neq0}\frac{\prod_j(1+|\nu_j|)^{-1}}{1+|\nu|} < \infty. \tag{FW11}$$

To check it, on the dyadic shell $|\nu|_\infty \asymp 2^m$ one coordinate contributes a bounded harmonic sum and the other three contribute at most $C(m+1)^3$. Multiplication by 2^{-m} makes the shell sum summable. Coarse differentiation does not worsen this majorant.

Remove the principal pole by writing

$$A_\mu(z) = \lambda_0(z)c_\mu(z) = |q_{\mu,0}(z)|^2 + \lambda_0(z)\sum_{\nu\neq0}|q_{\mu,\nu}(z)|^2\lambda_\nu(z)^{-1}. \tag{FW12}$$

All factors on the right have the analytic extension just described. On the real zone,

$$c \le A_\mu(z) \le C \tag{FW13}$$

uniformly in n: the principal box-and-line factor is bounded below, and the flat average is an L^2 contraction. The uniform derivative bounds allow one fixed narrower complex strip on which A_μ is nonzero.

After multiplying the numerator by λ_0, its principal term is $e^{iz\cdot\xi}\overline{q_{\mu,0}}$; its first fine difference has the same term multiplied by the principal fine difference symbol. This proves that the value and first fine-difference multipliers extend analytically with uniform bounds. They are 2π-periodic: at a coarse Brillouin boundary the finite aliases reindex, and a fine-zone wrap contributes $e^{2\pi i n\xi_j} = 1$.

Thus their coarse convolution kernels obey, in retained units,

$$|k_\xi(z)| + |k_\xi^{(1)}(z)| \le Ce^{-c|z|}. \tag{FW14}$$

At zero coarse momentum, H_0 reproduces constants and its first fine difference vanishes. Equivalently,

$$\sum_z k_\xi(z) = 1, \qquad \sum_z k_\xi^{(1)}(z) = 0. \tag{FW15}$$

For arbitrary physical r, the first fine-difference kernel is $r^{-1}k_\xi^{(1)}$. Subtract $B(y)$ using the second cancellation in (FW15), and telescope $B(y - rz) - B(y)$ along coarse coordinate bonds. The factor r from each difference cancels r^{-1}; the path length is paid by the exponential moment of (FW14). Minkowski's inequality and averaging over all fine offsets prove (FW7), including its L^p part.

On a finite torus, periodize the infinite coarse kernels. The same absolute exponential-moment bound applies before periodization, so the constants do not depend on the number of coarse sites.

59.4 4. The box biharmonic lift and the actual box minimizer

The box lift in (FW3) satisfies

$$\|E_{\mathrm{box}}c\|_{W_a^{2,p}} \le C_p\|c\|_{W_r^{2,p}}. \tag{FW20}$$

This follows directly from the already established mixed-lift theorem. In fact

$$E_{\mathrm{box}} = E_{\mathrm{mix}}\, C_{\mathrm{mix}}C_{\mathrm{box}}^{-1}, \quad C_{\mathrm{mix}} = NG_a^2R^*, \quad C_{\mathrm{box}} = RG_a^2R^*.$$

The coarse multiplier ratio is uniformly analytic on a fixed strip. Multiply numerator and denominator by λ_0^2: their principal terms are respectively q_0 and q_0^2, their nonprincipal tails are absolutely analytic, and their real lower bounds follow from LF11 and the positive box sum. The same strip argument as in (FW12)–(FW14) gives a uniformly summable coarse kernel for the ratio. It commutes with coarse differences. Combining this with LF29–36 proves (FW20), with constants preserved.

Now put

$$H_{\mathrm{bl}}B = H_0B' + d_aE_{\mathrm{box}}c. \tag{FW21}$$

The constraints give $QH_{\mathrm{bl}}B = B' + d_rc = B$. Moreover

$$\delta_aH_0B' = G_aR^*\delta_r(C^{-1}B') = 0$$

on nonzero modes. Its harmonic component is the constant part of B. Thus (FW21) belongs to the exact slice. The multiplier $\lambda = C^{-1}B'$, with its harmonic part set to zero, is co-closed, and

$$\delta_ad_aH_{\mathrm{bl}}B = Q^*\lambda.$$

This proves stationarity for the curl energy against every variation in $\ker Q$. Convexity gives minimality. The scalar slice and $RE_{\mathrm{box}} = I$ remove its remaining gauge freedom, so the representative is unique.

Combining (FW7), (FW17), (FW19), and (FW20) yields

$$\|H_{\mathrm{bl}}B\|_{W_a^{1,p}} \le C_{p,T}\|B\|_{W_r^{1,p}}, \quad 1 < p < \infty. \tag{FW22}$$

59.5 5. The accumulated raw connector in finite p

For the actual fixed-L composition,

$$\|K_{a,r}w\|_{L_r^p} \le C_{p,L}r^2\|D_aw\|_{L_a^p}, \qquad p > 2. \tag{FW23}$$

To see this directly, a one-step endpoint connector at scale ℓ_j has zero constant moment. Subtracting a constant link value and telescoping its fixed-L paths gives $\|\rho_jw_j\|_p \le C_L\ell_j^2\|Dw_j\|_p$. Here $w_j = Q_{\mathrm{bl}}^{(j)}w$,

whose derivative L^p norm is contracted by the positive normalized box-and-line filters. The exact connector recurrence in CR5 restricts each site factor to the final centers. That restriction costs $(r/\ell_j)^{4/p}$. Therefore the total bound is

$$C_L r^{4/p} \sum_j \ell_j^{2-4/p} \|D_a w\|_p \le C_{p,L} r^2 \|D_a w\|_p.$$

The exponent is positive precisely for the stated $p > 2$. No point trace on a limiting continuum Sobolev space is being asserted.

Two elementary coarse inverse differences and $r \le 1$ give

$$\|K_{a,r} w\|_{W_r^{2,p}} \le C_{p,L} \|w\|_{W_a^{1,p}}. \tag{FW24}$$

59.6 6. Exact conversion to the raw minimizer

Define

$$H_{\mathrm{raw}} = H_{\mathrm{bl}} + d_a E_{\mathrm{mix}} K_{a,r} H_{\mathrm{bl}}. \tag{FW25}$$

Both summands belong to the exact slice. By (FW2)–(FW3),

$$Q_{\mathrm{raw}} H_{\mathrm{raw}} = I - d_r K_{a,r} H_{\mathrm{bl}} + d_r N E_{\mathrm{mix}} K_{a,r} H_{\mathrm{bl}} = I.$$

The correction is a fine gradient, so it changes no curvature.

For minimality, take any v with $Q_{\mathrm{raw}} v = B$. Then

$$v' = v - d_a E_{\mathrm{box}} K_{a,r} v$$

has $Qv' = B$ and the same curl as v. The box minimality of $H_{\mathrm{bl}} B$ therefore implies raw minimality of (FW25). The raw gauge identity, $N E_{\mathrm{mix}} = I$, and the exact slice remove the zero-energy gauge freedom, proving uniqueness in that slice. The required injectivity of the mixed scalar normalizer is precisely LF11: on every nonzero coarse mode its symbol is bounded below by the positive principal alias.

Finally, (FW22), (FW24), and the $W^{2,p}$ mixed-lift bound give (FW5). Constant one-forms are reproduced because the connector annihilates constants.

59.7 7. The hidden inverse on the flat physical force space

There is also a positive hidden-inverse statement after imposing the flat gauge identity. Suppose

$$\delta_a f = 0, \qquad f \in W_a^{-1,p}.$$

Its harmonic component pairs to zero with $\ker Q_{\mathrm{raw}}$, because the raw map preserves the physical mean. Set

$$v = G_a(f - P_H f), \qquad \Gamma_{\mathrm{phys}} f = v - H_{\mathrm{raw}} Q_{\mathrm{raw}} v. \tag{FW26}$$

The uniform flat elliptic estimate gives $\|v\|_{W_a^{1,p}} \le C_{p,T} \|f\|_{W_a^{-1,p}}$. It is co-closed and hence belongs to the slice. The other term in (FW26) also belongs to the slice and has the same raw coarse datum, so $\Gamma_{\mathrm{phys}} f \in \ker Q_{\mathrm{raw}} \cap \ker(R_0 \delta_a)$.

For every variation z in that space, the energy-minimizing property of H_{raw} gives

$$\langle d_a \Gamma_{\mathrm{phys}} f, d_a z \rangle = \langle d_a v, d_a z \rangle = \langle f, z \rangle.$$

Thus (FW26) is precisely the hidden inverse restricted to these compatible forcings. The map $Q_{\mathrm{raw}} = Q - d_r K_{a,r}$ is uniformly bounded on $W^{1,p}$ by (FW24) and the flat average bound. Therefore

$$\boxed{\|\Gamma_{\mathrm{phys}} f\|_{W_a^{1,p}} \le C_{p,L,T} \|f\|_{W_a^{-1,p}}, \qquad \delta_a f = 0, \quad 2 < p < \infty.} \tag{FW27}$$

The theorem establishes the prescribed-data part and the flat compatible-force part of the proposed analytic setup. The unrestricted force choice is excluded by FI1–25. A nonlinear implicit-function argument still has to define a codomain that respects the exact covariant identity, and prove that its nonlinear map takes values there. Equation (FW27) alone does not supply that nonlinear identification.

60 A local augmented propagator and a compatible wall expansion

This chapter constructs a replacement wall expansion for a precise flat model. The projected gauge propagator is represented by a local saddle system, so its scalar and vector blocks can be weakened together. The construction gives both the usual weak field bounds and a fractional curvature estimate on protected rows.

The proved uniform estimates use the **box-and-line constraint**, whose gradient identity involves site averaging. They are not estimates for an arbitrary raw node-sampling constraint. In particular, they do not replace the separate analysis of the actual nonlinear raw average, the original background gauge, or mixed determining boundaries. Section 7 states exactly what replacing an interpolation would require in the renormalization argument.

60.1 1. A local Schur realization of the projected propagator

Rescale the retained lattice spacing to one. The fine spacing is $a = 1/n$, with centered odd n, and all periodic tori are compatible. Scalar products include their physical volume factors. Let d, δ be the flat lattice exterior differential and its transpose, and put

$$\Delta = \delta d \quad \text{on scalars}, \qquad \Delta_1 = \delta d + d\delta \quad \text{on one-forms}. \tag{AL1}$$

Let S be the normalized scalar site average and Q the normalized box-and-line one-form average. Thus

$$SS^* = I, \qquad Qd = d_c S. \tag{AL2}$$

The adjoint S^* is block hold. Let a_c be a positive coarse weight bounded above and below by fixed positive constants.

Define R to be the orthogonal projection onto $\Delta \ker S$. Scalar constants cause no difficulty: $\ker S$ contains no nonzero constant, and every inverse below either acts on mean-zero scalars or is part of the displayed constrained saddle system. The projected one-form operator is

$$L = \delta d + dR\delta + Q^* a_c Q. \tag{AL3}$$

The smallest local symmetric realization used here has one fine scalar and one coarse scalar in addition to the one-form:

$$\mathbb{L} = \begin{pmatrix} \delta d + Q^* a_c Q & d\Delta & 0 \\ \Delta\delta & -\Delta^2 & S^* \\ 0 & S & 0 \end{pmatrix}. \tag{AL4}$$

This is a three-type realization; no assertion of uniqueness or an abstract minimal-dimension theorem is intended.

For right side $(f, 0, 0)$, the last two rows say

$$S\lambda = 0, \qquad \langle \Delta v, \Delta\lambda \rangle = \langle \Delta v, \delta A \rangle \quad (v \in \ker S).$$

Hence $\Delta\lambda = R\delta A$, and the first row becomes $LA = f$. Consequently

$$\pi_A \mathbb{L}^{-1} \iota_f = L^{-1}. \tag{AL5}$$

Every entry of (AL4) has finite range at retained scale one. The nonlocal projection has not been discarded; it is the result of eliminating the two scalar types.

For comparison, the constrained biharmonic inverse is

$$B_S = \Delta^{-2} - \Delta^{-2} S^* (S\Delta^{-2} S^*)^{-1} S\Delta^{-2} = \Delta^{-1} R \Delta^{-1}. \tag{AL6}$$

It gives $\lambda = B_S \Delta \delta A$, another verification of the same Schur complement.

60.1.1 A second-order form

The three-type form is convenient for the Schur identity. Its equivalent second-order form is better for local regularity. Introduce

$$\psi = \delta A - \Delta\lambda.$$

The equations become

$$\begin{aligned} (\Delta_1 + Q^* a_c Q)A - d\psi &= f, \\ \Delta\psi + S^*\nu &= 0, \\ -\delta A + \psi + \Delta\lambda &= 0, \\ S\lambda &= 0. \end{aligned} \tag{AL7}$$

Write the local four-type matrix on the left as $\mathbb{A}$. In particular, this formulation does not put a nonlocal projection back into the first row. Its differential entries have order at most two.

For general right sides (f, g, h, k), elimination of ψ reduces (AL7) to (AL4) with right side

$$(f + dh,\ g - \Delta h,\ k). \tag{AL8}$$

This identity will also fix the function spaces in the estimates.

60.2 2. Uniform energy estimates for the boxed system

Here and below constants may depend on the fixed dimension, the positive bounds on a_c, and a fixed Hölder exponent, but not on the fine cutoff or the torus volume.

First,

$$\|A\|^2_{H^1_a} \le C \left(\|dA\|_2^2 + \|R\delta A\|_2^2 + \|QA\|_2^2 \right). \tag{AL9}$$

A short proof also identifies why the boxed constraint matters. Write the fine Hodge decomposition as $A = A_T + d\phi$, with $\delta A_T = 0$, retaining harmonic one-forms in A_T. Put

$$H_S = \Delta^{-2} S^* (S\Delta^{-2} S^*)^{-1}, \qquad \phi = \phi_0 + H_S S\phi, \qquad S\phi_0 = 0.$$

Then

$$\Delta\phi_0 = R\delta A, \qquad \|d\phi_0\|_{H^1} \le C\|\Delta\phi_0\|_2. \tag{AL10}$$

The second inequality follows from the cell Poincaré inequality on $\ker S$, followed by summation by parts. The boxed biharmonic lift, proved by the unsigned version of the LF alias calculation, gives

$$\|dH_S c\|_{H^1} \le C\|d_c c\|_2. \tag{AL11}$$

The coarse H^1 and L^2 norms of a coarse gradient are comparable at spacing one.

For completeness, A_T is controlled by dA and the coarse transverse part of QA. On a coarse alias fiber the principal box-and-line symbol Q_0 is diagonal, has uniformly positive smallest singular value, and satisfies

$$Q_0 d_0 = q_0 d_c, \qquad q_0 \ge c > 0.$$

Thus its map from $d_0^{\perp}$ to the quotient by the coarse gradient has a uniformly bounded inverse: the distance of $Q_0 v$ from $Q_0 \operatorname{span}(d_0)$ is at least its smallest singular value times the distance of v from $\operatorname{span}(d_0)$. All nonprincipal aliases have Laplacian eigenvalue at least a fixed positive constant, and their transverse components are controlled by dA. Since (AL2) makes $Qd\phi$ a coarse gradient,

$$\|A_T\|_{H^1} \le C(\|dA\|_2 + \|QA\|_2). \tag{AL12}$$

At zero coarse momentum, Q acts as the identity on the constant one-form and the same conclusion holds. Equations (AL10)–(AL12), and $QdH_S S\phi = QA - QA_T$, prove (AL9).

The scalar saddle block in (AL4) is uniformly invertible from $H^{-2} \oplus \ell_c^2$ to $H^2 \oplus \ell_c^2$. Indeed $S : H^2 \to \ell_c^2$ has a uniformly bounded right inverse. One may use the smooth site lift SG at retained spacing one. On $\ker S$,

$$\|\lambda\|_{H^2} \le C\|\Delta\lambda\|_2. \tag{AL13}$$

Decompose λ into that right inverse of its prescribed average and a member of $\ker S$, solve the latter by the coercive form $\langle \Delta v, \Delta\lambda\rangle$, and recover the coarse multiplier by testing against the right inverse. This proves the assertion including the scalar constant mode.

The Schur identity (AL5), (AL9), and the bounded maps $d\Delta : H^2 \to H^{-1}$ and $\Delta\delta : H^1 \to H^{-2}$ give a uniform inverse for (AL4). Equivalently,

$$\mathbb{A}^{-1} : \mathcal{Y}_E \longrightarrow \mathcal{X}_E, \quad \begin{cases} \mathcal{X}_E = H^1(A) \oplus L^2(\psi) \oplus H^2(\lambda) \oplus \ell_c^2(\nu), \\ \mathcal{Y}_E = H^{-1} \oplus H^{-2} \oplus L^2 \oplus \ell_c^2. \end{cases} \tag{AL14}$$

The equivalence follows directly from (AL8). These estimates hold on every compatible periodic box, with the same constants.

60.2.1 Adding the coarse constraint

Add a coarse one-form multiplier μ, a $Q^*\mu$ term in the first row of (AL7), and the fifth row $QA = B$. Denote this five-type local matrix by $\mathbb{K}$. The new source and solution spaces add one ℓ_c^2 factor.

The constraint has a uniform inf-sup constant on H^1. An explicit bounded right inverse is

$$E_Q = (I + \Delta_1)^{-1} Q^* [Q(I + \Delta_1)^{-1} Q^*]^{-1}. \tag{AL15}$$

The coarse Schur symbol is bounded below by the contribution of its principal alias: Q_0 is uniformly invertible and $\lambda_0 \le C$ in the coarse Brillouin zone. Therefore $E_Q : \ell_c^2 \to H^1$ is uniformly bounded. It is used here only to prove inf-sup stability; it is not inserted as a new nonlocal entry in $\mathbb{K}$.

The ordinary constrained coercivity argument applied to L, followed by the already solved scalar block, proves the same uniform energy inverse for $\mathbb{K}$. If $G = L^{-1}$, its two one-form output blocks are exactly

$$\begin{aligned} C &= G - GQ^*(QGQ^*)^{-1}QG, \\ H_0 &= GQ^*(QGQ^*)^{-1}. \end{aligned} \tag{AL16}$$

Thus $A = Cf + H_0 B$. The suppressed scalar and coarse multipliers remain blocks of the same inverse.

For $f = 0$, the boxed identity (AL2) also proves that the gauge auxiliary λ vanishes. Apply δ to the first row and use the next two rows to get

$$\Delta^2\lambda + S^*\delta_c(\mu + a_c B) = 0.$$

Pair with $\lambda \in \ker S$. It gives $\Delta\lambda = 0$, then $\lambda = 0$. Hence $H_0 B$ obeys the strict projected Landau condition and is the physical flat constrained minimizer. At this flat endpoint the original quadratic minimizer H and H_0 coincide. This argument is not asserted for the independently weakened maps.

Neither (AL9) nor (AL15) has used raw point restriction. Replacing $Qd = d_c S$ by $Q_{\mathrm{raw}} d = d_c N$ would require a different uniform constrained estimate. The algebraic Schur formula alone supplies no such estimate.

60.3 3. Local regularity and the old weak-map norm

Fix $0 < \beta < 1$. For the four-type system define

$$\begin{aligned} \|\mathbf{u}\|_{\mathcal{X}_\beta} &= \|A\|_{C^{1,\beta}} + \|\Delta_1 A\|_\infty + \|\psi\|_{C^{1,\beta}} + \|\Delta\psi\|_\infty \\ &\quad + \|\lambda\|_{C^{2,\beta}} + \|\nu\|_{\ell^\infty_c}, \\ \|\mathbf{f}\|_{\mathcal{Y}_\beta} &= \|f\|_\infty + \|g\|_\infty + \|h\|_{C^\beta} + \|k\|_{\ell^\infty_c}. \end{aligned} \tag{AL17}$$

Add $\|\mu\|_{\ell^\infty_c}$ and $\|B\|_{\ell^\infty_c}$ for the constrained system. All differences and Hölder quotients use physical lattice spacing. The vector Laplacian in this flat frame acts componentwise.

There is a uniform inverse

$$\|\mathbb{A}^{-1}\mathbf{f}\|_{\mathcal{X}_\beta} \le C_\beta \|\mathbf{f}\|_{\mathcal{Y}_\beta}, \qquad \|\mathbb{K}^{-1}\mathbf{f}\|_{\mathcal{X}_\beta} \le C_\beta \|\mathbf{f}\|_{\mathcal{Y}_\beta}. \tag{AL18}$$

Here is a derivation that does not suppose a volume-uniform L^∞ inverse merely from an L^2 gap.

Conjugate the local matrix by e^ω, using the same smooth physical weight on all types, sampled at their respective sites. Choose $\|D^j\omega\|_\infty \le C_j\gamma$ for the finitely many required orders, and $|\omega(x) - \omega(y)| \le C\gamma$ whenever an averaging entry connects x, y. The difference between the conjugated and original matrices has norm at most $C\gamma$ from $\mathcal{X}_E$ to $\mathcal{Y}_E$. For example the order-three commutator in the symmetric form maps H^2 to H^{-1}, while in (AL7) only second-order commutators are needed. The S, Q commutators cost the difference of the two weight values, over distance at most a fixed retained length.

Choose γ below a fixed multiple of the reciprocal energy-inverse bound. A Neumann argument gives the weighted energy inverse. Using smooth approximations to distance on the periodic box proves, for fixed unit neighborhoods D_i, D_j,

$$\|1_{D_i}\Gamma 1_{D_j}\|_{\mathcal{Y}_E \to \mathcal{X}_E} \le Ce^{-c\,\mathrm{dist}(D_i, D_j)}. \tag{AL19}$$

Local cutoffs are included in this notation. The result is uniform on periodic boxes of every size. Summing unit input neighborhoods in (AL19) gives a uniformly bounded local energy norm for

bounded data. Coarse ℓ^2 output controls each individual coarse value, so ν, and μ when present, are uniformly bounded.

Now use the literal equations in (AL7). The right side of $\Delta\psi = g - S^*\nu$ is bounded. Uniform interior lattice Poisson estimates give ψ in $C^{1,\beta}$, with its local L^2 norm included. The first row then has a bounded right side: $Q^*(a_cQA+\mu)$ is bounded because Q^* is normalized hold followed by line averaging, and the unit-block energy bound controls each value of QA. It gives the asserted bounds for A. Finally

$$\Delta\lambda = h + \delta A - \psi$$

has a C^β right side, giving the $C^{2,\beta}$ bound for λ. The local energy norm controls the additive scalar constant.

The constant-coefficient lattice Poisson estimates used here are cutoff-uniform. They follow, for example, by writing the localized solution with the lattice Green kernel: the second-difference kernel has order $|x|^{-d}$ and zero mean, so subtracting the local value of a C^β source makes its sum bounded by $\int_0^1 t^{\beta-1}dt$. The near/far split of the first-difference kernel gives every C^β modulus below one for a bounded source. Smooth physical cutoffs contribute only fixed local energy terms. This argument also applies on the periodically repeated tile, before restricting to its interior.

The same proof with input supported near D_j retains the exponential in (AL19). Thus (AL18) holds with a summable exponentially localized operator majorant. It proves the displayed weak-map norm, including the Laplacian supremum. No assertion about an additional unlisted norm in the background-field paper is being made.

60.5 5. Fractional regularity on the protected set

The common expansion gives more than a weak-core supremum estimate. Let $0 < s < 1/2$. At retained spacing one,

$$\|S^*v\|_{H^s_a} + \|Q^*b\|_{H^s_a} \le C_s(\|v\|_{\ell^2_c} + \|b\|_{\ell^2_c}). \tag{AL28}$$

For block hold,

$$\|\tau_h S^*v - S^*v\|_2^2 \le C\min(1, |h|)\|v\|_2^2.$$

Only a slab of relative volume $O(|h|)$ changes blocks. Integrating the fractional-difference formula gives (AL28) precisely for $s < 1/2$, uniformly down to the fine spacing. The adjoint line average is an average of translations, so it does not increase that norm.

Suppose the protected equations for a background column are

$$\begin{aligned} \delta dA + d\Delta\lambda + Q^*\mu' &= f, \\ \Delta(\delta A - \Delta\lambda) + S^*\nu &= 0. \end{aligned} \tag{AL29}$$

Here $\mu' = \mu + a_c QA$, and the other protected rows are retained. Let $B \Subset B_+$, with a fixed positive physical collar and with B_+ containing the required retained cells. From $\psi = \delta A - \Delta\lambda$,

$$\Delta\psi = -S^*\nu, \qquad \Delta_1 A = f - Q^*\mu' + d\psi. \tag{AL30}$$

Interior lattice elliptic estimates, first for ψ and then for A, give

$$\begin{aligned} &\|A\|_{H^{2+s}_a(B)} + \|dA\|_{H^{1+s}_a(B)} + \|\lambda\|_{H^{3+s}_a(B)} \\ &\qquad \le C_{s,B,B_+}\big[\|\mathbf{u}\|\chi_{E(B_+)} + \|f\|_{H^s_a(B_+)} + \|\mu'\|_{\ell^2_c(B^*_+)} + \|\nu\|_{\ell^2_c(B^*_+)}\big]. \end{aligned} \tag{AL31}$$

Use successively nested interior sets. The final bound for λ follows from $\Delta\lambda = \delta A - \psi$. Smooth cutoffs in the elliptic proof contribute lower-order terms controlled by the local energy norm; no derivative of a step-function hold is taken pointwise.

There is also a direct curvature proof. With $F = dA$, apply d to the first equation in (AL29):

$$\Delta_2 F = df - dQ^*\mu'. \tag{AL32}$$

The gauge auxiliary disappears because $d^2 = 0$. Equation (AL28), the L^2 curvature norm and interior elliptic regularity give $F \in H^{1+s}$. This proves the claimed cutoff-uniform fractional curvature regularity even when λ is nonzero in the weakened system.

This conclusion is stronger than multiplying an old supremum error by a small exponential. It follows from the actual protected equations. It is also specific: replacing Q^* in (AL28) by an uncontrolled raw endpoint adjoint would invalidate this proof.

60.5.1 A scalar residual bound away from its actual support

The augmented scalar variables can also give a concrete residual estimate when exact protected equations are unavailable on a smaller set. For example, suppose a scalar λ obeys

$$\Delta^2\lambda = 0 \quad \text{on } B_R, \qquad \|\lambda\|_{H^2(B_R)} \le \varepsilon.$$

Then interior biharmonic regularity gives

$$\|d\Delta\lambda\|_{H^s(B_{R/2})} \le C_{s,R}\varepsilon. \tag{AL33}$$

This is uniform in the fine cutoff when R is a fixed physical distance. After scaling, its derivative price is a fixed power of R^{-1}, not of the microscopic spacing. If the forcing is $S^*\nu$ rather than zero, add its actual H^{s-1} norm to the right side.

The support qualification is essential. A nonzero raw point embedding N^*q at a retained center inside B_R is not excluded by separation from the wall alone. Its high-frequency norm cannot be bounded by a fixed wall exponential without further compatibility. The later CR theorem addresses such terms when they are part of an **actual** H^s Wilson Euler residual. It does not turn an a priori low-norm scalar defect into an H^s defect.

60.6 6. What a nonlinear augmented construction would have to prove

The flat construction suggests using one nonlinear augmented equation

$$\mathbf{u}_s = \Gamma_s[\mathbf{b} - \mathbf{N}(\mathbf{u}_s)] \tag{AL34}$$

instead of independently weakening the several propagators in a chosen representation. If $\mathbf{N}$ is genuinely local, holomorphic on a common weak ball, and has the same small Lipschitz constant in the displayed typed norm, the usual fixed point gives:

1. holomorphy in the wall polydisc and in the field variables;
2. the old weak-map bounds, with fixed spare margins;
3. exact nonlinear augmented equations on protected rows, by (AL25);
4. component dependence at zero walls, by the same convergent iteration.

This implication is an elementary fixed-point theorem. To make it a theorem about the Wilson reconstruction, its premises must be checked for the **actual** constraint, gauge coordinate map, and nonlinear force.

For a fractional upgrade, an adequate quantitative test is the following local tame form after subtracting the flat principal matrix:

$$\begin{aligned} \Delta_1 A &= f - Q^*\mu' + d\psi + \mathcal{R}_1, \\ \Delta\psi &= -S^*\nu + \mathcal{R}_2, \\ \Delta\lambda &= \delta A - \psi + \mathcal{R}_3. \end{aligned} \tag{AL35}$$

On a common weak chart, the remainders must be controlled in H^s, H^{s-1}, and H^{1+s}, respectively, by the known incoming strong data plus a small multiple of the strong output norms in (AL31). The local elliptic estimates then absorb that small multiple. The weak $C^{1,\beta}$ norm by itself does not establish these three estimates. Covariant commutators, current terms of the Hessian, and nonlinear constraint adjoints must all be included.

If one obtains a physical field u_s and an actual coarse multiplier $\lambda_s^{\mathrm{phys}}$, the more general usable target is

$$E_s = J_{\mathrm{Wilson}}(u_s) - (D\mathcal{M}_r(U_s))^{\mathsf{t}}\lambda_s^{\mathrm{phys}} \quad \text{with a uniform } H^s \text{ bound.} \tag{AL36}$$

CR proves that its exact gauge compatibility controls the resulting nonzero coarse Ward defect. There is therefore no additional independent exact-Ward assumption in this target. The remaining work is to derive the H^s bound for the actual complete residual, and the common weak chart, from the proposed augmented construction.

For a different locally defined gauge projection, the algebraic counterpart of (AL4) is immediate once that projection is specified as the projection onto the image of a local scalar operator on a local

constraint kernel. It is not legitimate to identify this new gauge with the original background-dependent projection without a proof. A change of gauge at the physical endpoint must preserve the actual coarse target and the gauge frame of the term being localized, or be accompanied by its explicit compensating transformation.

60.7 7. Replacing an interpolation in the renormalization expansion

There is a useful separation between an interpolation used to decompose an already defined function and a change of physical integration variables.

Let F be an actual finite-cutoff integrand term. Suppose a proposed family $F(s)$ has the following verified properties:

- $F(1) = F$, including every fixed chart, gauge-frame factor, Jacobian and coarea factor present in that term;
- $F(s)$ is holomorphic and uniformly bounded on the required complex wall discs and on the same admitted field domain;
- at zero walls it depends only on the prescribed components, with the actual enlarged carriers and boundary collars;
- its absolutely summable differentiated carrier majorants have the decay and reserve used by the original cluster argument.

Then the finite fundamental theorem of calculus identity

$$F(1) = \sum_{Y \subseteq W} \int_{[0,1]^Y} (\partial_Y F)(s_Y, 0_{W \setminus Y})\, ds_Y \tag{AL37}$$

is exact. Component dependence organizes the surviving derivatives by their actual components; it does not make arbitrary mixed derivatives of a product vanish. For rooted connected or logarithmic quantities, disconnected attachments disappear only after the corresponding connected expansion is taken. Cauchy on the stated discs and the supplied carrier majorants then give the localized bounds. This proof uses no equality between the old and new intermediate values $F(s)$.

In particular, when (AL37) is applied to an **already constructed** integrand, no change of integration variables is performed at $s \neq 1$. A new Jacobian or coarea identity at every interpolated value is not required. The physical factors at $s = 1$, and their holomorphic extensions used inside $F(s)$, must nevertheless be exactly the factors of the original term. Interpolating an unrelated gauge representative while omitting its frame or coarea compensation would not satisfy the first condition.

This is the role of Bałaban II, CMP **116** (1988), (1.2)–(1.18): it treats the background solution as a functional of bounded analytic propagators and applies the fundamental theorem of calculus to obtain localized terms. Its (1.5) permits substitute operators with the required regularity bounds. That permission does not itself prove bounds for a new augmented nonlinear map; the explicit verification above covers only the flat boxed system.

A complete replacement proof must also address two consequences after the first localization:

- Absolute convergence allows linear marginal and vacuum extraction to be applied to the same total function. It does not automatically identify individual polymer activities or their historical origin records.
- If a later large-field preparation uses those activities or their supports, its generic incoming norm and carrier hypotheses must be checked for the new localized representation. The physical tests

and positive reference kernels should remain the original ones unless an additional exact change is proved.

Thus changing the interpolation is mathematically permissible in principle, and it avoids requiring equality to every old intermediate map. Endpoint equality, a common domain, factorization and quantitative local majorants are the actual obligations. The flat construction (AL4)–(AL32) supplies a concrete example satisfying these analytic obligations for its stated system.

60.8 8. Result and remaining identification

The flat result is complete within its scope:

- a local three-type Schur realization of the projected propagator;
- an equivalent second-order realization with uniform energy and old weak-map bounds;
- a common complete-carrier wall expansion for all scalar, vector and coarse blocks;
- exact protected augmented equations and uniform H^{1+s} curvature for $s < 1/2$;
- an explicit criterion allowing a replacement interpolation without new intermediate integration Jacobians.

What remains is not a formal carrier identity. It is the actual nonlinear and multiscale matching: the original gauge/constraint coordinates, their common complex domain, the complete nonlinear residual in the norms of (AL35) or (AL36), and the mixed preparation boundaries. The proven boxed system does not establish those statements for the raw hierarchy. The new real periodic weak-chart theorem WC helps at the physical real endpoint; it does not supply the auxiliary complex wall family.

61.1 1. A periodic value theorem without the anchor

Fix the physical torus and the common input ball from CX,

$$\mathcal{B} = \{A : \|A - A_0\|_{\mathcal{Y}_a} < \rho\}, \qquad m = \varepsilon + \rho. \tag{IR1}$$

CX5–20 gives a holomorphic representative $U_c(A)$ of the actual critical orbit with

$$U_c(A) = e^{av(A)}, \qquad \sup_{A\in\mathcal{B}} \Big(\|v(A)\|_{W_a^{1,p}} + \|F_a(U_c(A))\|_\infty \Big) \le Cm. \tag{IR2}$$

It need not have the originally prescribed raw coarse representative. The source supplies its complex gauge relation to that representative, but the norm of that gauge at an anchor has not yet been bounded.

Let P be a fixed homogeneous invariant trace polynomial of degree q, with all its curvature arguments based at the same site. For example,

$$P(F)(x) = \sum_\sigma c_\sigma \operatorname{tr}\big(F_{\sigma_1}(x) \cdots F_{\sigma_q}(x)\big). \tag{IR3}$$

The contractions are the holomorphic invariant contractions. A Hermitian expression such as $\operatorname{tr}(F^\dagger F)$ is not substituted for its holomorphic complexification.

For any complex fine gauge g, exact covariance gives

$$F(U^g)(x) = g(x)F(U)(x)g(x)^{-1}, \qquad P(F(U^g))(x) = P(F(U))(x). \tag{IR4}$$

There is no factor involving $\|g\|$, $\|g^{-1}\|$, or their derivatives in the second equality. Thus the scalar function

$$\mathcal{O}_P(A)(x) = P(F(U_c(A)))(x)$$

is the value of the same readout on the actual raw critical representative, whenever that readout is defined there by its invariant analytic branch. It obeys

$$\boxed{\sup_{A\in\mathcal{B}} \|\mathcal{O}_P(A)\|_\infty \le C_P m^q.} \tag{IR5}$$

The constant is independent of the fine cutoff, retained scale and blocking depth. It depends on the fixed polynomial, group and the fixed physical-domain constants already present in CX.

For logarithmic curvature, similarity preserves the plaquette spectrum. The branch may be defined by analytic matrix functional calculus on the conjugacy-invariant spectral neighborhood corresponding to the small branch in the controlled representative. One need not require a large conjugate matrix to remain in a small matrix-norm ball. An entire lattice curvature such as the appropriately normalized traceless part of $(U_p - U_p^{-1})/(2i)$ has no logarithm issue and obeys the same conclusion, with the fixed small-branch comparison constant.

The same argument permits curvature factors at different sites if they are transported to one common basepoint along specified paths. In the controlled representative,

$$\|U_\gamma\| + \|U_\gamma^{-1}\| \le C \exp(Cm \operatorname{length}(\gamma)).$$

Consequently (IR5) acquires only the corresponding physical path-length factor. A bound for an arbitrary growing carrier is not inferred from a bound for fixed path lengths.

The polynomial coefficients in this statement are fixed. A cutoff-dependent normalization or mixing matrix must have its own norm included; (IR5) does not establish that normalization.

All colored arguments of the readout must participate in the same gauge transformation. A pair expression such as $\operatorname{tr}(F(U)J_{\mathrm{ref}})$ is invariant under a joint transformation of U, J_{ref}, but need not be invariant when only U is replaced by its controlled representative and the reference current is held fixed. General pair activities therefore do not enter (IR5) merely because they are called gauge invariant.

61.1.1 Parameter derivatives and integration

If a holomorphic parameter family $A(z)$ remains in $\mathcal{B}$ on $|z_i| < \rho_i$, ordinary Banach-valued Cauchy gives

$$\|\partial_z^{\gamma}\mathcal{O}_P(A(z))|_{z=0}\|_{\infty} \le C_P m^q\, \gamma! \prod_i \rho_i^{-\gamma_i}. \tag{IR6}$$

The gauge relating representatives may depend on z. Its derivative is not estimated or discarded: the scalar functions are identical for every z, so their derivatives agree.

For a fixed lattice test f,

$$\left| a^4 \sum_x f(x)\mathcal{O}_P(A)(x) \right| \le C_P m^q \|f\|_{L^1_a}. \tag{IR7}$$

Likewise, if an original integration kernel evaluates this readout on verified members of the periodic orbit family, its pointwise bound can be used inside that **unchanged** kernel. No integration variable is transformed, and no new Jacobian is needed for this inequality. This does not cover arbitrary fine fields inside a live island or an auxiliary background whose membership in that orbit family has not been proved.

61.2 2. What gauge invariance does not supply

There are three distinct derivative statements.

61.2.1 Ordinary derivatives of matrix curvature

An ordinary derivative of gFg^{-1} contains derivatives of g. Even a constant unbounded complex conjugation can enlarge the matrix norm without altering any invariant trace. Therefore (IR5) does not bound the raw representative's componentwise H^{1+s}, C^{β}, or other matrix-valued curvature norm.

CX22 is part of the route to a controlled same-target representative. It is unnecessary for the scalar equality (IR4), but scalar equality does not prove that representative theorem.

61.2.2 Spatial derivatives of the scalar readout

The scalar functions in (IR4) agree at every site. Their ordinary finite differences, and any spatial norm of those scalar functions, therefore agree as well. No anchor is needed to **transfer an already proved scalar derivative estimate**.

However, CX5–20 provides only the displayed weak bound. It does not itself bound those scalar derivatives. The distinction is concrete. In a commuting color direction T, take

$$A_2(x) = \frac{\epsilon}{N}\sin(Nx_1)T, \qquad A_\mu = 0 \ (\mu \neq 2), \tag{IR8}$$

on meshes resolving N. Then the weak potential norm and curvature supremum stay $O(\epsilon)$, whereas

$$\|D_1 \operatorname{tr}(F_{12}^2)\|_{L^p} \asymp N\epsilon^2.$$

For commuting links the plaquette logarithm is exactly the lattice curl, so the example is also a lattice calculation. It demonstrates the failure of the asserted weak-to-strong norm implication, not a failure of a reconstruction estimate with stronger input data.

61.2.3 Covariant derivative insertions

The exact lattice covariant difference satisfies

$$\nabla_\mu^U F(x) = a^{-1}\left[U_\mu(x)F(x+ae_\mu)U_\mu(x)^{-1} - F(x)\right],$$

and

$$\nabla_\mu^{U^g} F(U^g)(x) = g(x)\nabla_\mu^U F(U)(x)g(x)^{-1}. \tag{IR9}$$

Repeated covariant differences have the same covariance, with the required orientations and transports retained. Completed trace polynomials in $F, \nabla_U F, \dots$ are therefore invariant under an arbitrary complex gauge, just as in (IR4).

Their scalar value equality also needs no anchor. Their **uniform estimates** require bounds for the covariant derivatives in some representative. CX20 does not supply them: its direct finite-cutoff estimate for a first covariant difference costs a^{-1}. A separately proved H^{1+s} curvature estimate in a controlled representative would, for example, bound suitable integrated quadratic first-derivative readouts. Pointwise high derivatives and arbitrary products need the corresponding additional regularity and integrability estimates.

For real physical inputs, WC together with the qualified periodic reconstruction theorem provides such a stronger representative when the input has the stated strong norm. That is a separate theorem. Real bounds alone do not extend the stronger estimate to a complex parameter domain.

61.4 4. Coarea, charts and complete integrands

For the readout factor alone, equality of the physical **orbit** at the endpoint suffices. There is no need to bound the connecting complex gauge in order to use (IR4).

A complete formation integrand can also contain fluctuation coordinates, frame-dependent boundary data, source-free tests, determinants and coarea factors. Scalar invariance of one readout does not identify those factors. Replacing a representative inside the complete integrand still requires its actual coordinate/gauge identities, or the exact endpoint equality specified in AL Section 7.

A field-dependent choice of gauge is not automatically a measure-preserving change of all integration variables. That issue is absent when one merely uses the pointwise invariant value bound inside the original kernel, as in (IR7). It returns if the coordinate formula or the kernel itself is replaced.

The conclusion is therefore precise: **CX22 is eliminated for the periodic value theorem (IR5), and for Cauchy derivatives of that scalar value.** It is not eliminated from the same-target representative theorem by this argument. Nor does the argument prove uniform covariant derivative insertion bounds, wall factorization, prepared-domain membership, or a full coarea/integrand comparison.

61.5 Inputs

The only physical-orbit inputs in (IR1)–(IR7) are WC and CX5–20, with their explicitly imported periodic regular-orbit theorem and analytic reconstruction. AL1–32 is used only for the separate flat statements in Section 3. All gauge identities in this note are exact matrix identities; no new source theorem or nonlinear wall identification is imported.

62 A compatible nonlinear KKT formulation with a uniform weak analytic inverse

The unrestricted hidden inverse in FI is false. A nonlinear implicit-function argument must remove the endpoint current before posing its off-solution Euler equation. This chapter gives such a formulation. Its additional Ward equation, together with the exact Noether identity, restores the original Euler equation at every solution.

The conclusion is a holomorphic same-target $W^{1,p}$ critical branch. No uniform curvature supremum or strong fractional norm is asserted here.

62.1 1. Spaces and statement

Fix a physical four-torus, an odd blocking factor L, and $8 < p < \infty$. Put $q = p' = p/(p-1)$. All spaces use physical volume. The negative norm $W^{-1,p}$ is dual to $W^{1,q}$; similarly $W^{-2,p}$ is dual to $W^{2,q}$. Mean-zero inverses retain the fixed physical torus constants. Every transpose in a complex nonlinear identity is the invariant bilinear transpose; absolute Hermitian norms are used only for estimates.

Let

$$P_a = I - d_a G_a \delta_a \tag{CK1}$$

be the flat projection onto co-closed one-forms, including harmonic one-forms. Let P^0_{sc} remove scalar constants on the coarse lattice.

The unknowns are

$$\begin{aligned} &u \in W^{1,p}_a, && \delta_a u = 0, \\ &\chi \in W^{2,p}_r, && \overline{\chi} = 0, \\ &\lambda \in W^{-1,p}_r. \end{aligned} \tag{CK2}$$

The coarse scalar χ is a gauge parameter. It will restore the exact raw target after solving in the fine Coulomb representative.

Theorem CK. There is a fixed neighborhood of zero in $W^{1,p}_r$, independent of a, r, k, on which the actual raw target e^{rB} has a holomorphic critical branch

$$B \longmapsto (w(B), \lambda_B(B)), \qquad w(0) = 0,$$

with

$$\begin{aligned} &\mathcal{M}_r(e^{aw(B)}) = e^{rB}, \\ &J_a(w(B)) = D\mathcal{M}_r(e^{aw(B)})^{\mathsf{t}} \lambda_B(B), \\ &d^{\mathsf{t}}_{e^{rB}} \lambda_B(B) = 0, \\ &\|w(B)\|_{W^{1,p}_a} + \|\lambda_B(B)\|_{W^{-1,p}_r} \le C \|B\|_{W^{1,p}_r}. \end{aligned} \tag{CK3}$$

The branch is real on real inputs. Its uniqueness is initially the uniqueness of the small coordinates (CK2) constructed below. The theorem does not, by itself, identify every stationary orbit or prove that an arbitrary stationary field is in this neighborhood.

62.2 2. Remove endpoints before taking the multiplier adjoint

Along the actual raw hierarchy let $U_{j+1} = M_j(U_j)$, with spacings h_j and $\ell_j = h_{j+1} = Lh_j$. Denote its physical differential by $\mathcal{Q}_j$. The exact full-space endpoint decomposition PG15d is

$$\mathcal{Q}_j = \mathcal{B}_j - d_{U_{j+1}}\rho_j + \mathcal{R}_j.$$

Define

$$\mathcal{A}_j := \mathcal{Q}_j + d_{U_{j+1}}\rho_j = \mathcal{B}_j + \mathcal{R}_j, \qquad S_j := N_j + \rho_j d_{U_j}. \tag{CK4}$$

These are analytic finite-step operators. The raw gauge chain gives

$$\mathcal{A}_j d_{U_j} = d_{U_{j+1}} S_j. \tag{CK5}$$

The scalar S_j is an actual covariant site average. For a pure gauge variation of an endpoint transport $P(y,x)$, the endpoint logarithmic variation is $\operatorname{Ad} P(y,x)\phi(x) - \phi(y)$, with the common sign fixed by the convention for d_U. Averaging this identity in the definition of ρ_j gives $S_j\phi$ as the normalized average of the transported endpoint values. If an endpoint transport is itself an admitted nonlinear group mean, its gauge variation uses the adjoint of that actual mean transport, not an arithmetic mean of the path adjoints. The remaining normalized endpoint/path averages are exactly those in the actual definition of ρ_j. Every resulting transport coefficient is $I + O(\ell_j M)$, which is the property used below.

Compose the endpoint-removed and scalar operators:

$$\mathcal{A}(u) = \mathcal{A}_{k-1}\cdots\mathcal{A}_0, \qquad S(u) = S_{k-1}\cdots S_0, \qquad V(u) = \mathcal{M}_r(e^{au}).$$

Then

$$\mathcal{A}(u)d_U = d_{V(u)}S(u). \tag{CK6}$$

At zero, $\mathcal{A}(0) = Q_{\mathrm{bl}}$ and $S(0) = R$, the direct scalar site average.

If $d^{\mathsf{t}}_{V(u)}\lambda = 0$, the complete intermediate currents obtained by propagating λ through $\mathcal{A}^{\mathsf{t}}_j$ satisfy their Ward equations by (CK5). The removed endpoint term then vanishes at every step. Consequently

$$\mathcal{A}(u)^{\mathsf{t}}\lambda = D\mathcal{M}_r(e^{au})^{\mathsf{t}}\lambda \quad \text{when } d^{\mathsf{t}}_{V(u)}\lambda = 0. \tag{CK7}$$

This identity is asserted only after the Ward equation has been established.

For a small $W^{1,p}$ physical chart of radius M, the actual intermediate charts have radius CM, by WP. A path at step j has physical length $O_L(\ell_j)$; its transport differs from the identity by $O_L(\ell_j M)$. Since the nested site blocks partition the final block, $S(u)^{\mathsf{t}}$ sends a coarse constant color c to transported values satisfying

$$\|S(u)^{\mathsf{t}}c - c\|_\infty \le C_L r M |c|. \tag{CK8}$$

In particular its finite-dimensional physical mean map on constant colors is $I + O(rM)$.

62.3 3. The endpoint-removed operator has the correct negative-norm bound

The required estimate is

$$\|\mathcal{A}_j - Q_{\mathrm{bl},j}\|_{W^{1,q}_{h_j}\to W^{1,q}_{\ell_j}} \le C_{p,L,T}\ell_j(M + M^2). \tag{CK9}$$

It does not require a curvature supremum.

First, $W^{1,p}$ coefficients multiply $W^{1,q}$, uniformly on the lattice:

$$\|cv\|_{W^{1,q}} \le C_{p,T}\|c\|_{W^{1,p}}\|v\|_{W^{1,q}}. \tag{CK10}$$

For the derivative falling on c, choose $1/\gamma = 1/q - 1/p$. The discrete Sobolev embedding $W^{1,q} \subset L^\gamma$ holds since $p > 4$; the other derivative term uses $W^{1,p} \subset L^\infty$.

Each middle-line transport coefficient in $\mathcal{B}_j - Q_{\mathrm{bl},j}$ differs from the identity in $W^{1,p}$ by $C_L\ell_j M$. A discrete difference of a path product is the exact telescoping sum of differences of its link factors. Their physical lengths add to $O_L(\ell_j)$, so no inverse spacing is lost. The same assertion holds for an admitted finite analytic endpoint mean. Apply (CK10) to its finite shifted input values. Fixed-L sampling has only a fixed norm cost. This proves the middle-line part of (CK9).

For the full-space error $\mathcal{R}_j$, its coefficient matrices have L^p norm bounded by

$$C_L\ell_j^2\|F_{h_j}(U_j)\|_{L^p} \le C_L\ell_j^2(M + M^2). \tag{CK11}$$

The first bound is the same exact relative-contour expansion used in PG15b–d, with the plaquette term kept in L^p. The second follows from the exact plaquette BCH expansion and the intermediate $W^{1,p}$ chart. The expansion is on a uniform branch: fixed-L point inversion bounds its dimensionless loop argument by $C_L\ell_j^{2-4/p}(M + M^2)$, which is small uniformly for $\ell_j \le 1$.

Pair a coefficient in L^p with a shifted input in L^γ, and use the embedding in (CK10). This bounds $\mathcal{R}_j v$ in L^q by $C\ell_j^2(M + M^2)\|v\|_{W^{1,q}}$. One elementary coarse inverse difference costs ℓ_j^{-1}, proving its $W^{1,q}$ bound in (CK9). No derivative of curvature is used.

Use the componentwise $W^{1,q}$ norm given by the q-sum of the L^q norms of every spatial component and every first physical difference. The flat normalized filters contract this norm with literal coefficient one. Iterate (CK9), using $\sum_j \ell_j \le C_L r$, to obtain

$$\begin{aligned} &\|\mathcal{A}(u)\|_{W_a^{1,q}\to W_r^{1,q}} \le C, \\ &\|\mathcal{A}(u) - Q_{\mathrm{bl}}\|_{W_a^{1,q}\to W_r^{1,q}} \le Cr(M + M^2). \end{aligned} \tag{CK12}$$

All finite-step maps and intermediate fields are holomorphic on a common $W^{1,p}$ ball. The preceding bounds hold there for absolute norms, hence also give uniform Banach-valued Cauchy estimates. In particular

$$\mathcal{A}(u)^{\mathsf{t}} : W_r^{-1,p} \longrightarrow W_a^{-1,p} \tag{CK13}$$

is a bounded holomorphic operator family. This is the precise adjoint estimate that an anchored quotient constraint alone would not supply.

62.4 4. The compatible nonlinear map

Let $\mathcal{E}_a(u)$ be the pointwise physical exponential differential: a logarithmic tangent v corresponds to physical tangent $\mathcal{E}_a(u)v$. It is $I + O(au)$, and its coefficients and inverse are uniformly $W^{1,p}$ on the small chart.

Define the following map on (CK2):

$$\mathcal{F}(u,\chi,\lambda) = \begin{pmatrix} P_a\mathcal{E}_a(u)^{\mathsf{t}}[J_a(u) - \mathcal{A}(u)^{\mathsf{t}}\lambda] \\ P_{\mathrm{sc}}^0 d_{V(u)}^{\mathsf{t}}\lambda \\ r^{-1}\log(V(u)^{e^\chi}) \end{pmatrix}. \tag{CK14}$$

Its target space is

$$(P_a W_a^{-1,p}) \;\times\; (P_{\mathrm{sc}}^0 W_r^{-2,p}) \;\times\; W_r^{1,p}. \tag{CK15}$$

We solve $\mathcal{F}(u,\chi,\lambda) = (0,0,B)$.

All three components are bounded holomorphic maps on a common small ball. The actual raw map and coarse gauge action are uniformly analytic in these $W^{1,p}$, $W^{2,p}$ spaces by WP and GS. The covariant-minus-flat divergence is a zero-order shifted multiplier with small $W^{1,p}$ coefficients; (CK10), by duality, bounds it on $W^{-1,p}$, hence into $W^{-2,p}$.

For the Wilson force, the exact plaquette expansion gives

$$\|F_a(u) - d_a u\|_{L^p} \le C\|u\|_{W^{1,p}}^2.$$

The flux correction is uniform without an F_∞ premise:

$$a^4\|F_a(u)^3\|_{L^p} \le a^{\,4-8/p}\|F_a(u)\|_{L^p}^3 \le C\|u\|_{W^{1,p}}^3.$$

Its analytic higher terms have the same small dimensionless branch. The incidence differs from the flat divergence by a zero-order coefficient $O(u)$. Thus

$$\|J_a(u) - \delta_a d_a u\|_{W_a^{-1,p}} \le C\|u\|_{W_a^{1,p}}^2. \tag{CK16}$$

The same uniform holomorphic bounds give the corresponding derivative estimates. Together with (CK12)–(CK13), these prove the asserted mapping property of (CK14).

62.5 5. The full flat block derivative is uniformly invertible

At the origin the derivative of (CK14) is

$$\mathcal{L}(u,\chi,\lambda) = \begin{pmatrix} \Delta_a u - P_a Q_{\mathrm{bl}}^* \lambda \\ \delta_r \lambda \\ Q_{\mathrm{raw}} u - d_r \chi \end{pmatrix}, \qquad \delta_a u = 0, \quad \overline{\chi} = 0. \tag{CK17}$$

It is an isomorphism between (CK2) and (CK15), with

$$\|\mathcal{L}^{-1}(f,q,b)\| \le C(\|f\|_{W_a^{-1,p}} + \|q\|_{W_r^{-2,p}} + \|b\|_{W_r^{1,p}}). \tag{CK18}$$

Here is an explicit inverse argument. First set $\lambda_L = d_r G_r q$, bounded in $W_r^{-1,p}$. Move $P_a Q_{\mathrm{bl}}^* \lambda_L$ to the forcing side. This is bounded because the flat box map is bounded on $W^{1,q}$. The mean Euler equation determines the harmonic multiplier as minus the mean of this corrected forcing; the mean constraint determines $u_H = b_H$.

For the remaining mean-zero transverse system, write $v = G_a f_T$. The fixed-torus flat elliptic estimate gives $v \in W_a^{1,p}$. With $c = b - Q_{\mathrm{bl}} v$, the transverse multiplier is

$$\lambda_T = C^{-1} P_C \Pi_r^0 c, \tag{CK19}$$

using the C and weighted coarse Hodge projector of FW; Π_r^0 denotes the mean-zero flat coarse co-closed projection. The operator C^{-1} is uniformly of order two: FW12 writes it as $\lambda_0 A^{-1}$, and the quotient λ_0/λ_r has uniform coarse multiplier bounds. Hence $C^{-1} : W_r^{1,p} \to W_r^{-1,p}$ is bounded. The transverse potential is $v + H_0 P_C \Pi_r^0 c$, controlled by FW7 and FW17.

The remaining difference $Q_{\mathrm{raw}}u - b$ has zero transverse and harmonic components. Set

$$\chi = G_r\delta_r(Q_{\mathrm{raw}}u - b).$$

FW24 bounds Q_{raw} on $W^{1,p}$, and the coarse elliptic estimate gives $\chi \in W^{2,p}_r$. This proves (CK18). The explicit Hodge decomposition also proves uniqueness.

The uniform holomorphic bounds in Section 4 imply

$$\|D\mathcal{F} - \mathcal{L}\| \le C\mathfrak{m}$$

on a coordinate ball of radius $\mathfrak{m}$, independently of the cutoffs. Applying $\mathcal{L}^{-1}$ and choosing one small radius gives a contraction, or equivalently the quantitative holomorphic implicit-function theorem. It constructs a unique small triple solving (CK14) with

$$\|u(B)\|_{W^{1,p}_a} + \|\chi(B)\|_{W^{2,p}_r} + \|\lambda(B)\|_{W^{-1,p}_r} \le C\|B\|_{W^{1,p}_r}. \tag{CK20}$$

62.6 6. Noether restores both omitted equations exactly

Put

$$R = J_a(u) - \mathcal{A}(u)^{\mathsf{t}}\lambda, \qquad q = d^{\mathsf{t}}_{V(u)}\lambda.$$

The second equation in (CK14) makes q a coarse constant color. The first gives

$$\mathcal{E}_a(u)^{\mathsf{t}}R = d_a\psi \tag{CK21}$$

for a mean-zero $\psi \in L^p_a$; here $\psi = G_a\delta_a\mathcal{E}_a(u)^{\mathsf{t}}R$. The exact Wilson gauge identity and (CK6) give

$$d^{\mathsf{t}}_U R = -S(u)^{\mathsf{t}}q. \tag{CK22}$$

Pair (CK22) with an arbitrary constant fine gauge color ζ. In logarithmic coordinates a constant gauge acts exactly by a commutator with u, with the sign fixed by the gauge convention. Therefore the left side is, up to that irrelevant common sign,

$$\langle d_a\psi, [\zeta, u]\rangle = \langle \psi, [\zeta, \delta_a u]\rangle = 0.$$

This is an exact equality, not a small-error bound. It follows that the physical mean of $S(u)^{\mathsf{t}}q$ is zero. By (CK8) its mean map on constants is $I + O(rM)$, hence invertible on the chosen radius. Thus

$$q = 0. \tag{CK23}$$

Now define the logarithmic gauge differential $\mathfrak{d}_u = \mathcal{E}_a(u)^{-1}d_U$. It has the exact form

$$\mathfrak{d}_u = d_a + \mathcal{B}_u,$$

where $\mathcal{B}_u$ is a zero-order sum of shifted multipliers with $W^{1,p}$ coefficient norm $O(M)$. Indeed, $\mathcal{E}_a(u)^{-1} - I = O(au)$; multiplying this by d_a uses $ad_a = \mathrm{shift} - I$, while $d_U - d_a$ is already zero order. No derivative of u beyond its $W^{1,p}$ norm is introduced.

In one oriented bond component this can be checked without a symbolic order argument. Put $z = a\,\mathrm{ad}\,u$ and $\mathcal{E}_a(u)^{-1} = z/(e^z - 1)$. Then

$$\mathfrak{d}_u\psi = d_a\psi + \frac{\mathcal{E}_a(u)^{-1} - I}{a}(\psi_+ - \psi) + [u, \psi_+].$$

The quotient coefficient is analytic in u, vanishes linearly at zero, and has $W^{1,p}$ norm $O(M)$. The shift difference in parentheses is unscaled. This also verifies the exact constant-gauge commutator used above.

Equations (CK21)–(CK23) imply

$$\Delta_a\psi = -\mathcal{B}_u^{\mathsf{t}} d_a\psi.$$

By (CK10) and duality, $\mathcal{B}_u^{\mathsf{t}}$ has $W^{-1,p}$ operator norm $O(M)$. Hence the mean-zero inverse gives

$$\|\psi\|_{L^p} \le C\|G_a\mathcal{B}_u^{\mathsf{t}} d_a\psi\|_{W^{1,p}} \le CM\|d_a\psi\|_{W^{-1,p}} \le CM\|\psi\|_{L^p}.$$

The fixed small radius forces $\psi = 0$, and therefore $R = 0$. Equation (CK7) now applies and proves the genuine physical KKT equation, together with the full coarse Ward equation.

62.7 7. Restore the actual coarse target

Let

$$g = \exp(E_{\mathrm{mix}}\chi), \qquad e^{aw} = (e^{au})^g. \tag{CK24}$$

Since $Ng = e^{\chi}$, the last equation of (CK14) gives exactly $\mathcal{M}_r(e^{aw}) = e^{rB}$. The mixed $W^{2,p}$ lift and the analytic gauge-map estimate give $\|w\|_{W^{1,p}} \le C\|B\|_{W^{1,p}}$. The coarse multiplier transforms as

$$\lambda_B = \mathrm{Ad}_{e^{\chi}}\,\lambda. \tag{CK25}$$

The invariant bilinear transpose and raw covariance give the physical KKT equation at w with this multiplier. Coarse $W^{1,p}$ multiplier bounds, by duality as above, preserve its $W^{-1,p}$ norm. All operations are holomorphic on the same uniform ball. This proves (CK3).

If desired, GS subsequently puts w into the exact $R_0\delta$ slice by a gauge fixing every retained center. This preserves the target and the weak analytic norm. It does not by itself create a curvature supremum.

62.8 8. Identification with the genuine real regular orbit

The real source orbit can now be identified without assuming a bound for its multiplier. Let A satisfy the small real weak hypotheses of WC, put

$$B = q_{r,\mathrm{raw}}(A), \qquad V = e^{rB},$$

and let U belong to the original genuine periodic regular critical orbit with this target.

First, the finite-dimensional constraint qualification is uniform. The raw coarsening map is holomorphic on the common $W^{1,p}$ ball, has derivative Q_{raw} at zero, and has uniformly bounded quadratic remainder there. FW therefore gives

$$Dq_{r,\mathrm{raw}}(w)\, H_{\mathrm{raw}} = I + O(M) \tag{CK26}$$

on coarse $W^{1,p}$ for every admitted small w. Its right side is invertible on one fixed radius. Multiplication by the fine and coarse exponential differentials converts this to a right inverse for the physical group differential. Thus ordinary finite-dimensional Lagrange multiplication applies to a genuine constrained critical field in this chart. The physical multiplier is unique. The Wilson gauge identity then gives its exact Ward equation, since the retained-center embedding N^{t} is injective.

WC provides a real same-target representative w_0 with small $W^{1,p}$ norm and the scale-improved curvature bound. Put it into ordinary fine Coulomb gauge by the strong version of the usual fixed point:

$$G_a\delta_a : W_a^{1,p} \longrightarrow W_a^{2,p}, \qquad \overline{\phi} = 0, \qquad (e^{aw_0})^{e^{\phi}} = e^{au}, \quad \delta_a u = 0.$$

This is the GS contraction with the ordinary mean-zero inverse in place of the retained-node operator. It has the same uniform quadratic gauge estimate. In particular $\|\phi\|_{W^{2,p}} + \|u\|_{W^{1,p}} \le C\|w_0\|_{W^{1,p}}$. The node restriction estimate GS5 gives

$$h = Ne^{\phi} \in W_r^{2,p}, \qquad \|h - I\|_{W_r^{2,p}} \le C\|w_0\|_{W_a^{1,p}}, \qquad \mathcal{M}_r(e^{au}) = V^h.$$

Choose a constant real group element c near the identity so that

$$\chi = \log(h^{-1}c^{-1}), \qquad \overline{\chi} = 0. \tag{CK27}$$

It exists uniquely by the finite-dimensional implicit-function theorem: the derivative with respect to $\log c$ at the identity is minus the identity. The normalized coarse mean has norm one, so the constants are independent of the number of sites. The bound is $\|\chi\|_{W_r^{2,p}} + |\log c| \le C\|h - I\|_{W_r^{2,p}}$. Set $u' = \mathrm{Ad}_c\, u$. It is still co-closed, and the multiplication order is exact:

$$\mathcal{M}_r(e^{au'})^{e^{\chi}} = (V^{ch})^{h^{-1}c^{-1}} = V.$$

It remains to put the actual multiplier into the small coordinate ball. The flat block estimate CK18 applied to $(0, 0, \lambda)$ gives

$$\|\lambda\|_{W_r^{-1,p}} \le C\left(\|P_a Q_{\mathrm{bl}}^{*}\lambda\|_{W_a^{-1,p}} + \|\delta_r\lambda\|_{W_r^{-2,p}}\right). \tag{CK28}$$

The corresponding operator at u', with $P_a\mathcal{E}_a(u')^{\mathsf{t}}\mathcal{A}(u')^{\mathsf{t}}\lambda$ and $P_{\mathrm{sc}}^{0}d_{V(u')}^{\mathsf{t}}\lambda$, differs by $O(M)$ in these same norms. At this genuine critical field the second component is zero, while the first is $P_a\mathcal{E}_a(u')^{\mathsf{t}}J_a(u')$. Absorbing the perturbation in (CK28), and using CK16, proves

$$\|\lambda\|_{W_r^{-1,p}} \le C\|u'\|_{W_a^{1,p}}. \tag{CK29}$$

The multiplier is finite at every cutoff before absorption. No initial uniform current or multiplier estimate is being assumed.

Thus (u', χ, λ) lies in the CK uniqueness ball when the real WC input radius is chosen sufficiently small. It solves the modified equations by its actual KKT and Ward identities. It must be the branch constructed in CK20. Restoring the raw target by CK24 therefore puts $w(B)$ in the same genuine regular critical orbit. All gauges used in this real identification are compact, so the source curvature improvement transfers unchanged:

$$\|F_a(w(q_{r,\mathrm{raw}}(A)))\|_{\infty} \le C\|F_a(A)\|_{\infty} \tag{CK30}$$

for the stated small real WC inputs.

This is a real curvature conclusion. It does not extend CK30 to a complex ball by assertion.

62.9 9. Complex gauge-invariant identities

On a common small complex input ball, the same-target CK family and the source analytic family in CX are holomorphic. For real inputs, Section 8 identifies their genuine critical orbit. Therefore every finite-lattice holomorphic gauge-invariant observable has the same value on the two families throughout the common complex ball. This follows from the finite-dimensional holomorphic identity theorem on the open real input set, applied at each cutoff.

Examples include traces of closed parallel transports and complete gauge-invariant traces formed from plaquette logarithms and transports on their common branch. This equality can transfer an invariant estimate established in the CX representative. It does not compare ordinary matrix Sobolev or supremum norms through an unbounded complex gauge.

62.10 10. What this does and does not resolve

The formulation proves a same-target holomorphic $W^{1,p}$ critical branch without the false arbitrary-force inverse and without assuming a bounded anchor for an independently chosen source gauge. If the input is $B = q_{r,\mathrm{raw}}(A)$, the actual raw $W^{1,p}$ coarsening bound gives the corresponding holomorphic branch on a cutoff-independent fine $W^{1,p}$ ball.

Section 8 identifies the branch with the original regular orbit for the small real WC inputs and supplies the real curvature supremum there. A uniform complex matrix curvature supremum and the complex strong PR hypotheses remain separate conclusions to establish. The theorem claims neither a uniform F_∞ bound from $W^{1,p}$ alone nor a completed Yang–Mills construction.